

System Analysis Models

Food Donation and Lifecycle Management System

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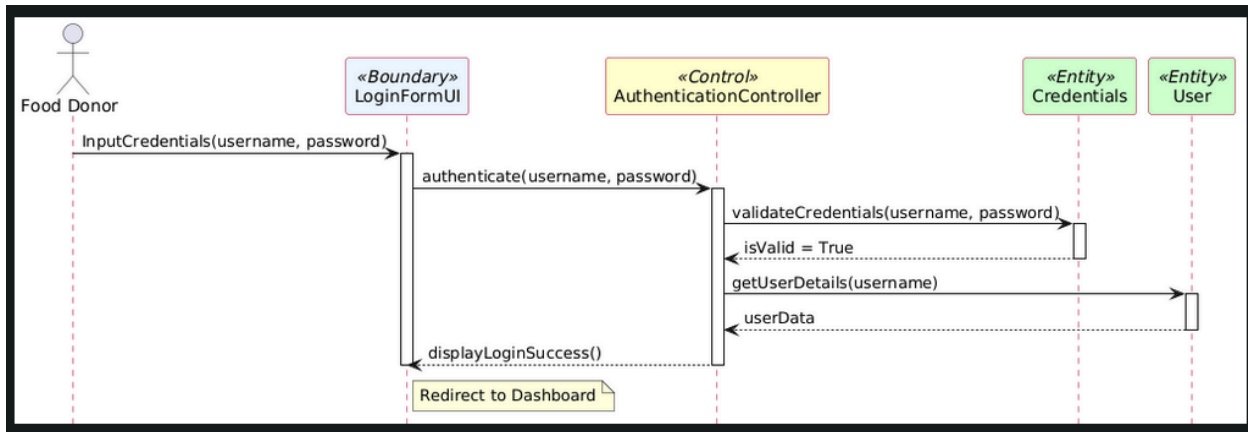
NIHIRA NERALWAR (241071049)

SE Lab 4B - Group C

1. Food Donor Use Cases

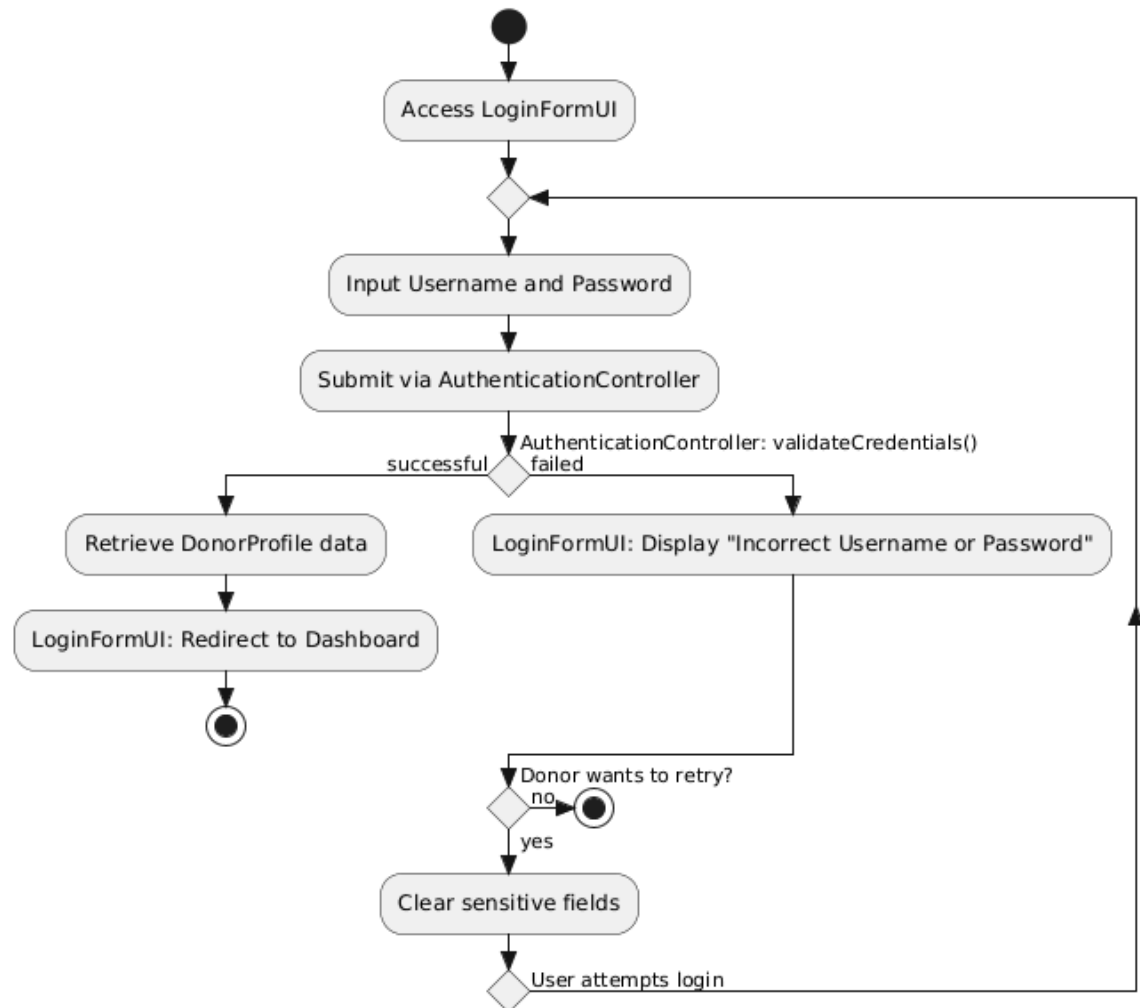
1.1 Register / Login (Food Donor)

SEQUENCE DIAGRAM -



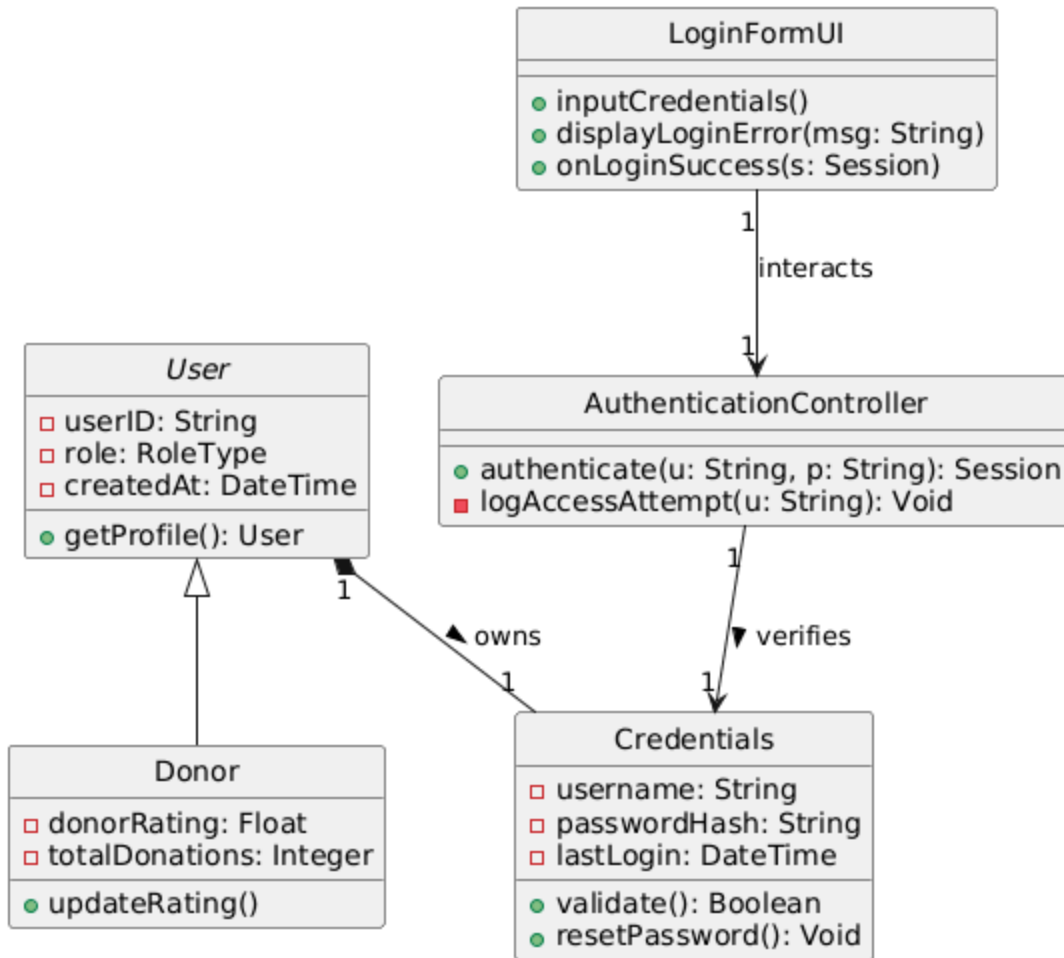
ACTIVITY DIAGRAM -

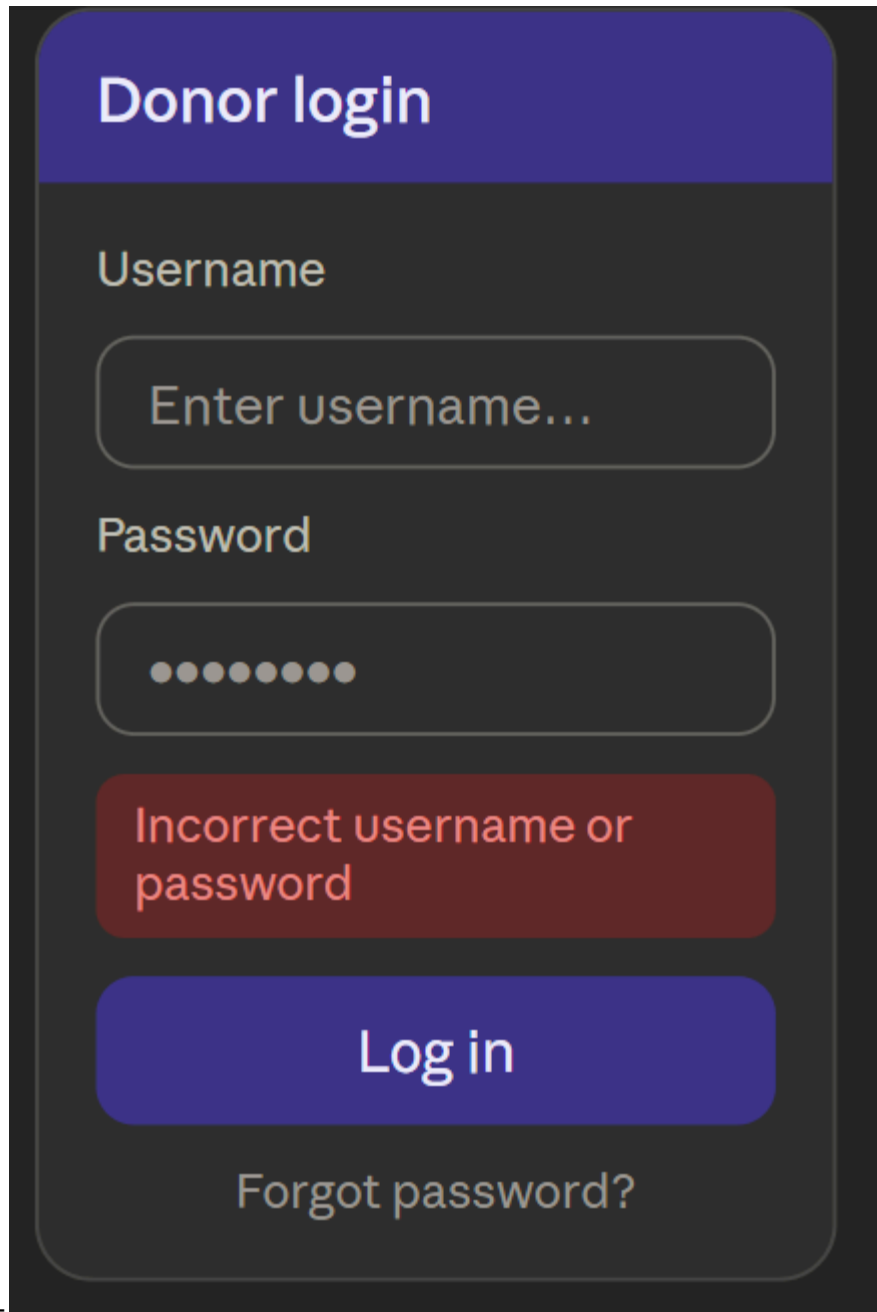
Activity Diagram - Food Donor Login



CLASS DIAGRAM -

Class Diagram: Login Use Case (Detailed)





The image shows a mobile app login screen for donors. It has a dark grey background. At the top is a purple header with the text 'Donor login' in white. Below the header, the word 'Username' is displayed in a light grey font. Underneath is a rounded rectangular input field with the placeholder text 'Enter username...'. Below the username field, the word 'Password' is displayed in a light grey font. Underneath is another rounded rectangular input field, but it contains eight dots, indicating a password field. Below the password field is a red rectangular box with rounded corners containing the text 'Incorrect username or password' in a light red font. Below this error message is a large purple button with rounded corners and the text 'Log in' in white. At the bottom of the login area is the text 'Forgot password?' in a light grey font.

Donor login

Username

Enter username...

Password

Incorrect username or password

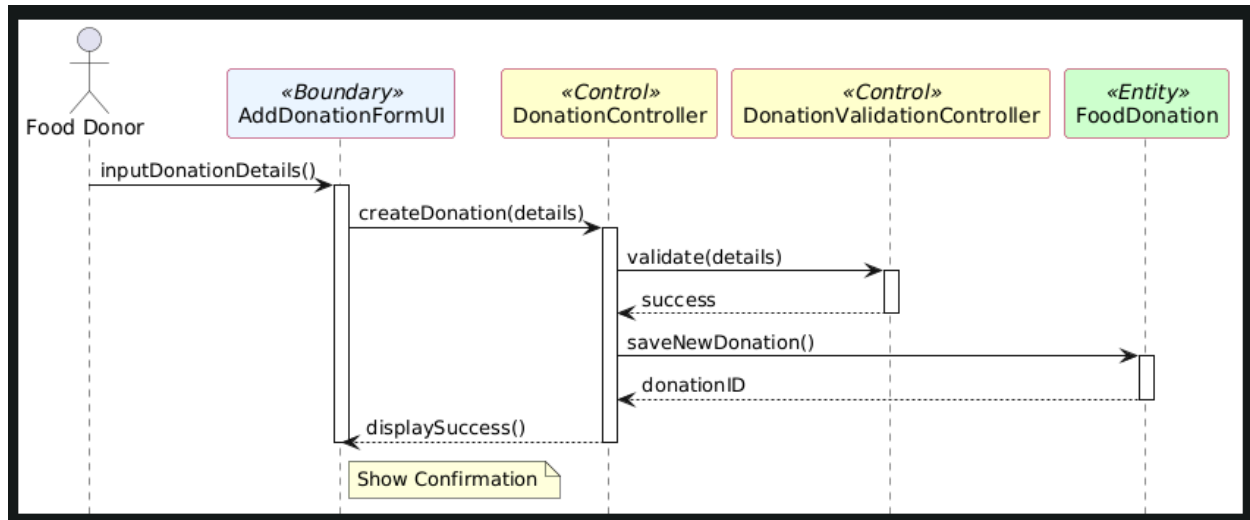
Log in

Forgot password?

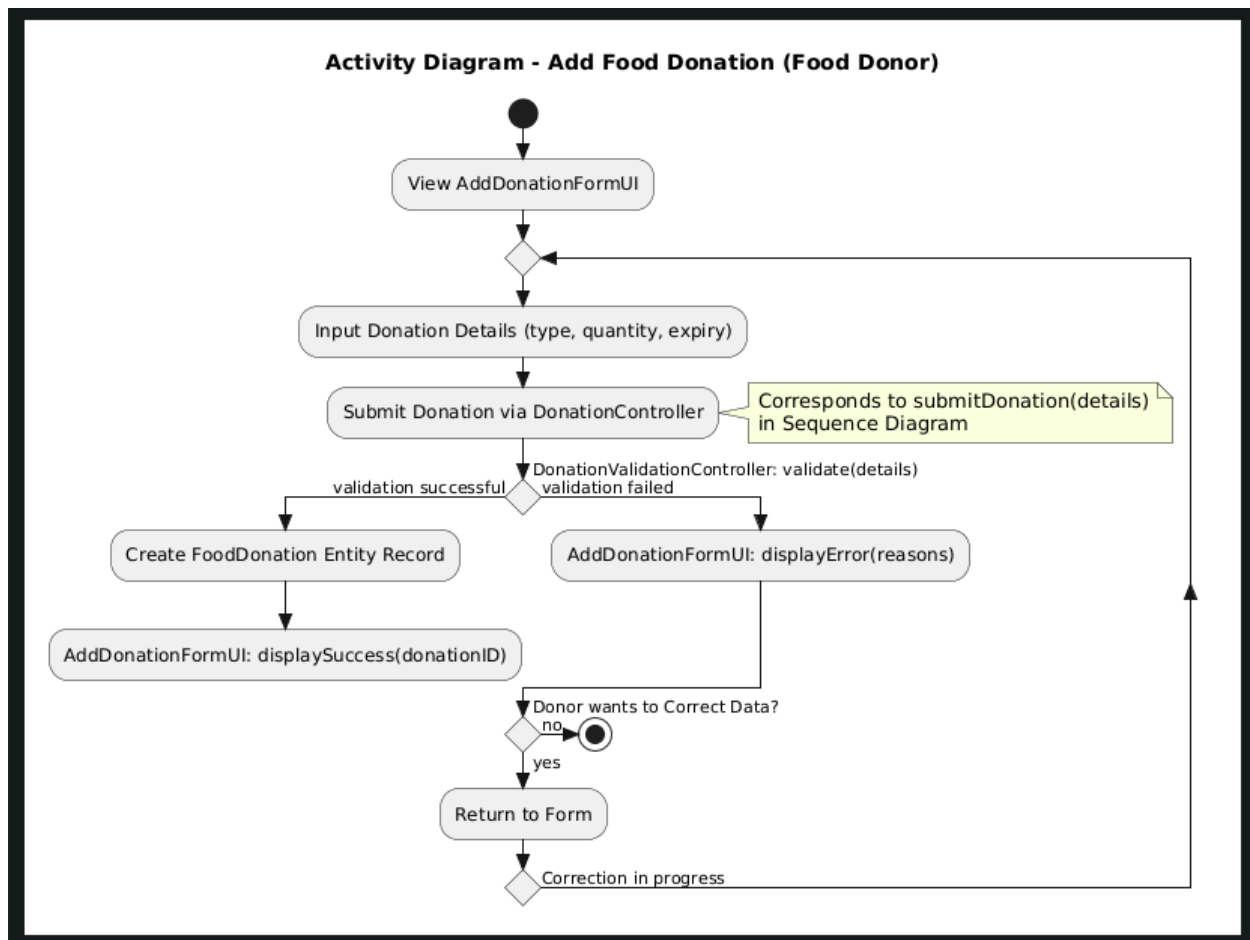
UI -

1.2 Add/Edit/Cancel/View Food request

SEQUENCE DIAGRAM -

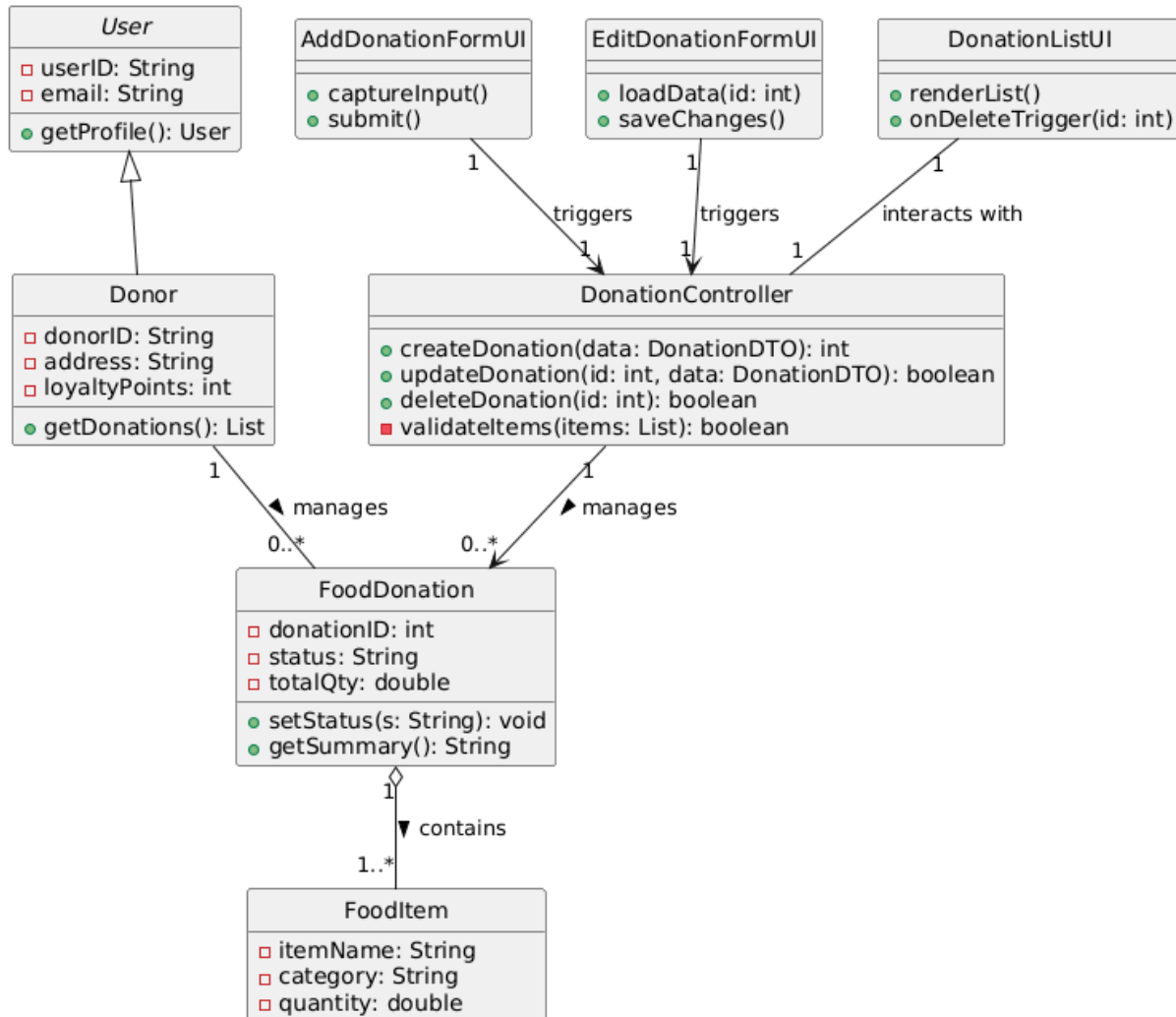


ACTIVITY DIAGRAM -



CLASS DIAGRAM -

Class Diagram: Food Request Management (Add/Edit/Delete)



UI -

The image shows a mobile application interface for adding a food donation. It features a dark grey background with a green header bar at the top containing the text "Add food donation". Below the header, there are three input fields: "Food type" with a dropdown arrow, "Quantity (kg)", and "Expiry date". A red error message "Expiry date is required" is displayed below the "Expiry date" field. At the bottom, there is a green "Submit donation" button and a green confirmation message "Donation added · ID #4421".

Add food donation

Food type ▼

Quantity (kg)

Expiry date

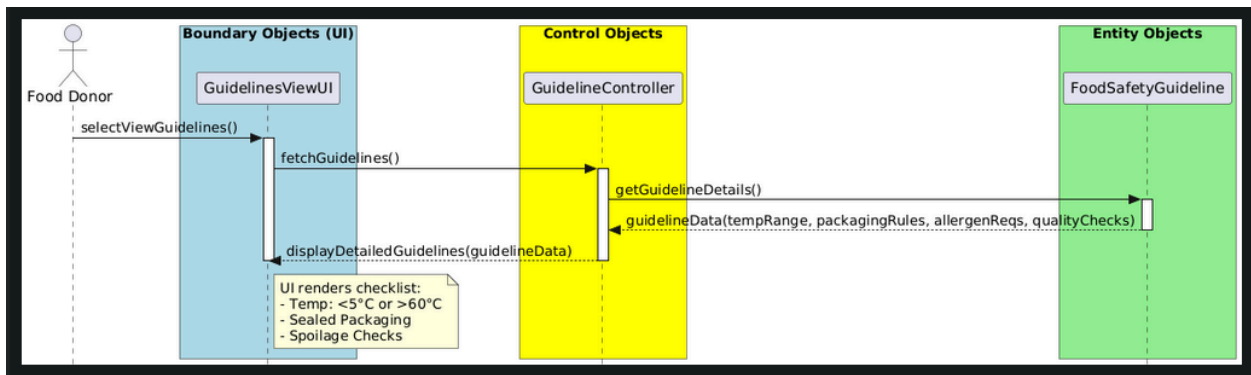
Expiry date is required

Submit donation

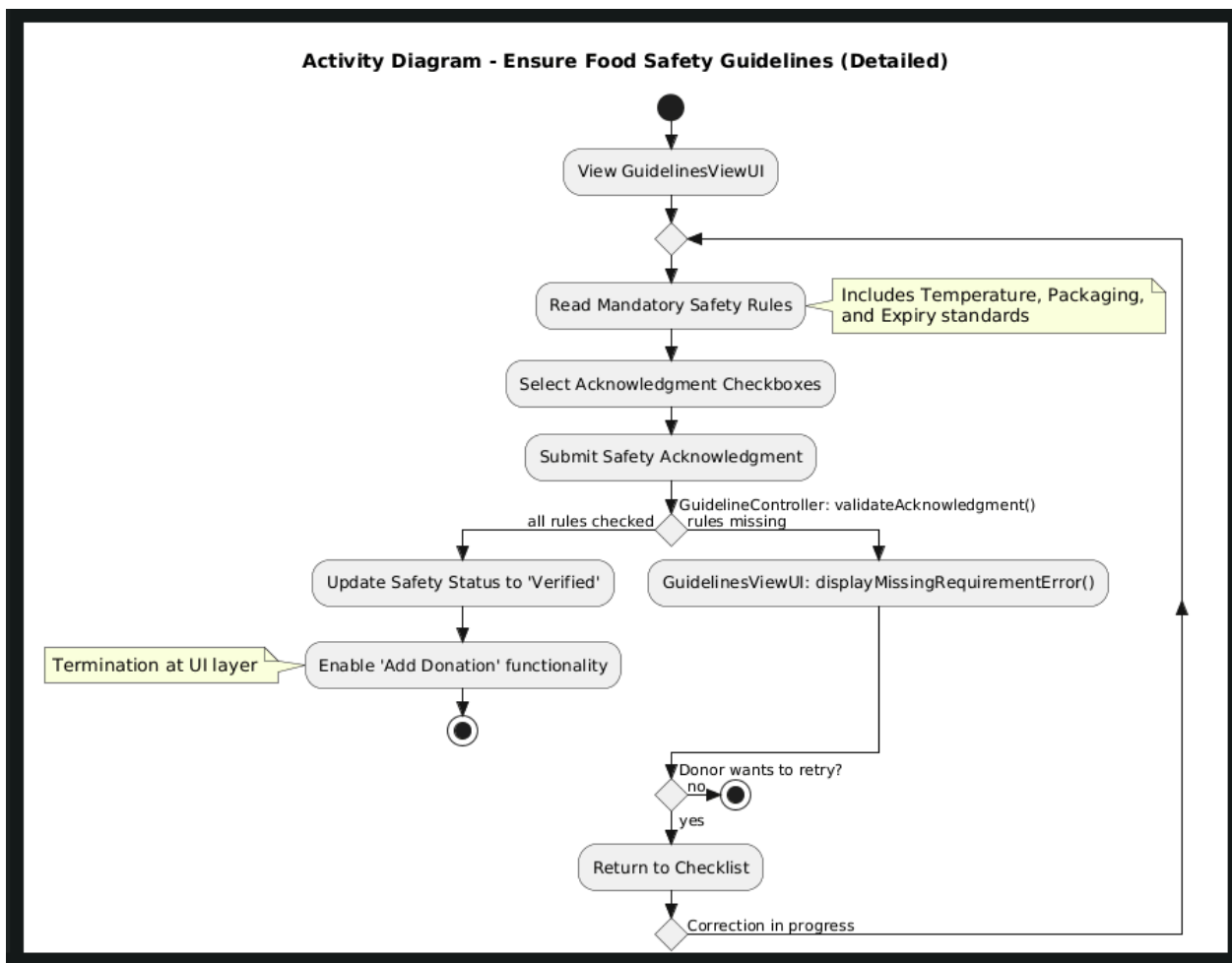
Donation added · ID #4421

1.3 Follow Food Safety Guidelines

SEQUENCE DIAGRAM -

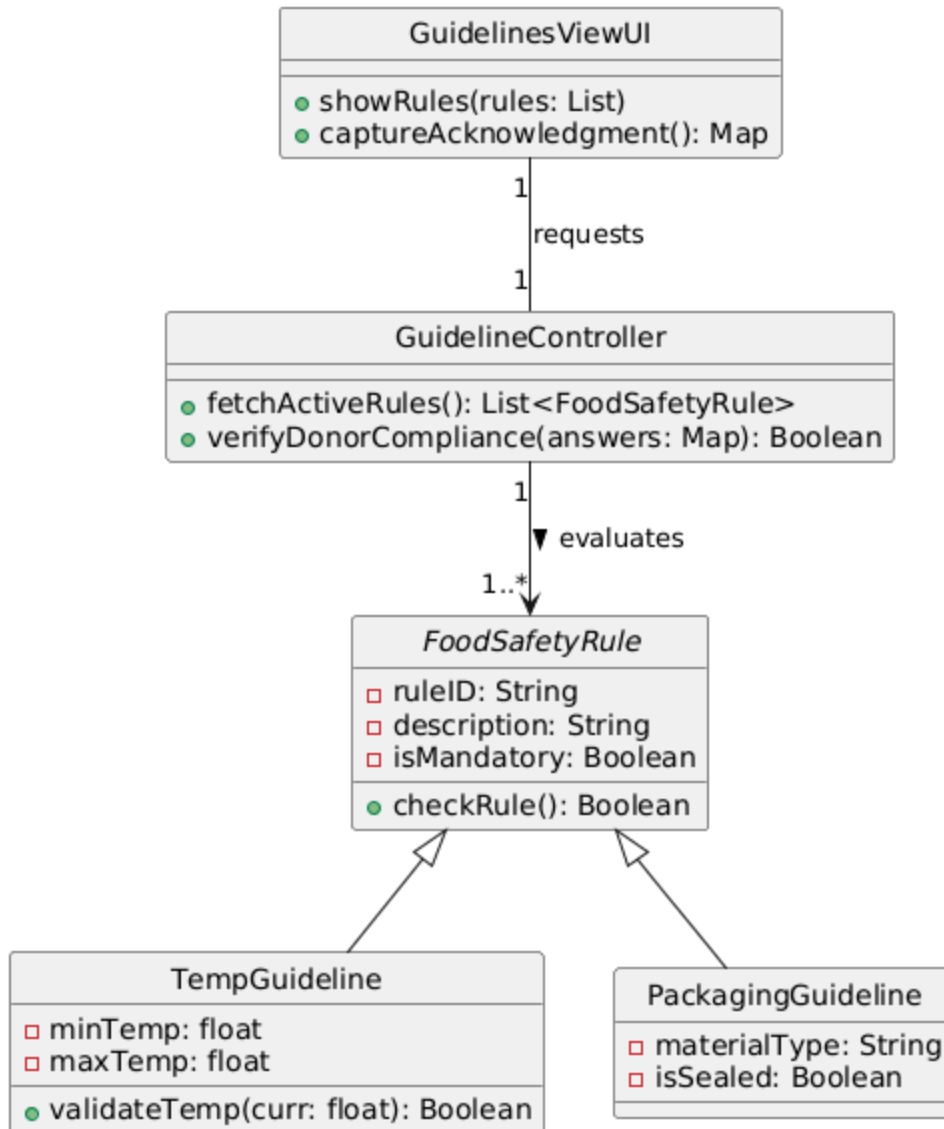


ACTIVITY DIAGRAM -



CLASS DIAGRAM -

Class Diagram: Food Safety Guidelines (Detailed)



The image shows a mobile application interface for food safety rules. It features a dark grey background with a central rounded rectangle containing the form elements. At the top is a brown header with the title 'Food safety rules'. Below this is a section titled 'Read and acknowledge all rules' in light grey. This section contains three checklist items, each with a white square checkbox: 'Temperature standards', 'Packaging requirements', and 'Expiry date policy'. Below the checklist is a brown button with the text 'All boxes must be checked'. Underneath that is a larger brown button labeled 'Submit acknowledgment'. At the bottom is a green button with the text 'Verified · Add Donation unlocked'.

Food safety rules

Read and acknowledge all rules

- ☐ Temperature standards
- ☐ Packaging requirements
- ☐ Expiry date policy

All boxes must be checked

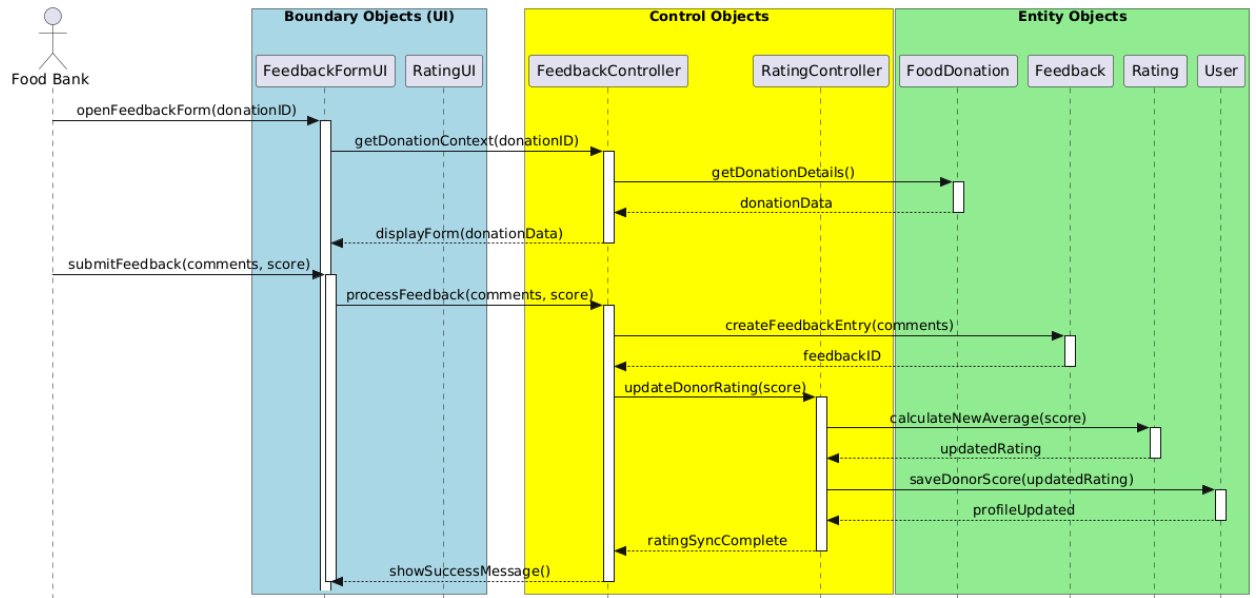
Submit acknowledgment

Verified · Add Donation unlocked

UI -

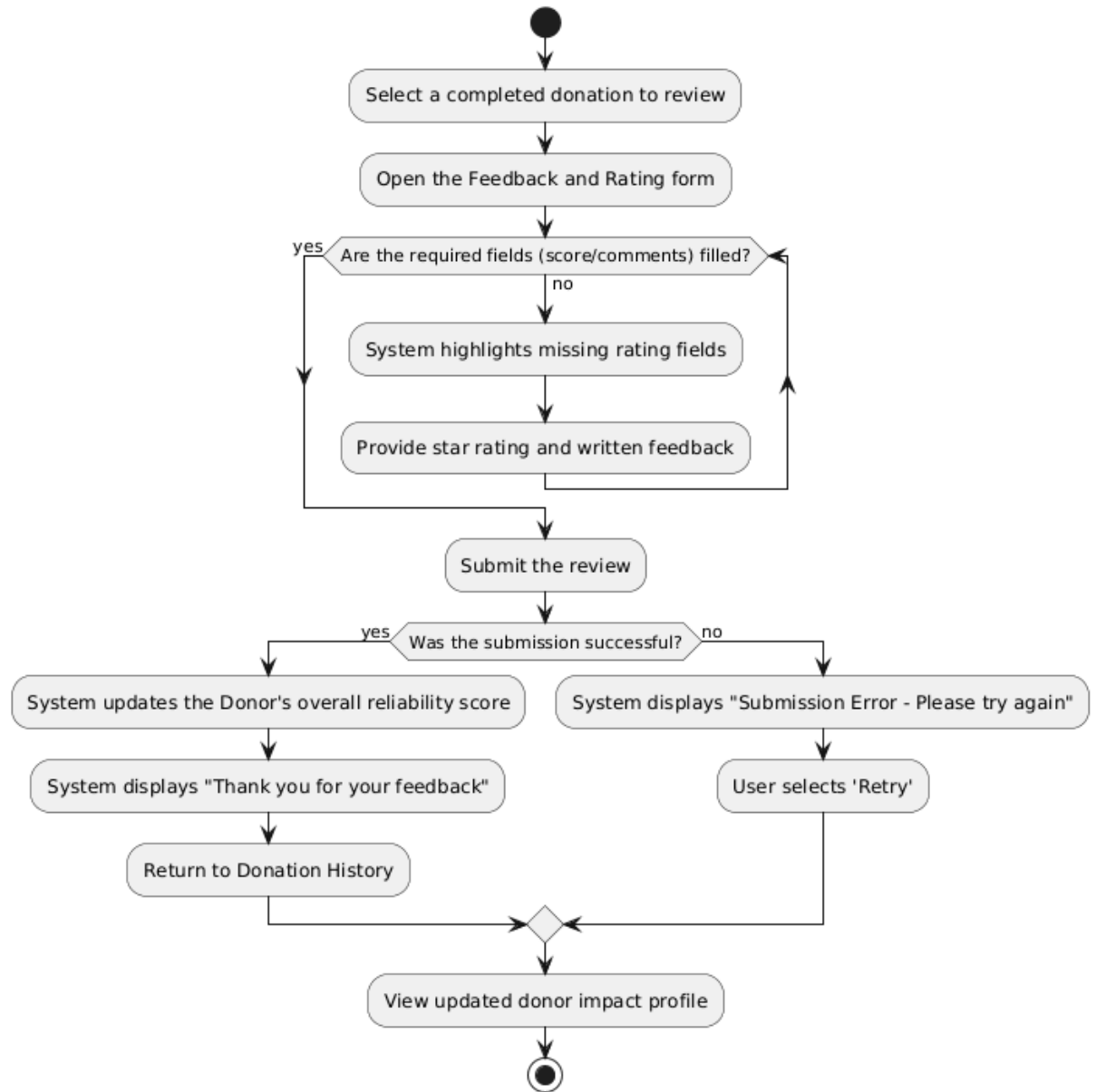
1.4 Feedback & Rating System

A. Sequence Diagram



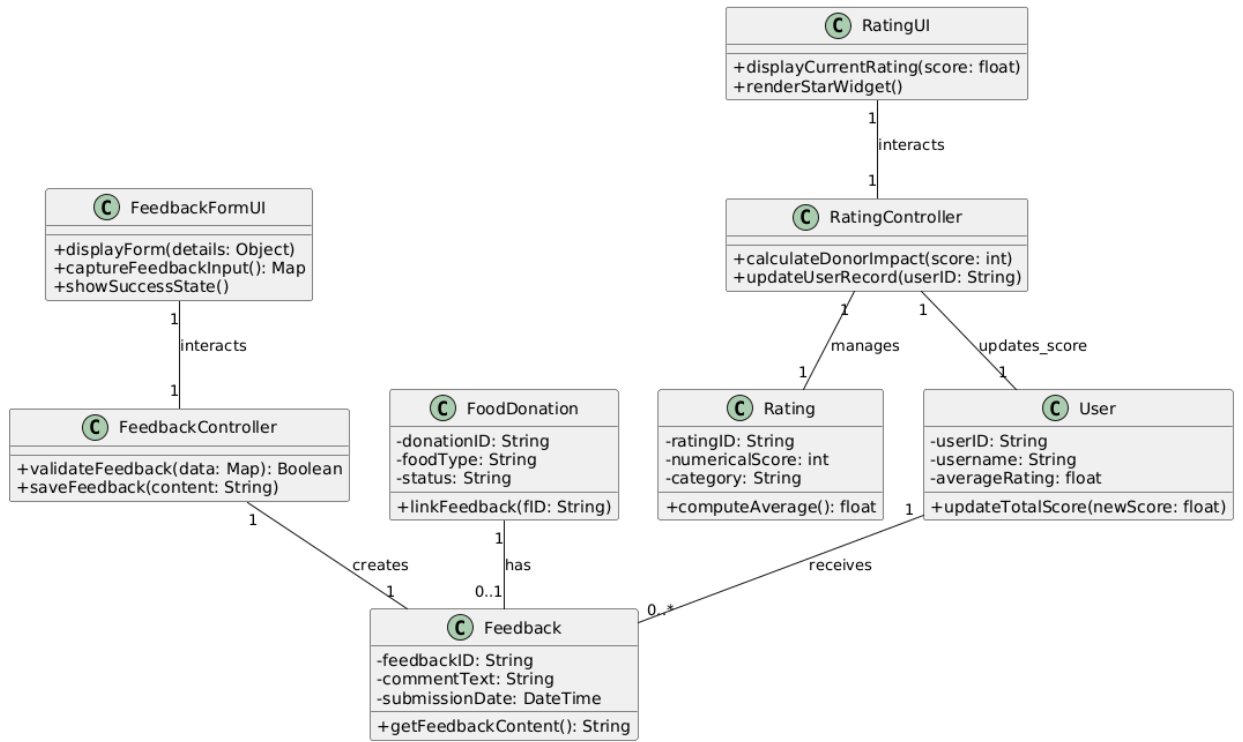
B. Activity Diagram

Activity Diagram - Feedback & Rating System (User Perspective)



C. Class diagram

Class Diagram: Feedback & Rating System (Detailed)



D. UI

Leave feedback

Donation #4401

Completed · 2 Apr

Star rating

★ ★ ★ ★ ★

Written comments...

Rating is required

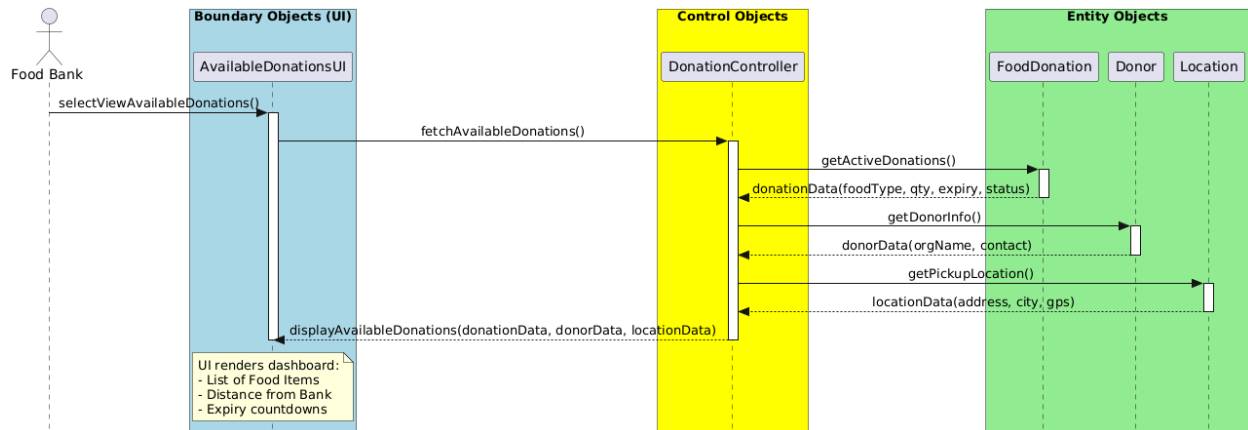
Submit review

Thanks for your feedback!

2. NGO / Food Bank Use Cases

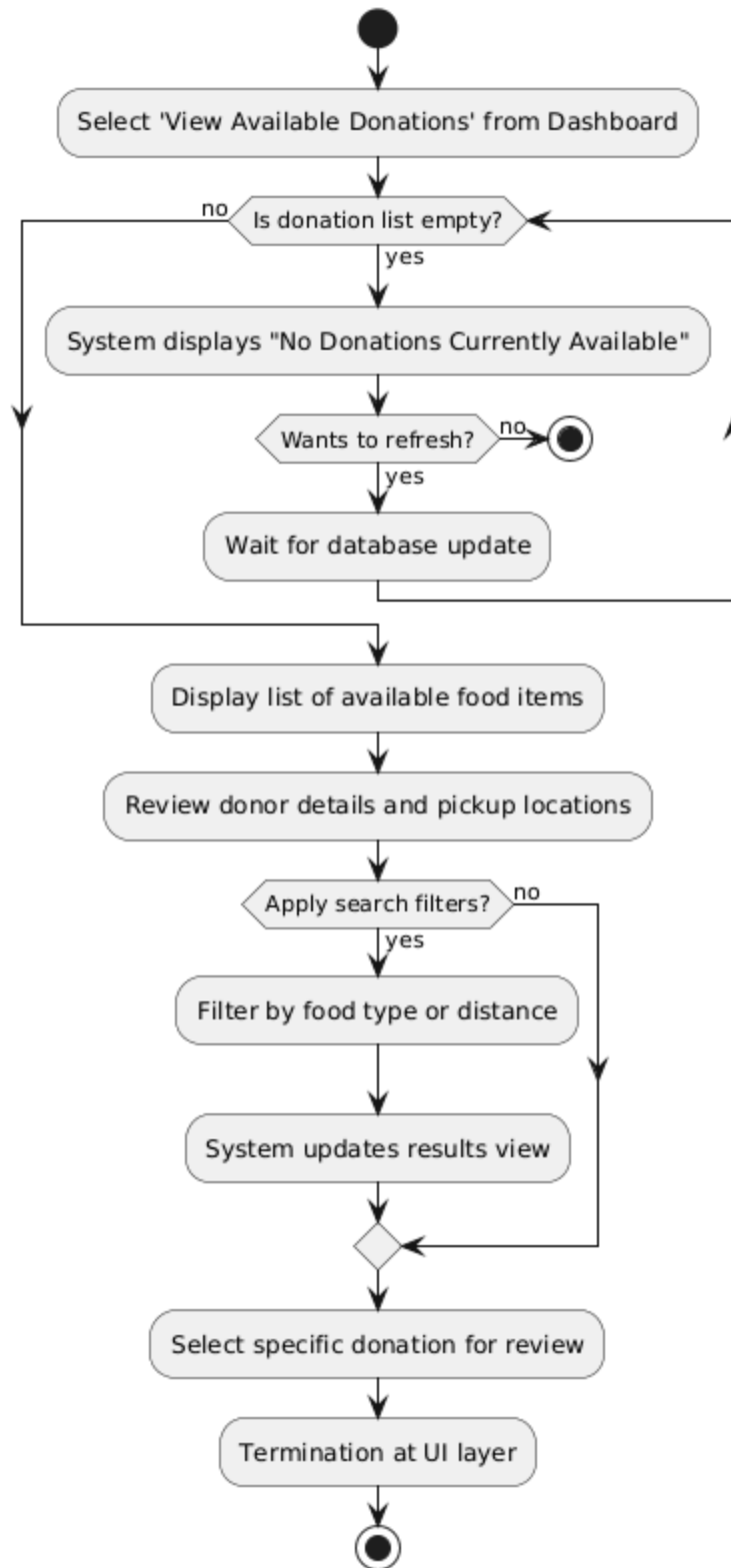
2.1 View Available Donations

A. Sequence diagram



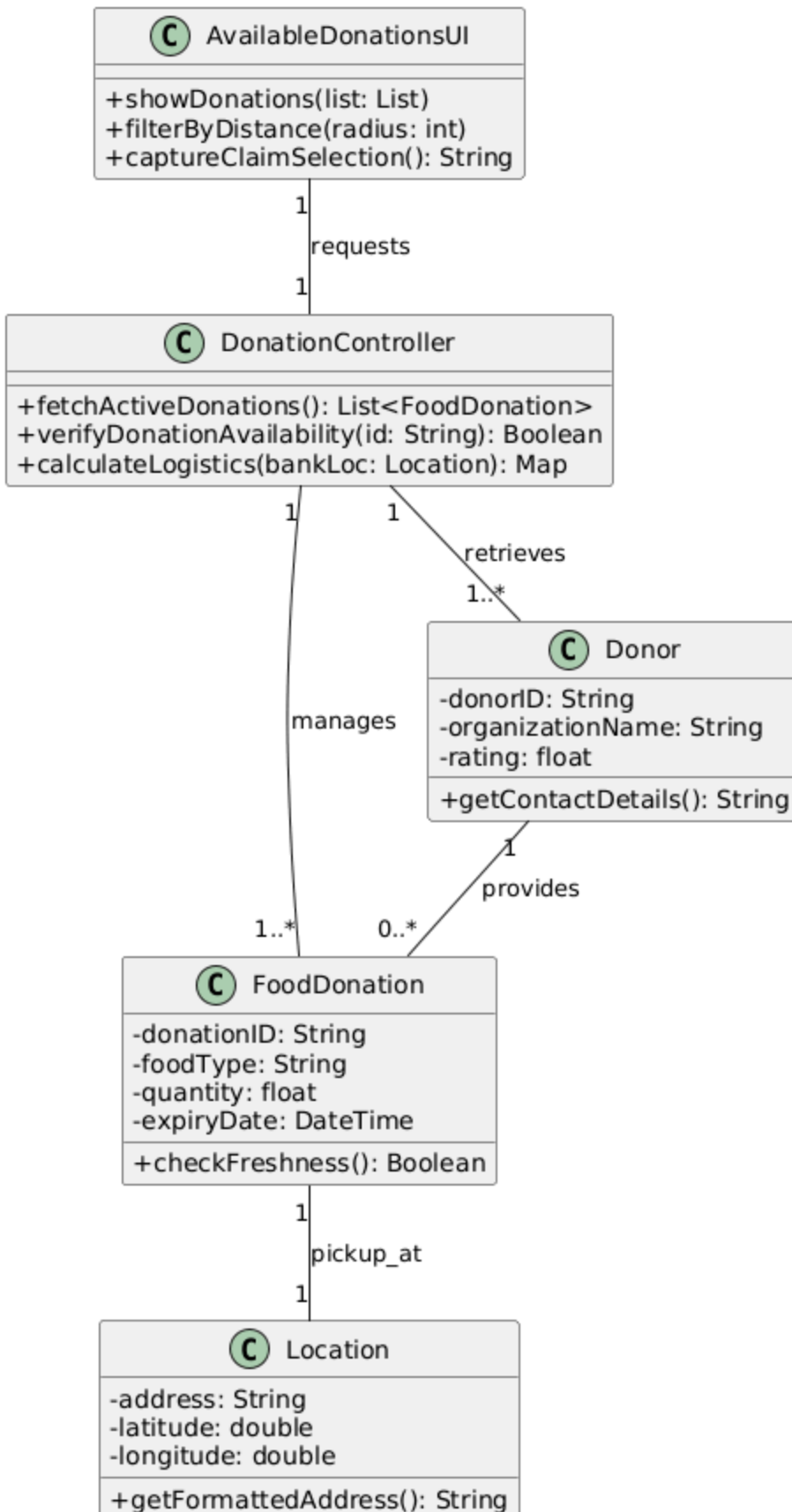
B. Activity diagram

Activity Diagram - View Available Donations (Food Bank Perspective)



C. Class diagram

Class Diagram: View Available Donations (Detailed)



D. UI

Available donations

Food type

Distance

**Batch #441 — Dry
goods**

Donor: Sharma · 2.1 km

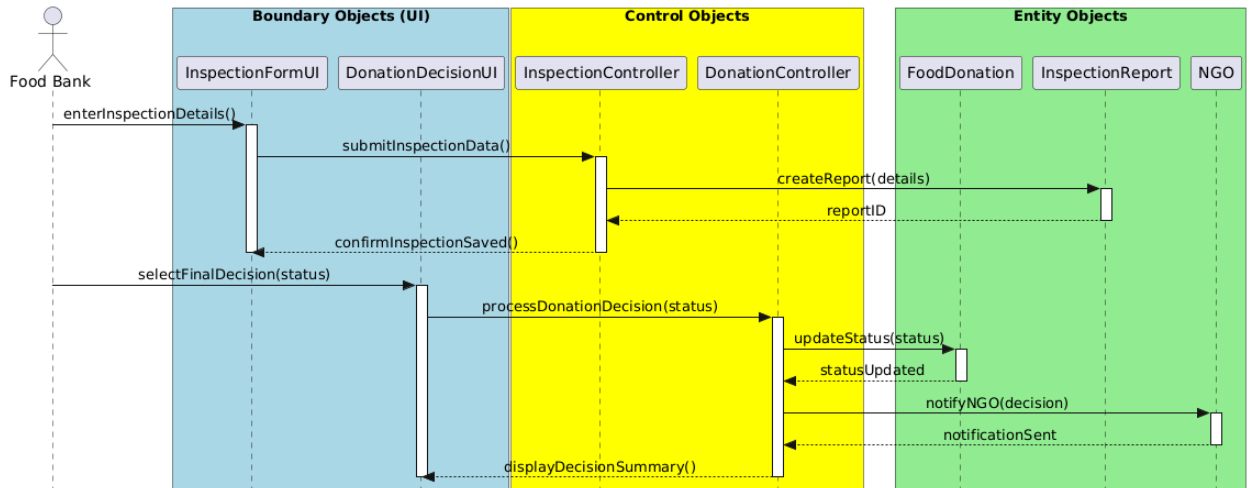
**Batch #442 —
Perishables**

Donor: Patel · 3.8 km

No more donations ·
Refresh

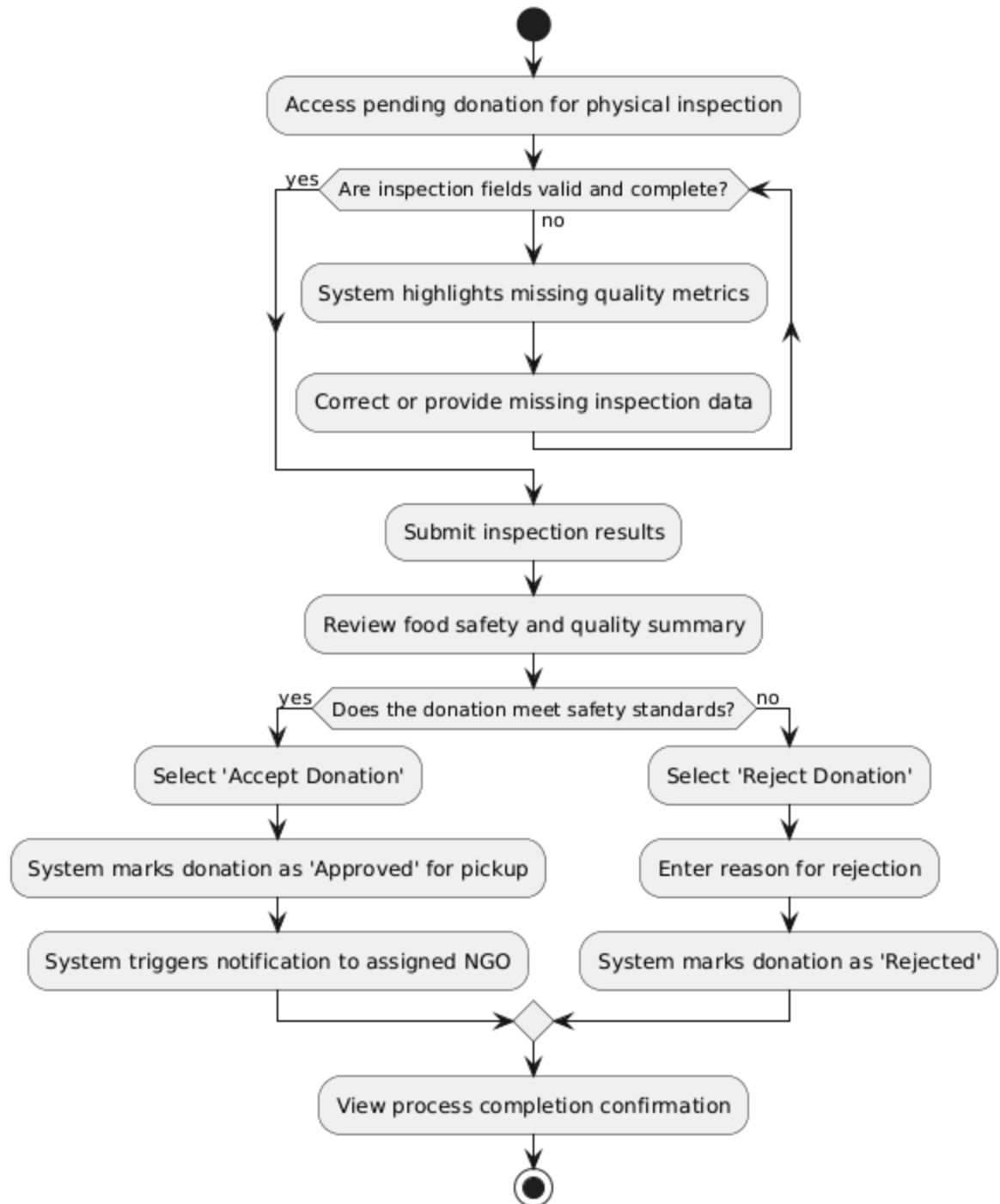
2.2 Accept/Reject Food Donation (Includes Inspect Food Donation)

A. Sequence diagram



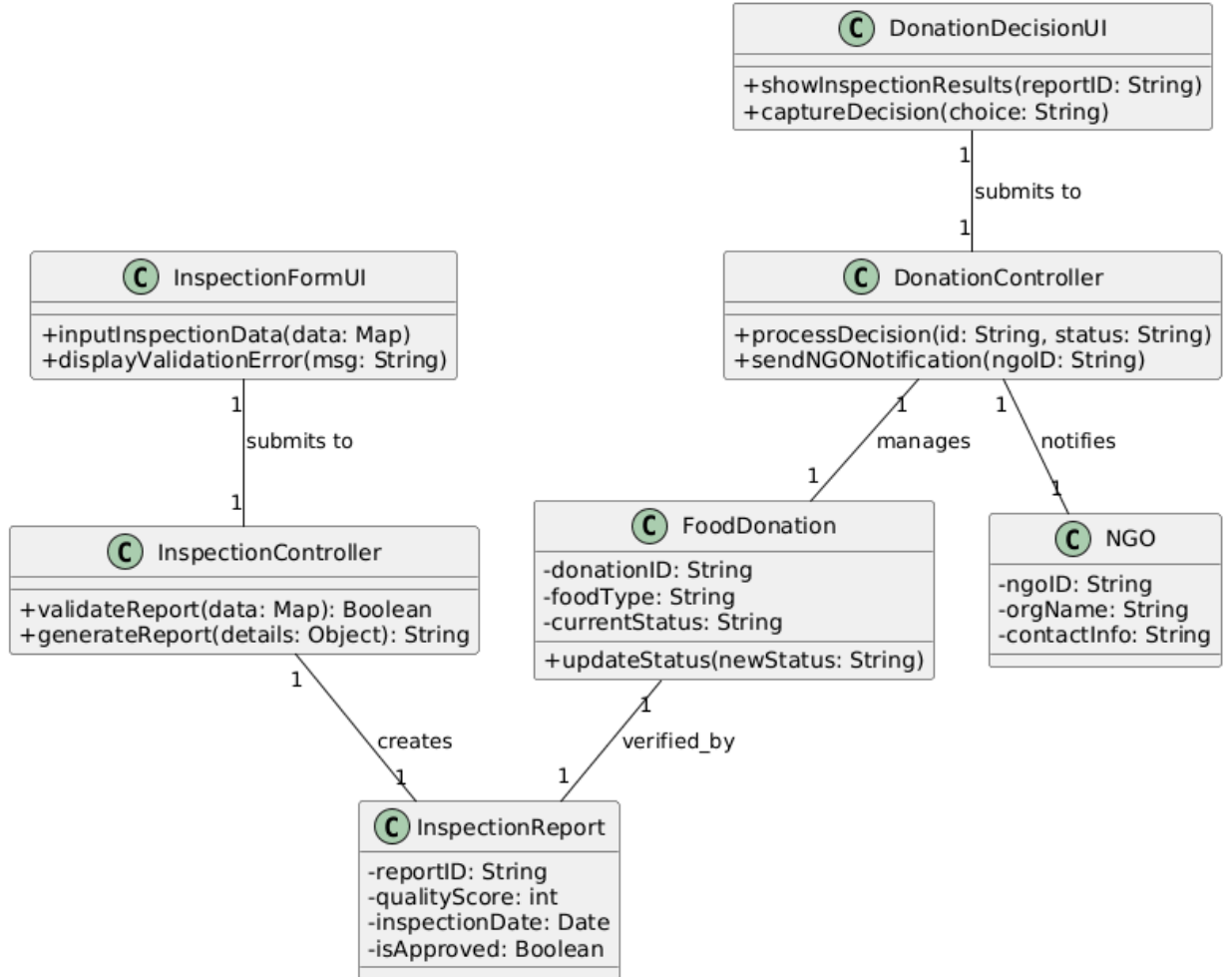
B. Activity diagram

Activity Diagram - Accept/Reject Food Donation (User Perspective)



C. Class diagram

Class Diagram: Accept/Reject Food Donation (Detailed)



D. UI

Donation review

Batch #441 · Dry goods

Qty: 20 kg ✓

Expiry: missing X

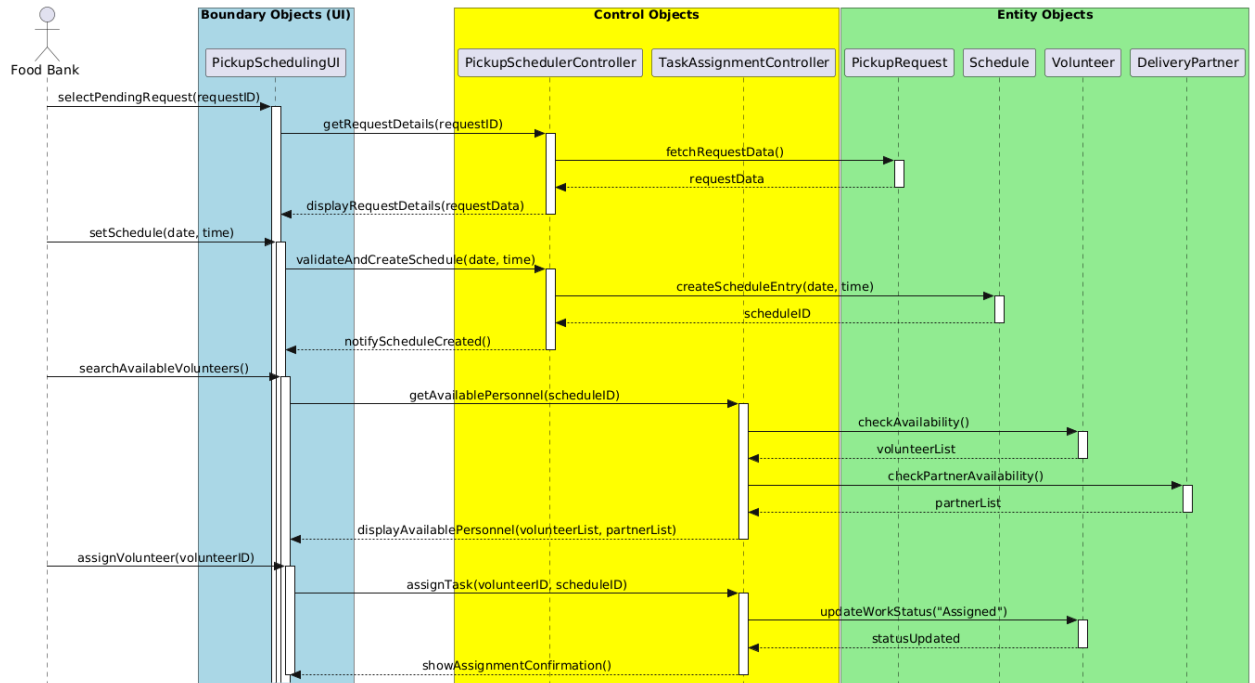
Complete missing fields
before submitting

Accept Reject

Rejection reason...

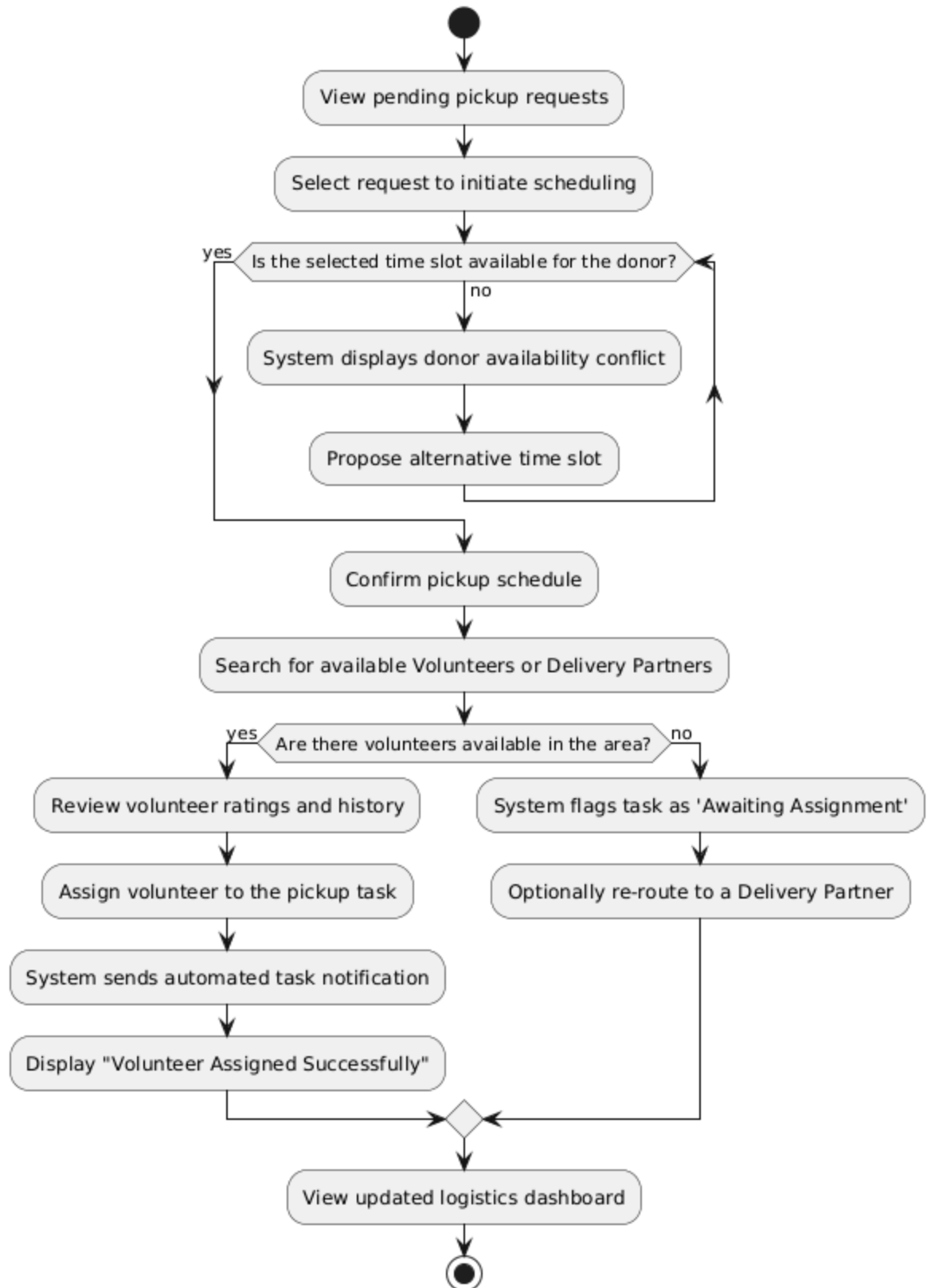
2.3 Manage Volunteers

A. Sequence Diagram

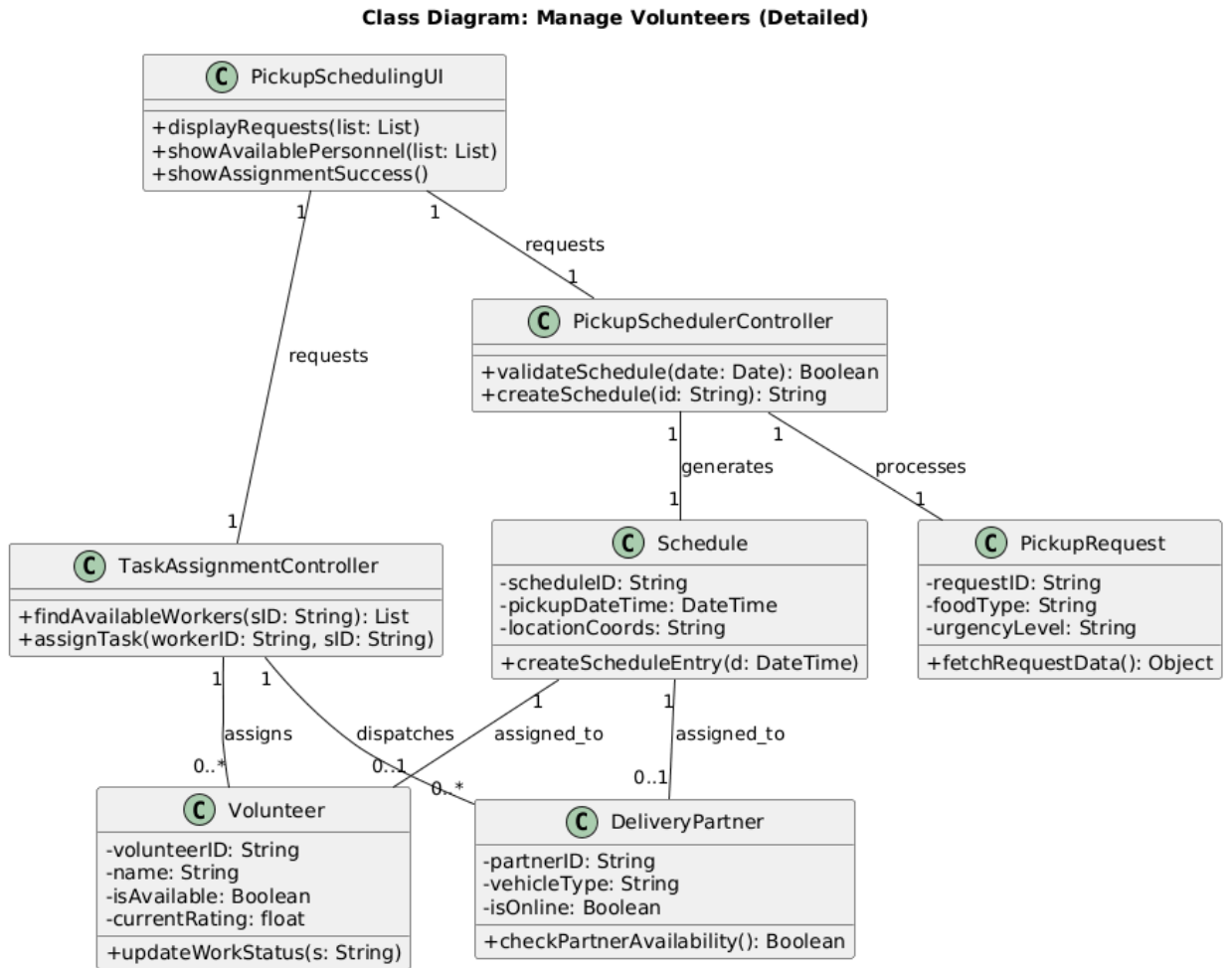


B. Activity Diagram

Activity Diagram - Manage Volunteers (User Perspective)



C. Class diagram



D. UI

Volunteer assignment

Pickup #88 · 10:00 AM

Food Bank A → NGO
Centre

Time conflict — propose
alternative

Assign volunteer



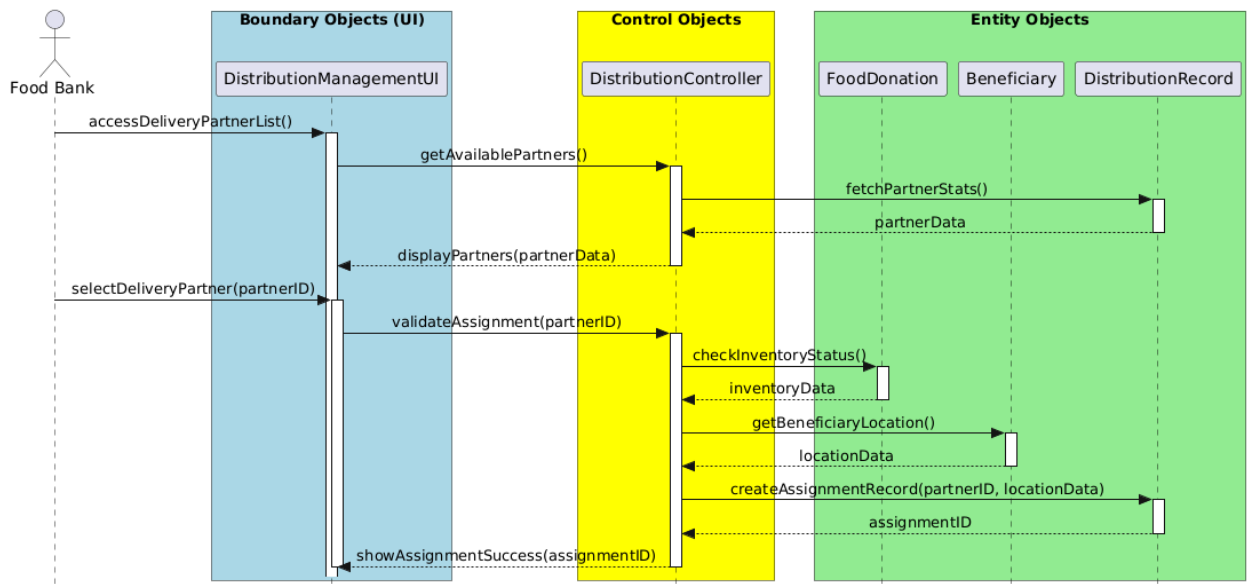
Arjun · ★ 4.8

12 tasks · 1.2 km away

Confirm assignment

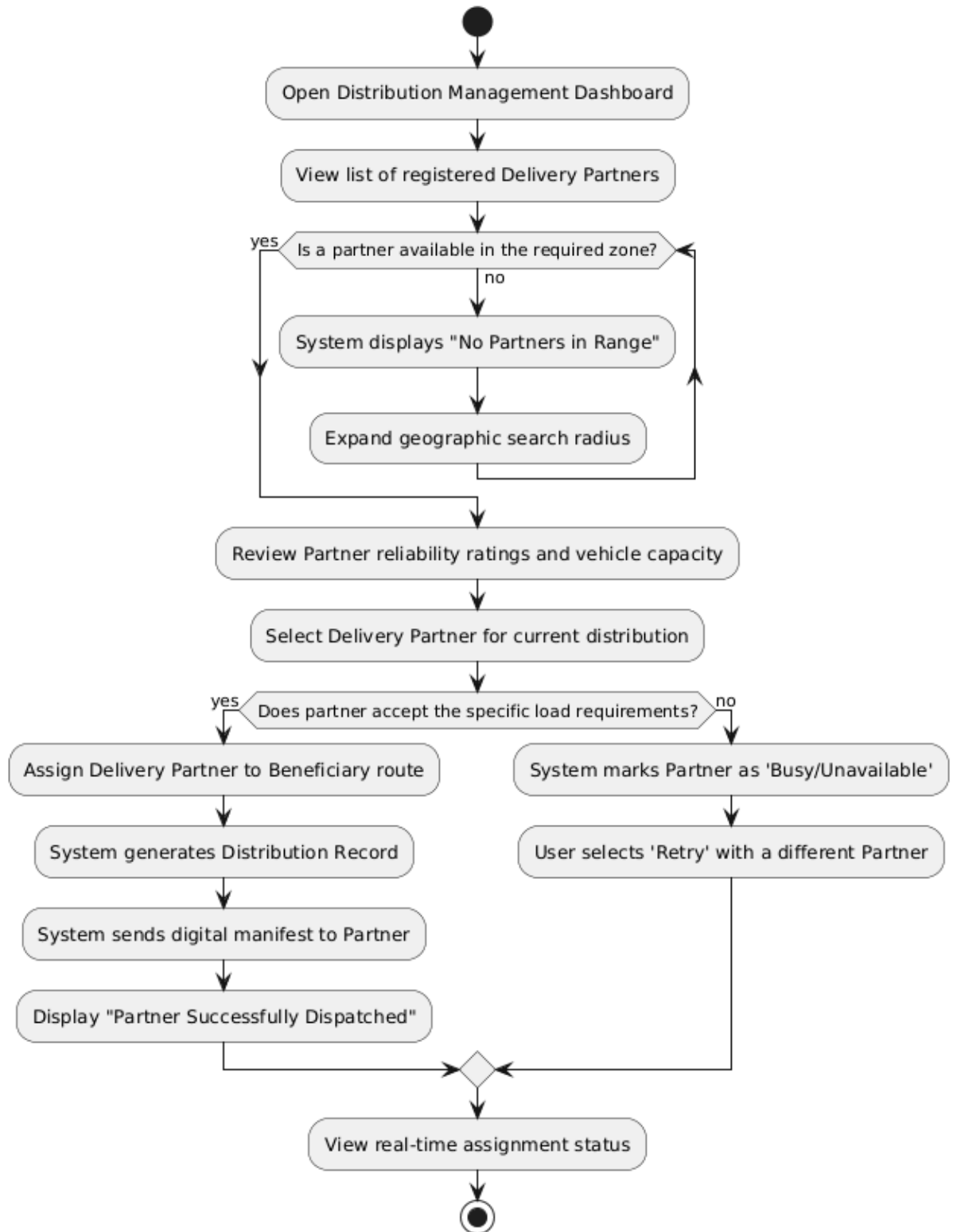
2.4 Manage Delivery Partners

A. Sequence Diagram



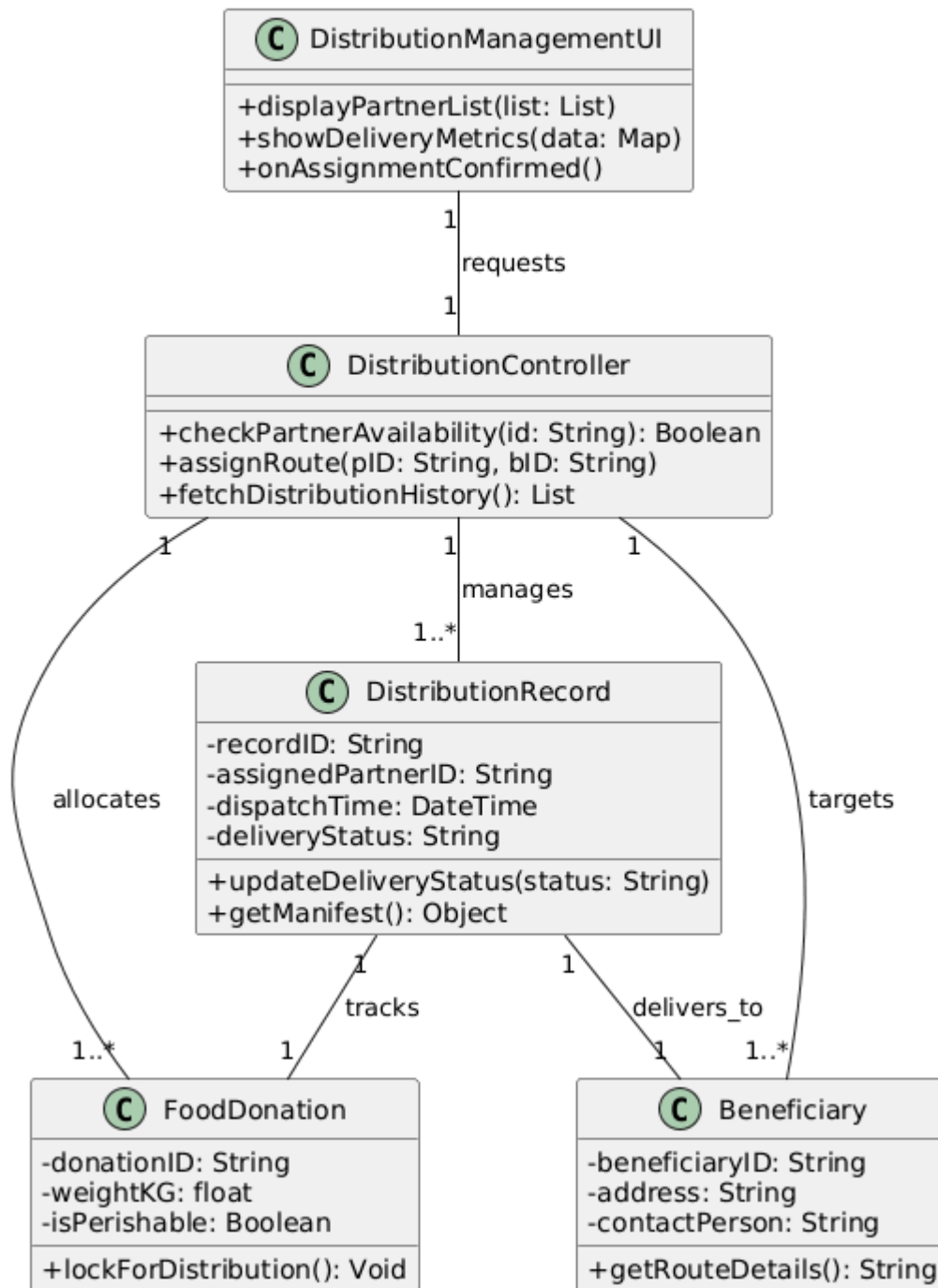
B. Activity Diagram

Activity Diagram - Manage Delivery Partners (User Perspective)



C. Class diagram

Class Diagram: Manage Delivery Partners (Detailed)



D. UI

Delivery partners

FastDeliver Co.

Van · ★ 4.6 · Zone A

QuickHaul Ltd.

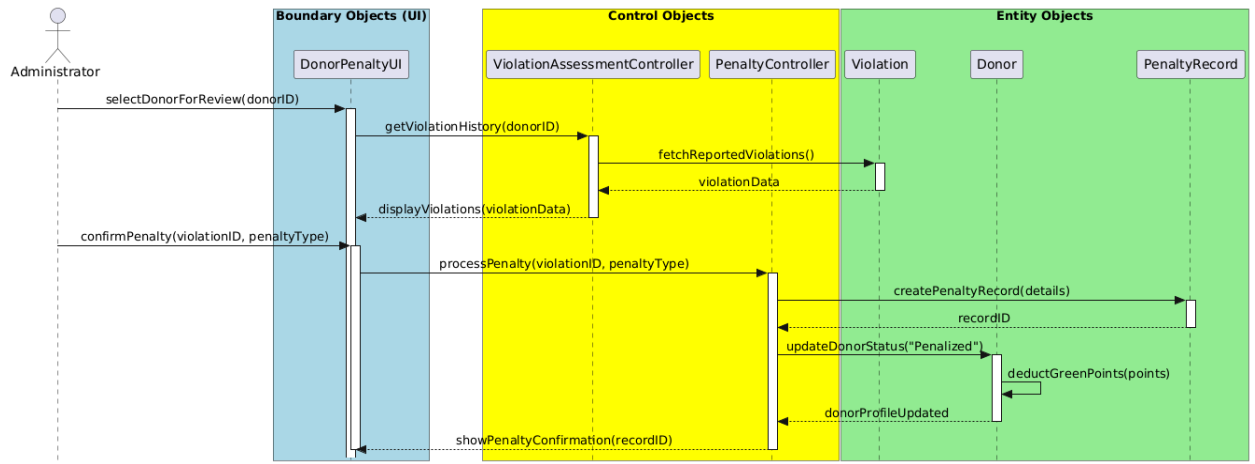
Busy · unavailable

No partners in range —
expand radius?

Dispatch partner

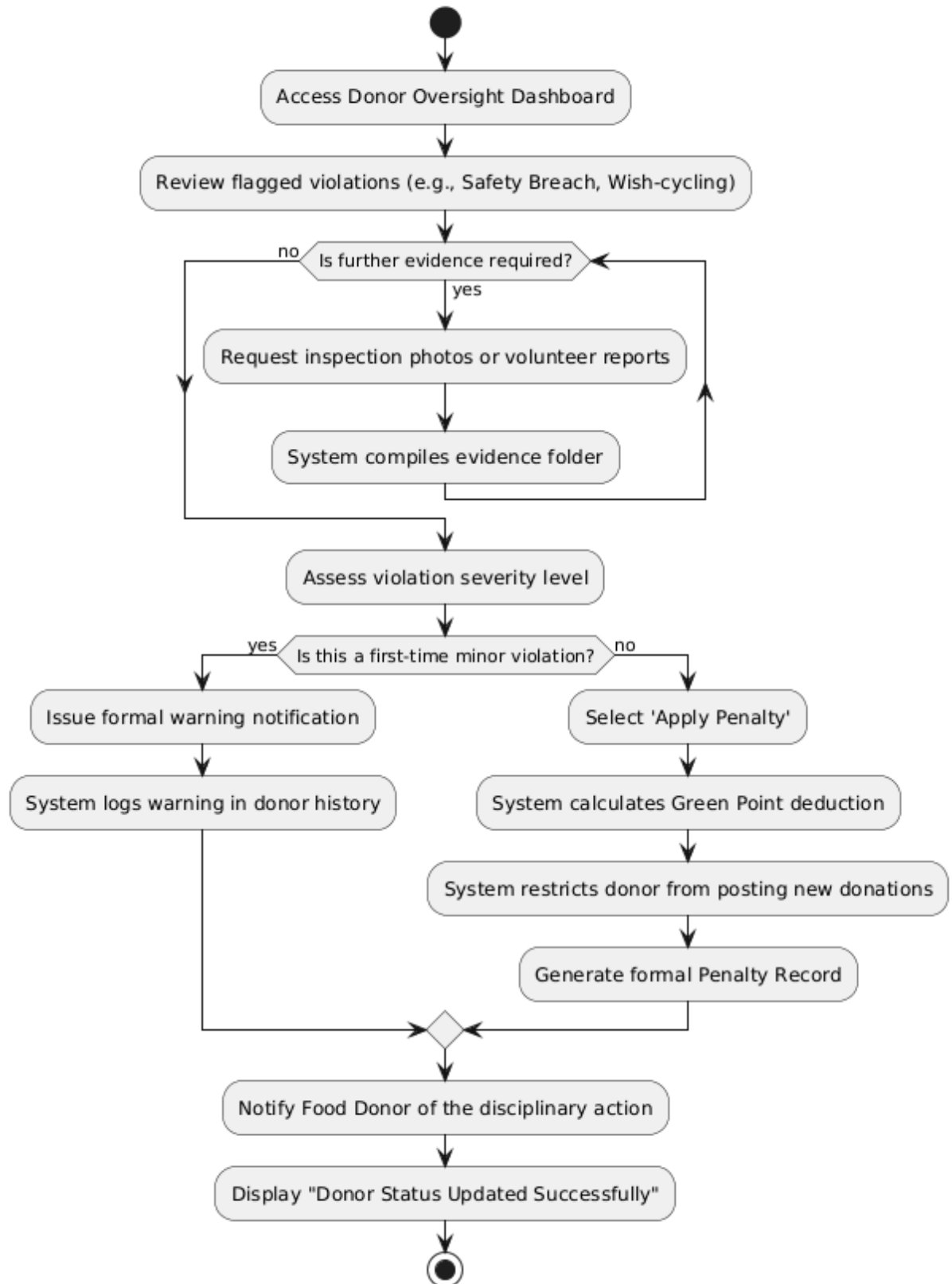
2.5 Penalize Food Donor

A. Sequence Diagram



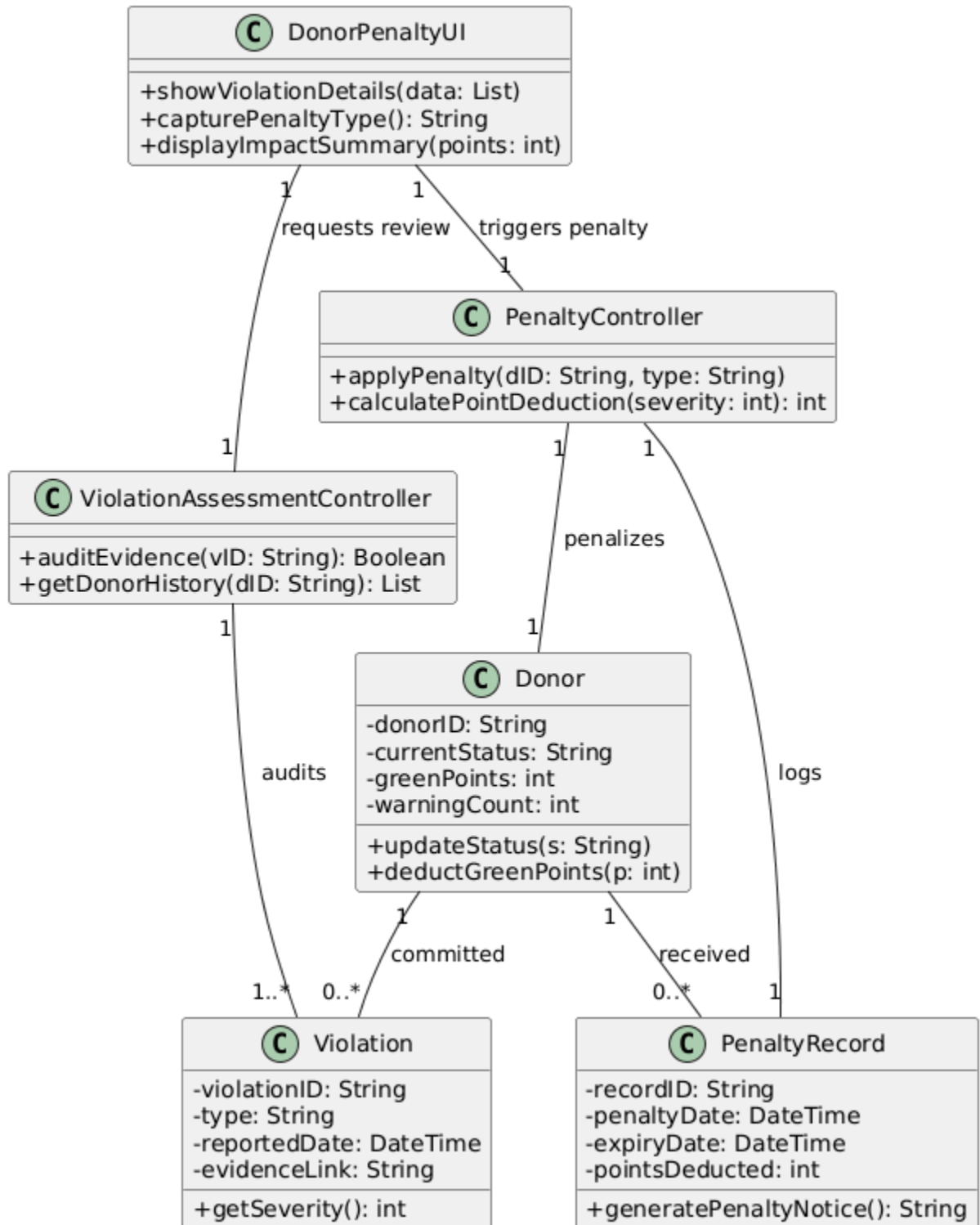
B. Activity Diagram

Activity Diagram - Penalize Food Donor (User Perspective)

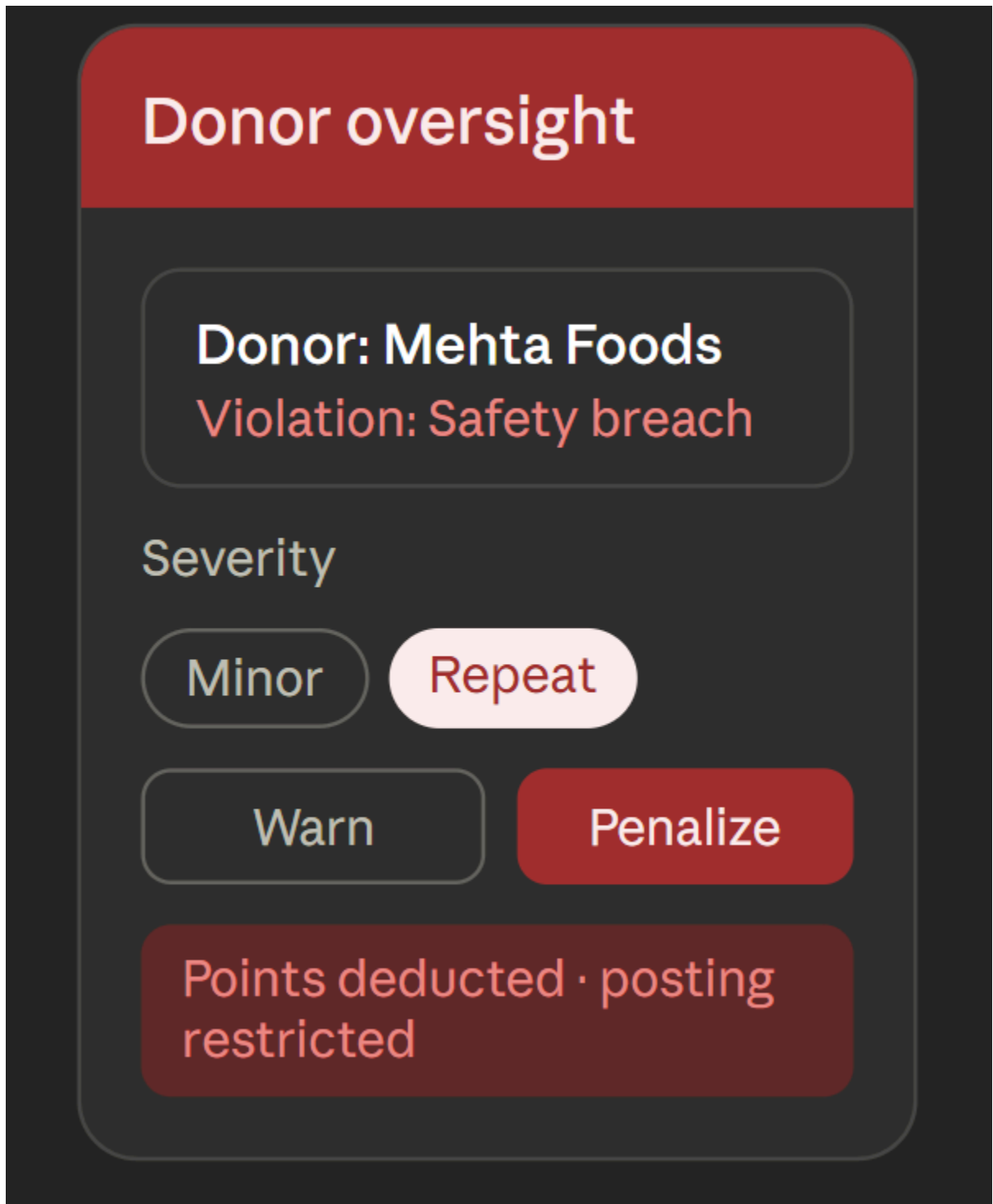


C. Class diagram

Class Diagram: Penalize Food Donor (Detailed)

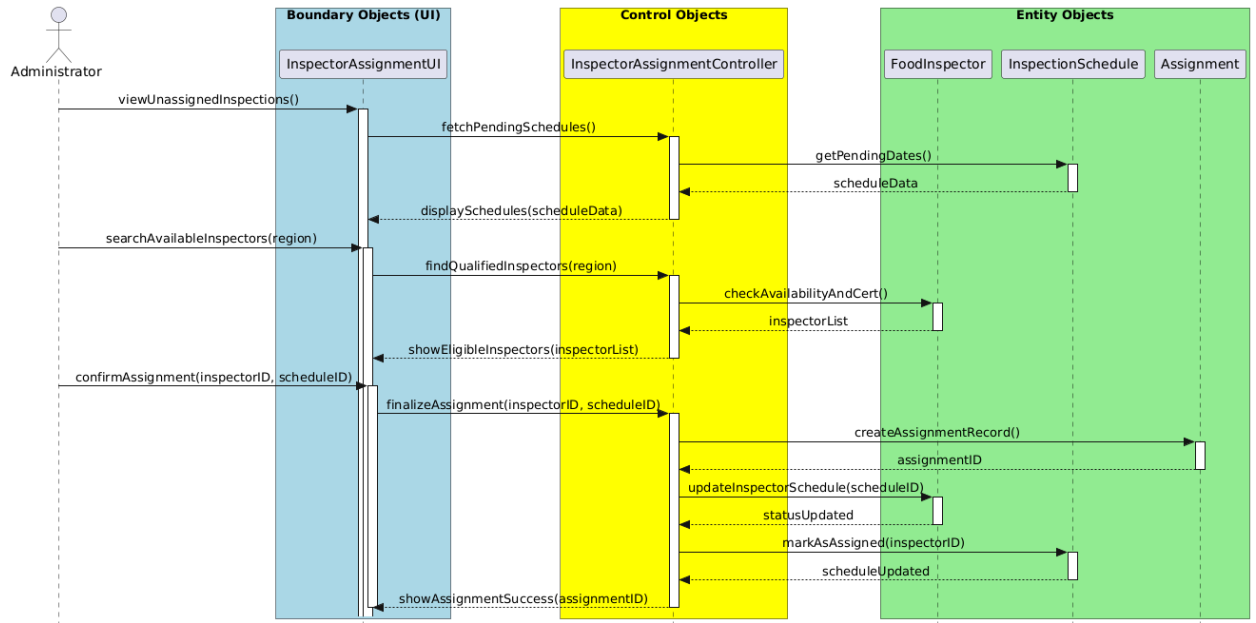


D. UI



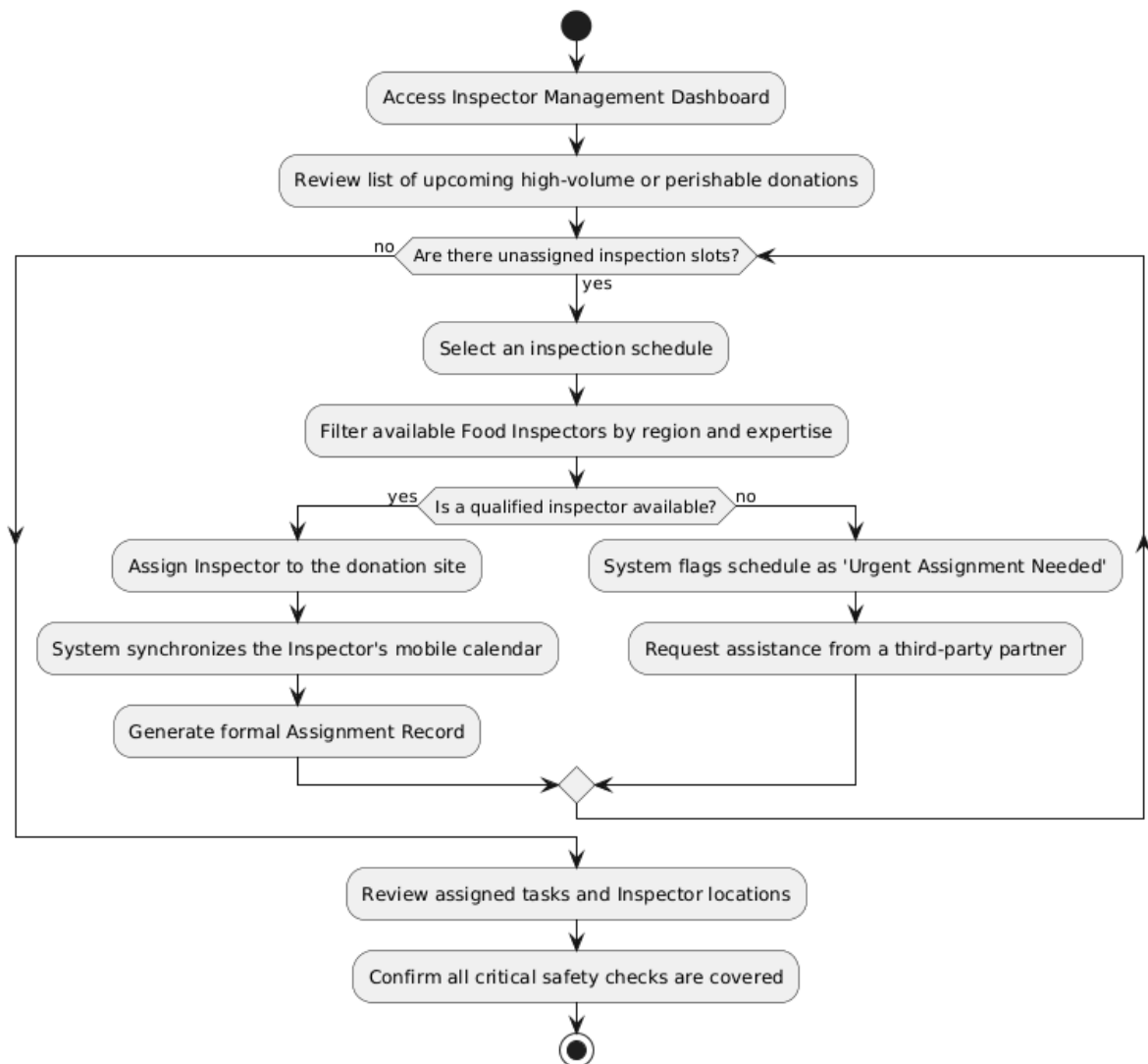
2.6 Assign/Manage Food inspector

A. Sequence Diagram



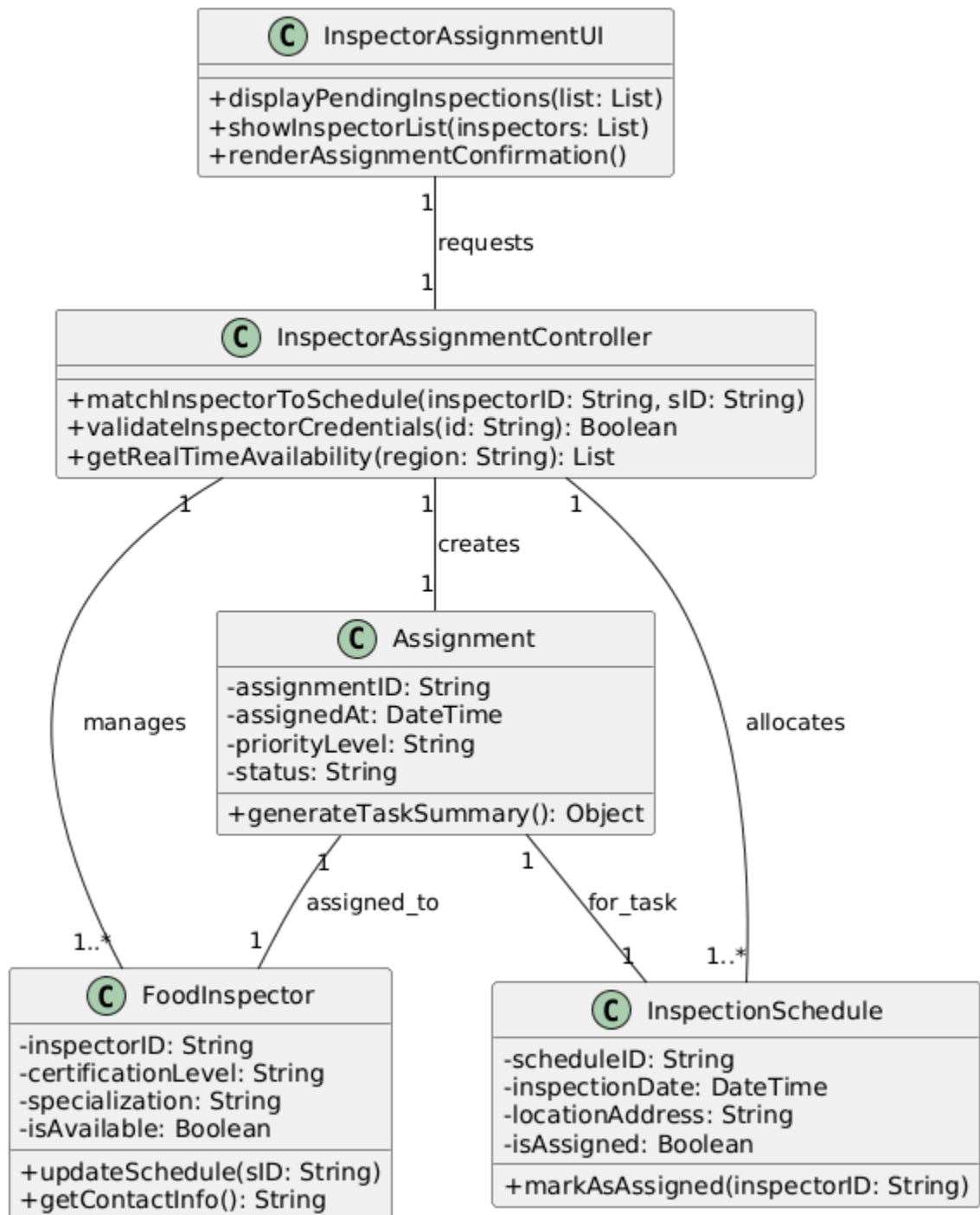
B. Activity Diagram

Activity Diagram - Assign / Manage Food Inspector (User Perspective)



C. Class diagram

Class Diagram: Assign / Manage Food Inspector (Detailed)



D. UI

Inspector management

**Batch #550 —
Perishables**
Unassigned · urgent

Select inspector



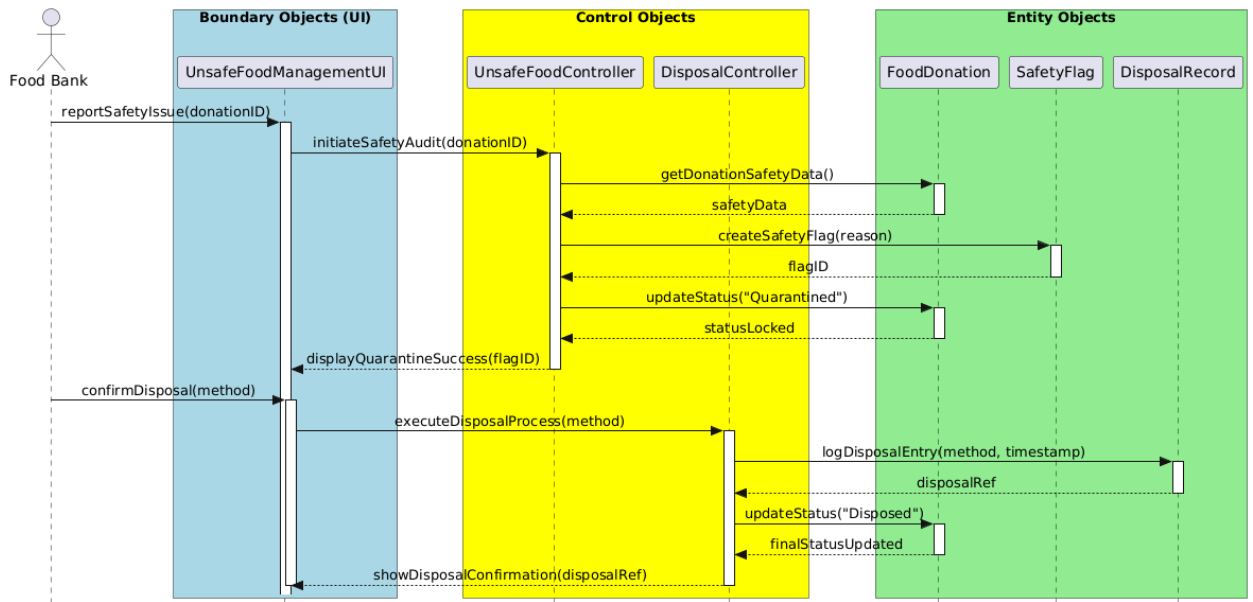
Kavita R. · Zone B
Food safety · available

Assign inspector

No inspector — flagged for
3rd party

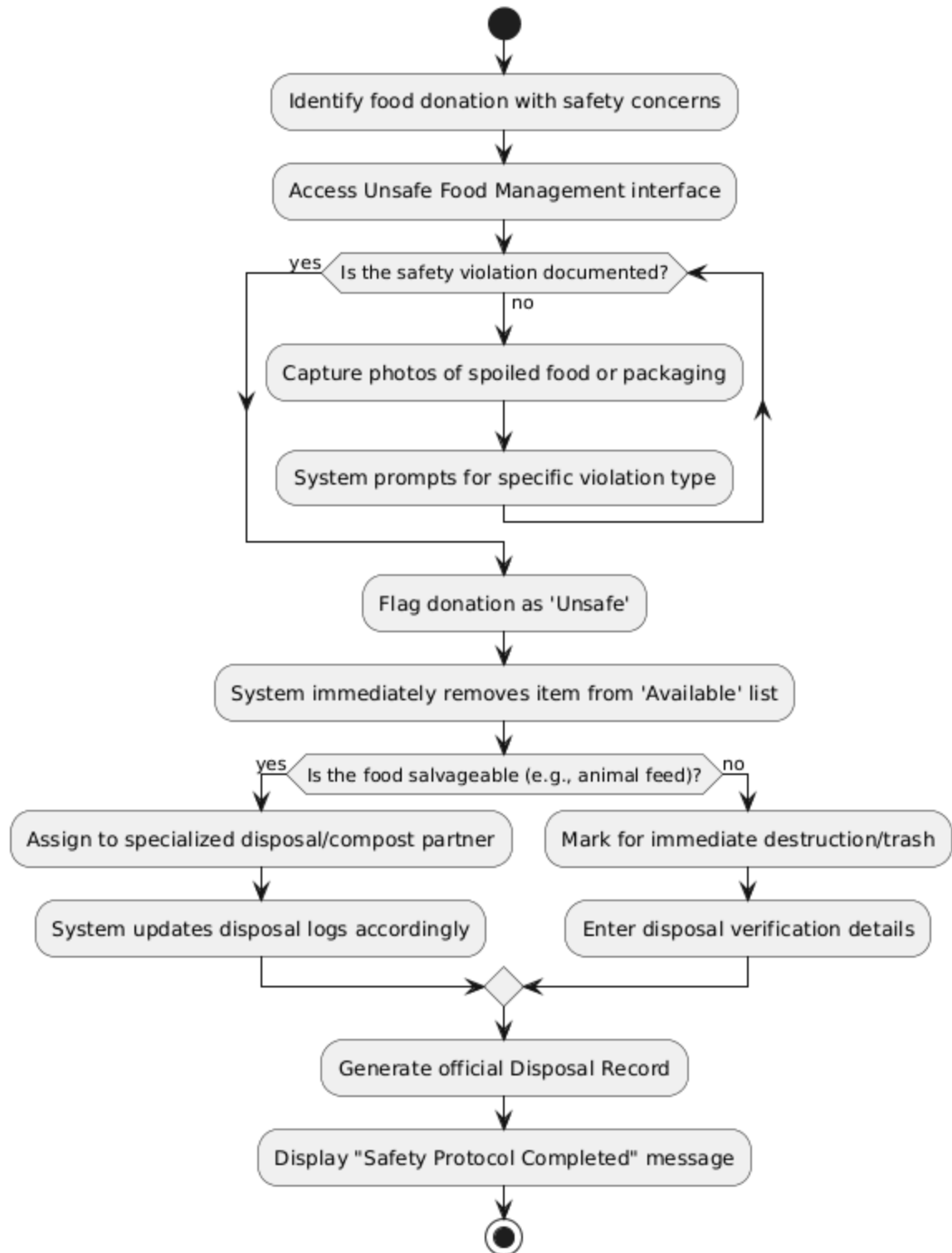
2.7 Manage Unsafe food

A. Sequence Diagram



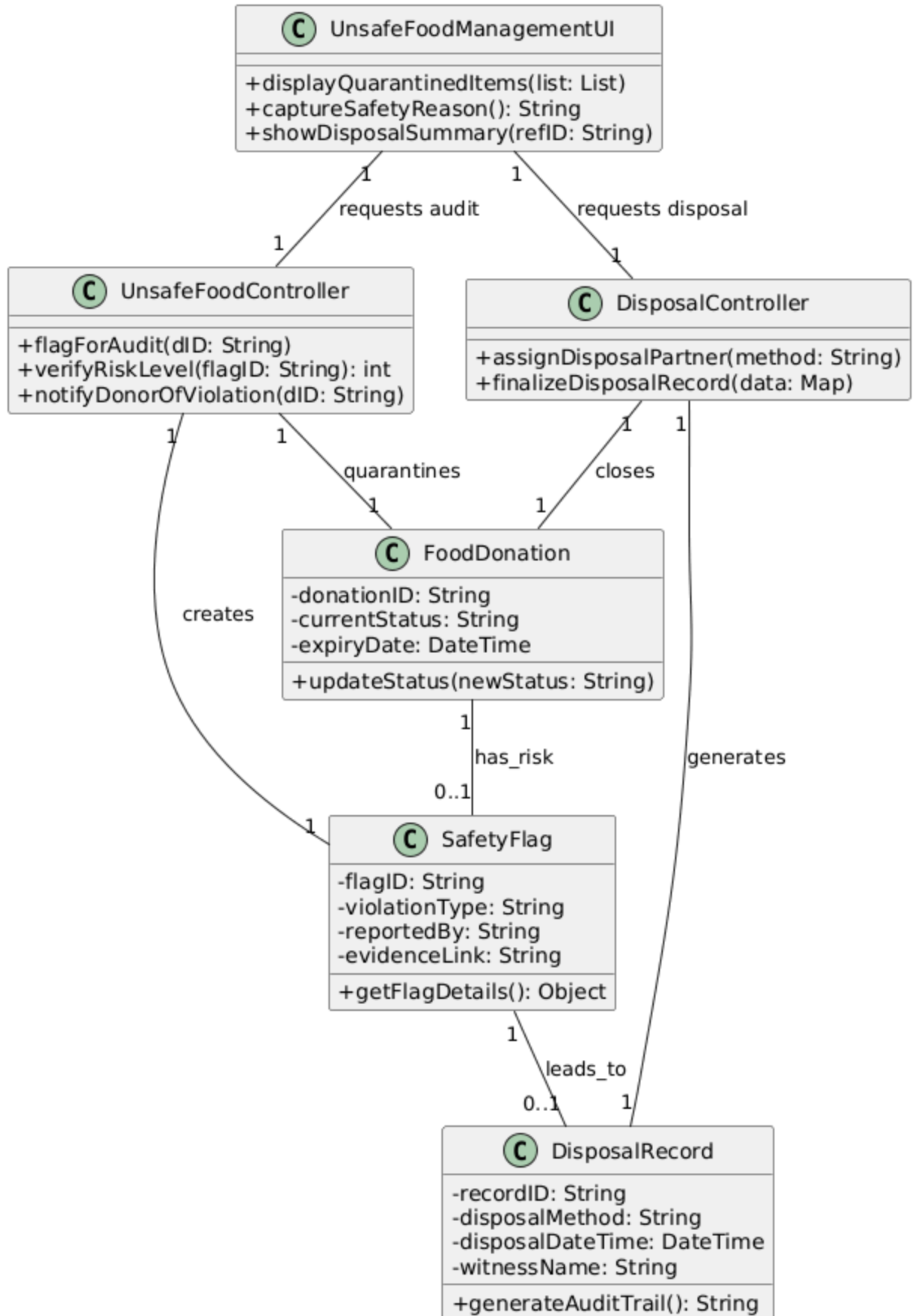
B. Activity Diagram

Activity Diagram - Manage Unsafe Food (User Perspective)



C. Class diagram

Class Diagram: Manage Unsafe Food (Detailed)



D. UI

Unsafe food mgmt.

Batch #331 — Spoiled
Removed from available
list

 Photo of violation

Disposal method

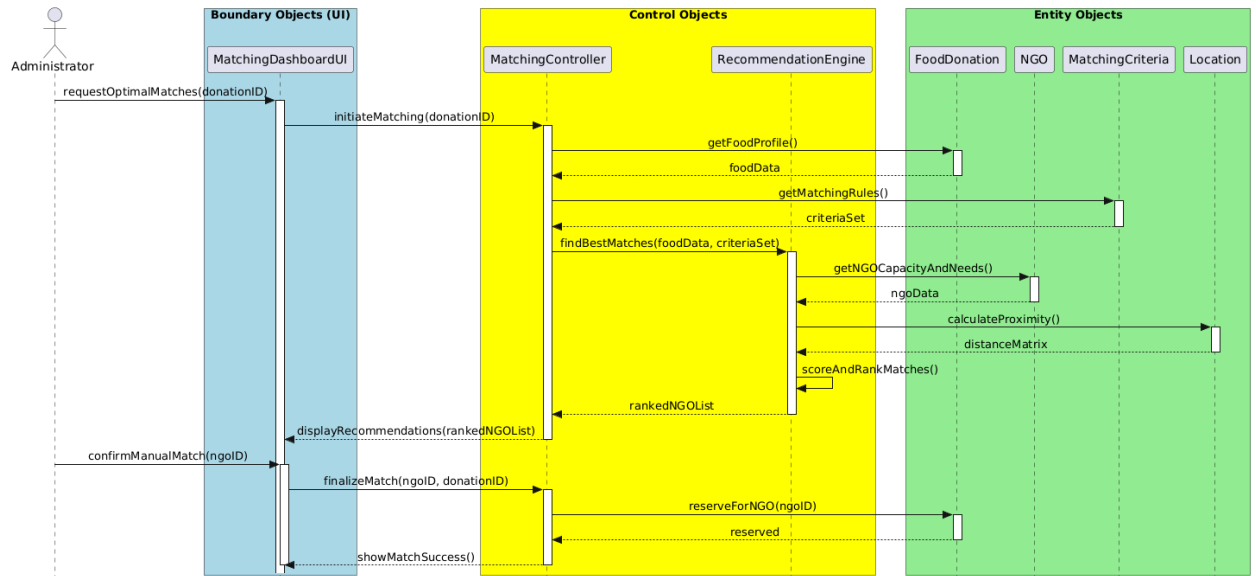
Compost

Destroy

Generate disposal
record

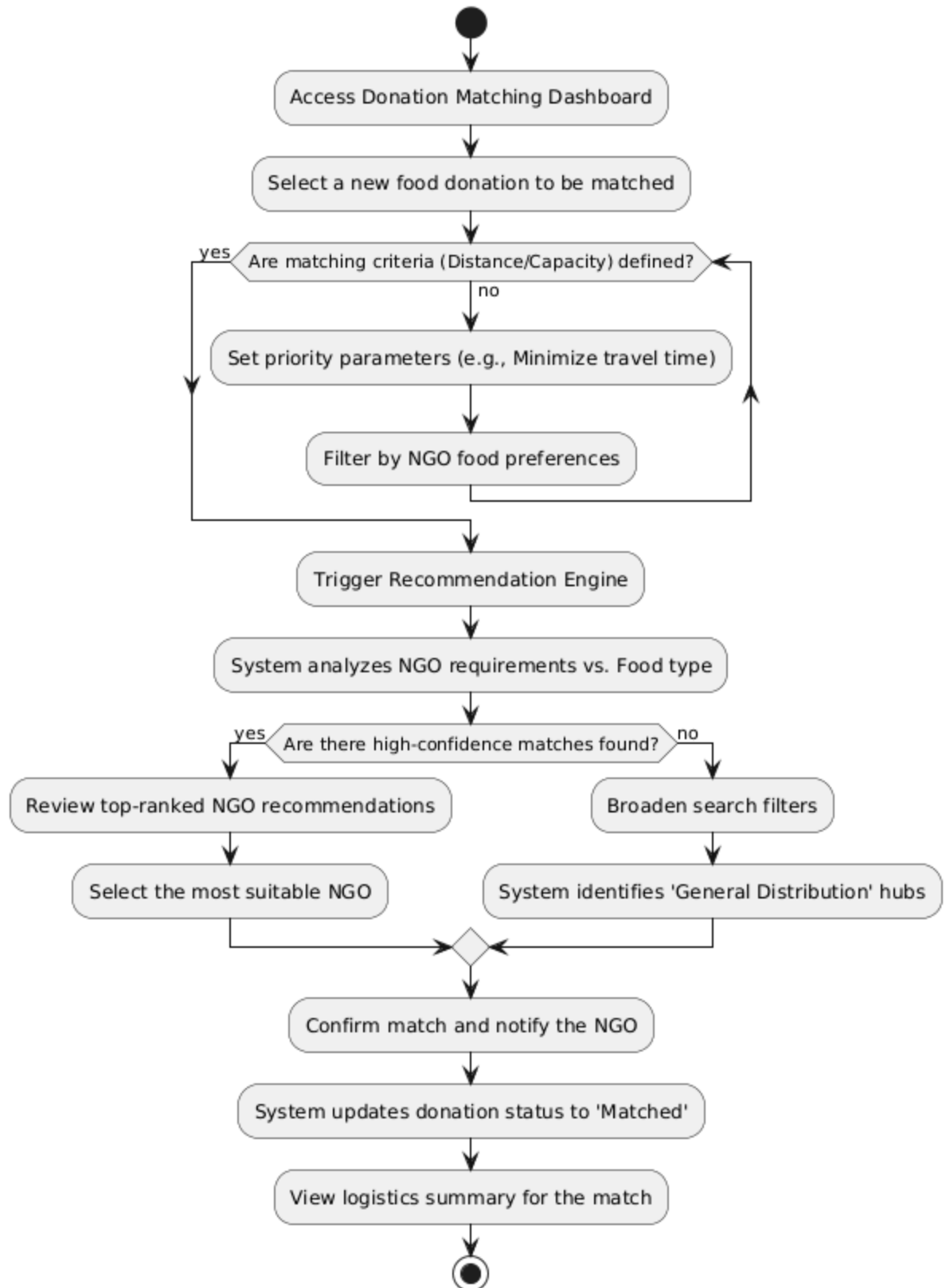
2.8 Donation matching system

A. Sequence Diagram



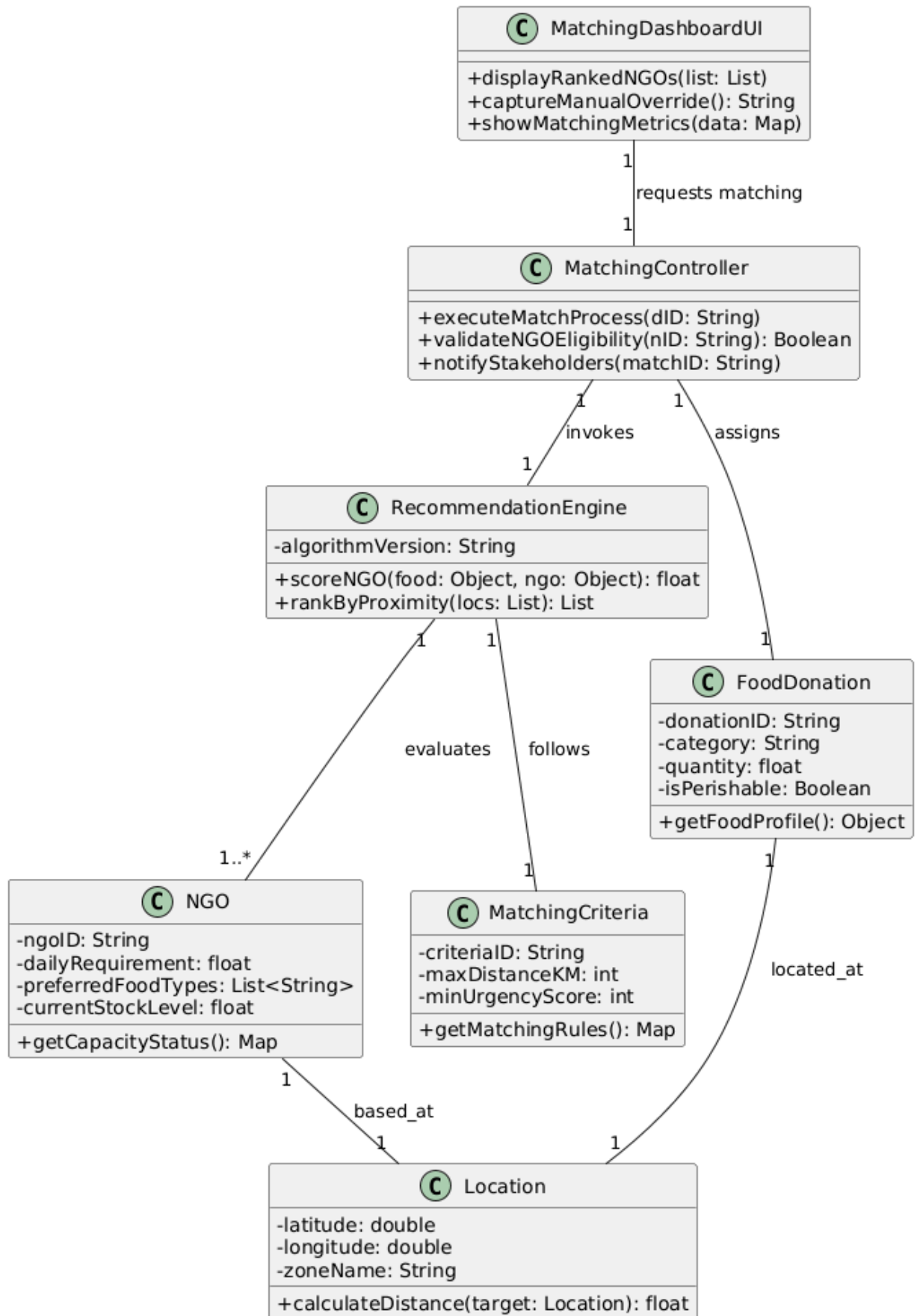
B. Activity Diagram

Activity Diagram - Donation Matching System (User Perspective)



C. Class diagram

Class Diagram: Donation Matching System (Detailed)



D. UI

Donation matching

Batch #441 · Dry goods
20 kg · ready to match

Min distance

NGO prefs

Top matches

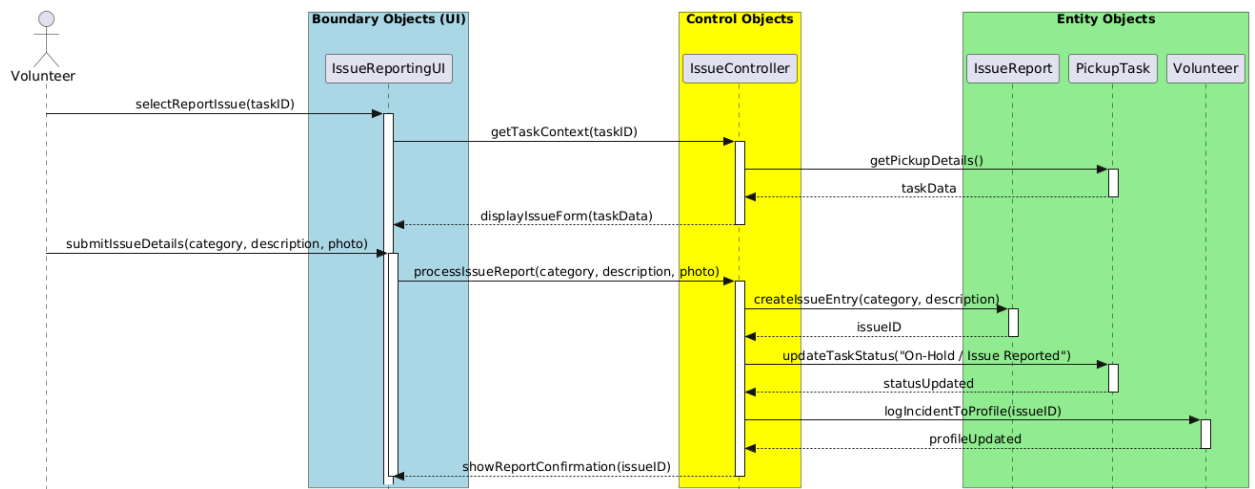
Hope NGO · ★ best fit
1.4 km · accepts dry goods

Confirm match

3. Volunteer Use Cases

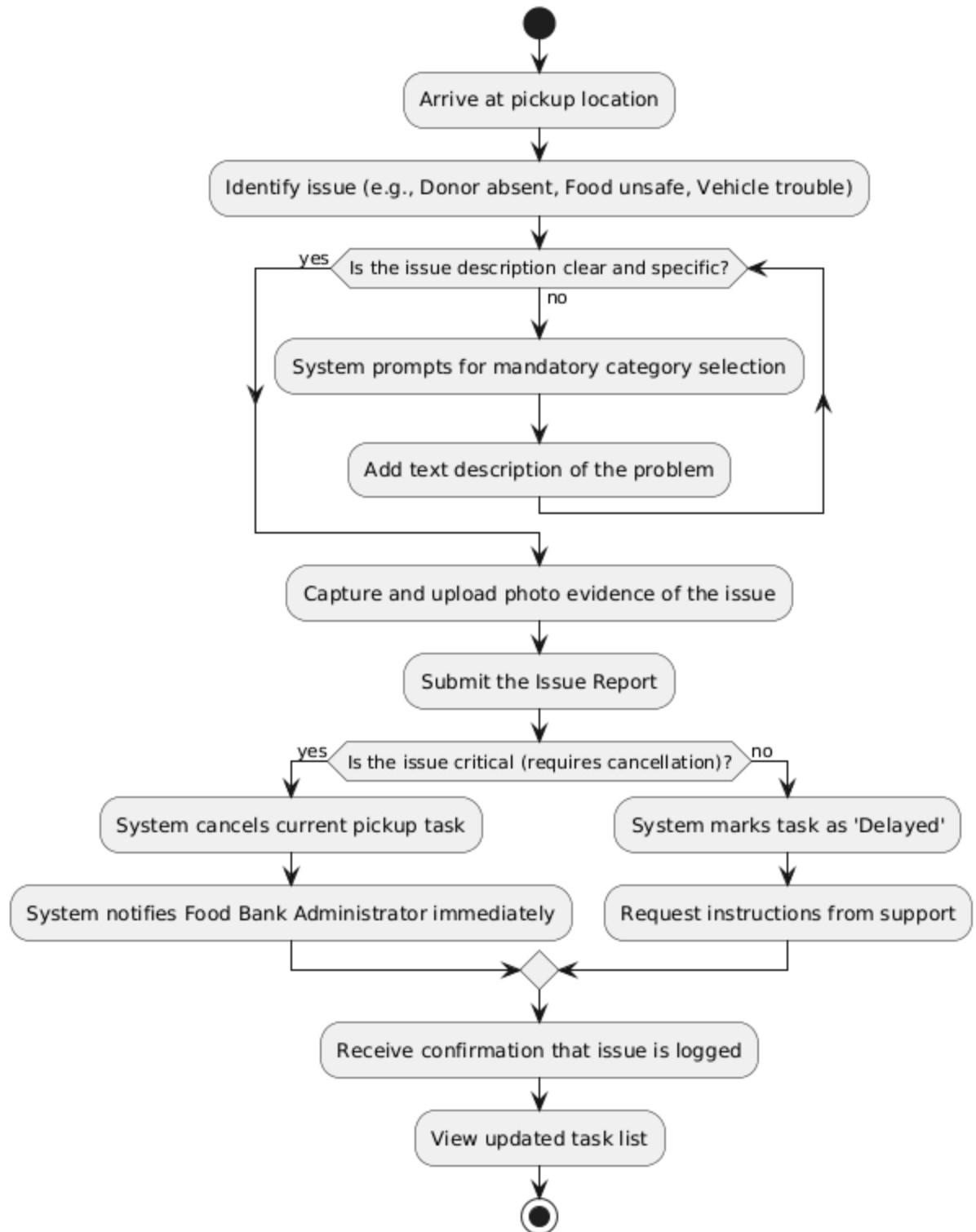
3.1 Report Issues in Pickup

SEQUENCE DIAGRAM -



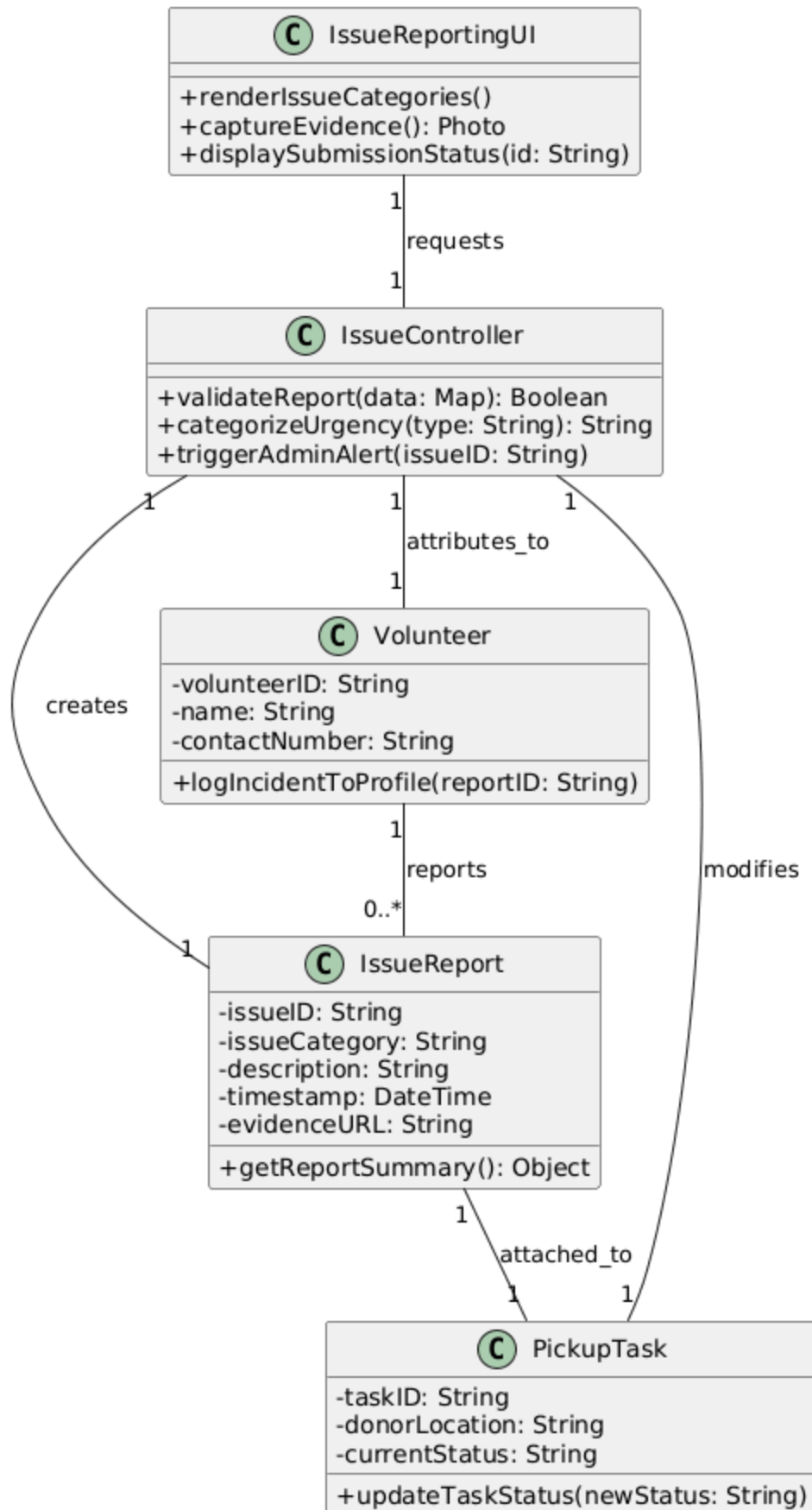
ACTIVITY DIAGRAM -

Activity Diagram - Report Issues in Pickup (User Perspective)



CLASS DIAGRAM -

Class Diagram: Report Issues in Pickup (Detailed)



Report an issue

Issue type

Select category ▼

Donor absent

Food unsafe

Vehicle trouble

Description

Add details...

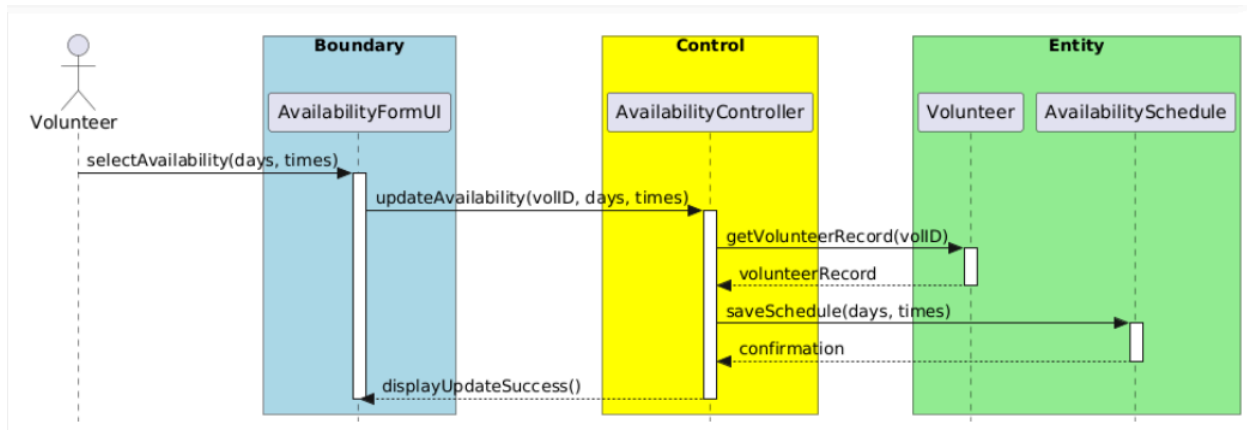


Upload photo

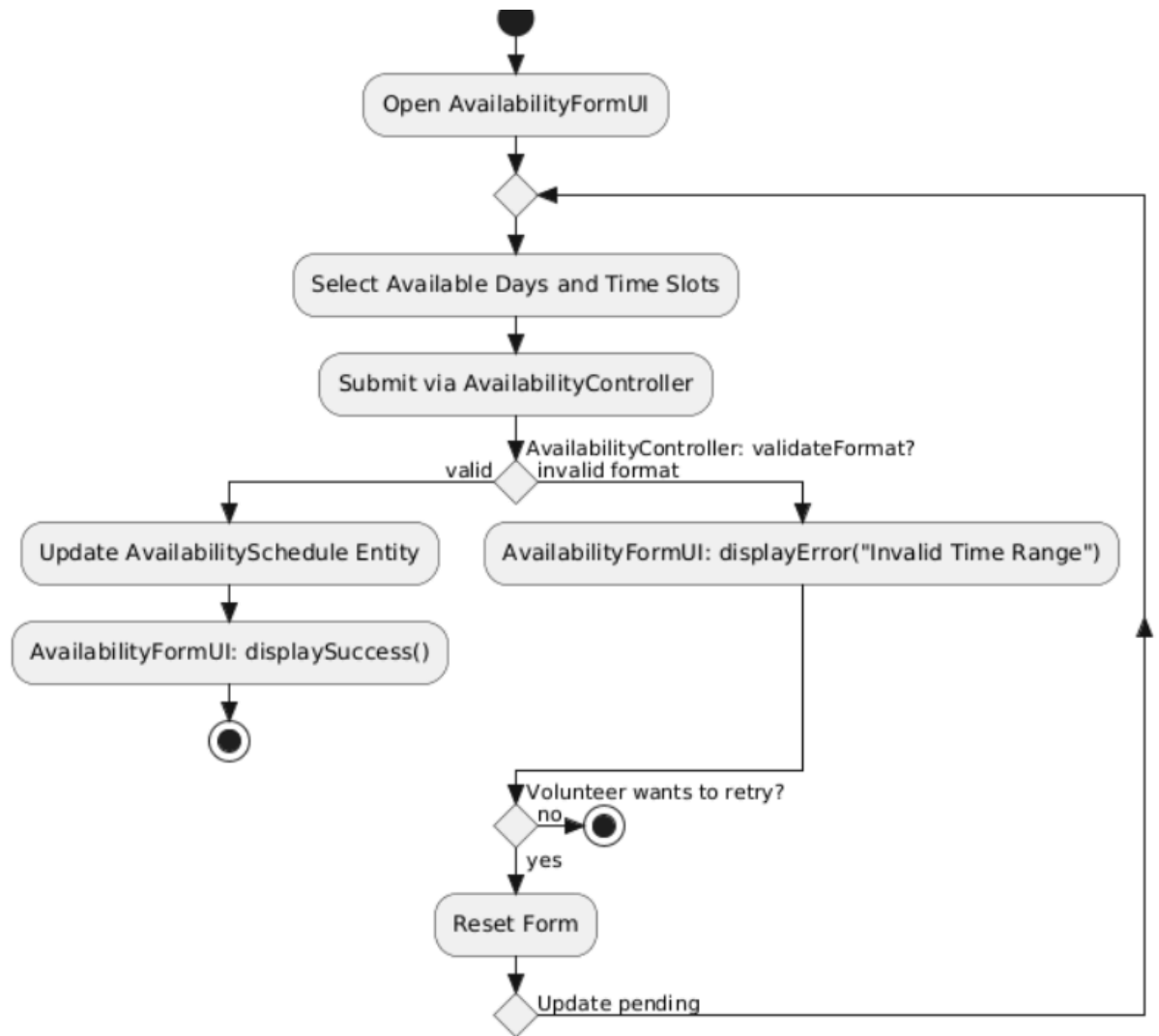
Submit report

3.2 Set Availability

- SEQUENCE DIAGRAM -

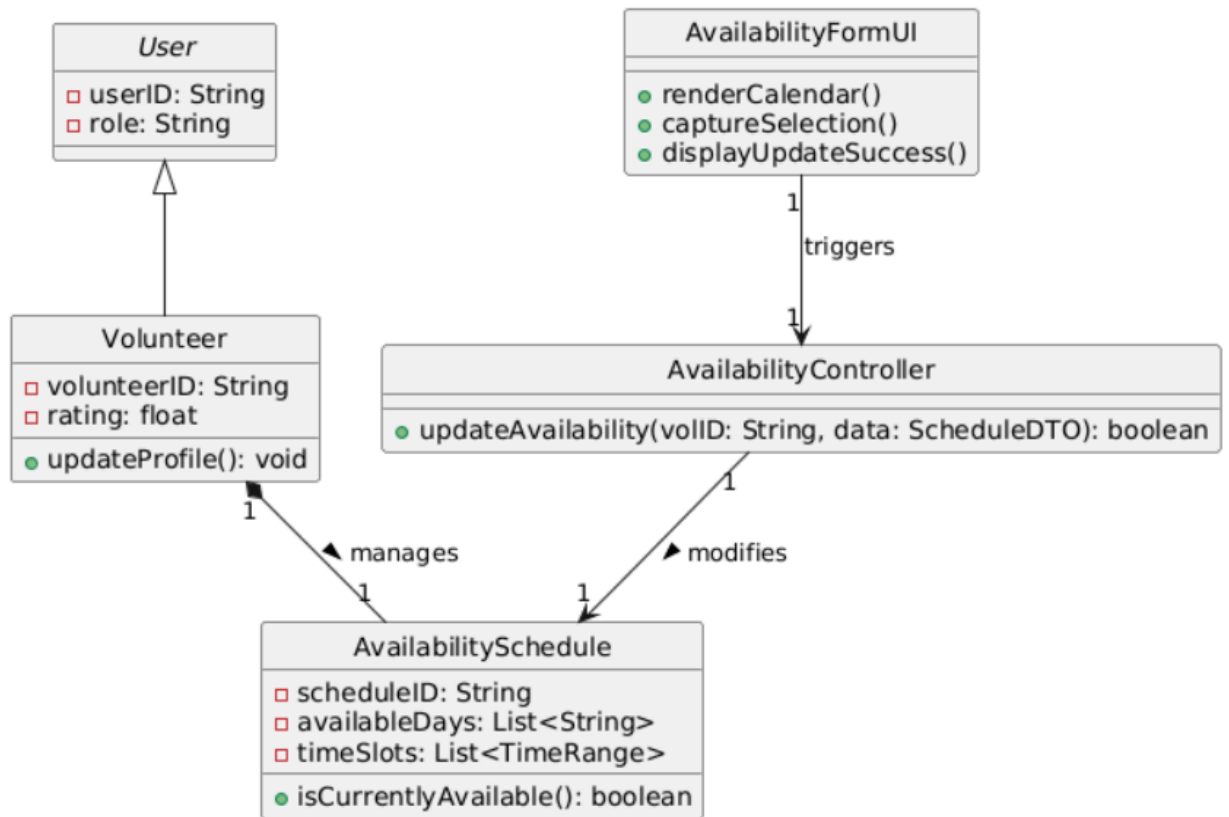


- ACTIVITY DIAGRAM -



- **CLASS DIAGRAM -**

Class Diagram - Set Availability



- UI -

My availability

Select days

Mon

Wed

Fri

Sat

Time slot

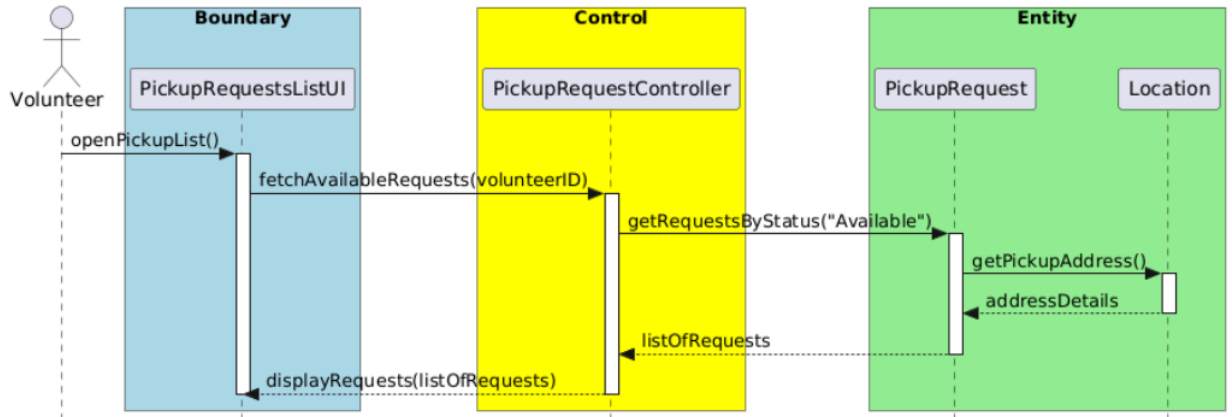
9:00 AM → 1:00 PM

Invalid time range —
please fix

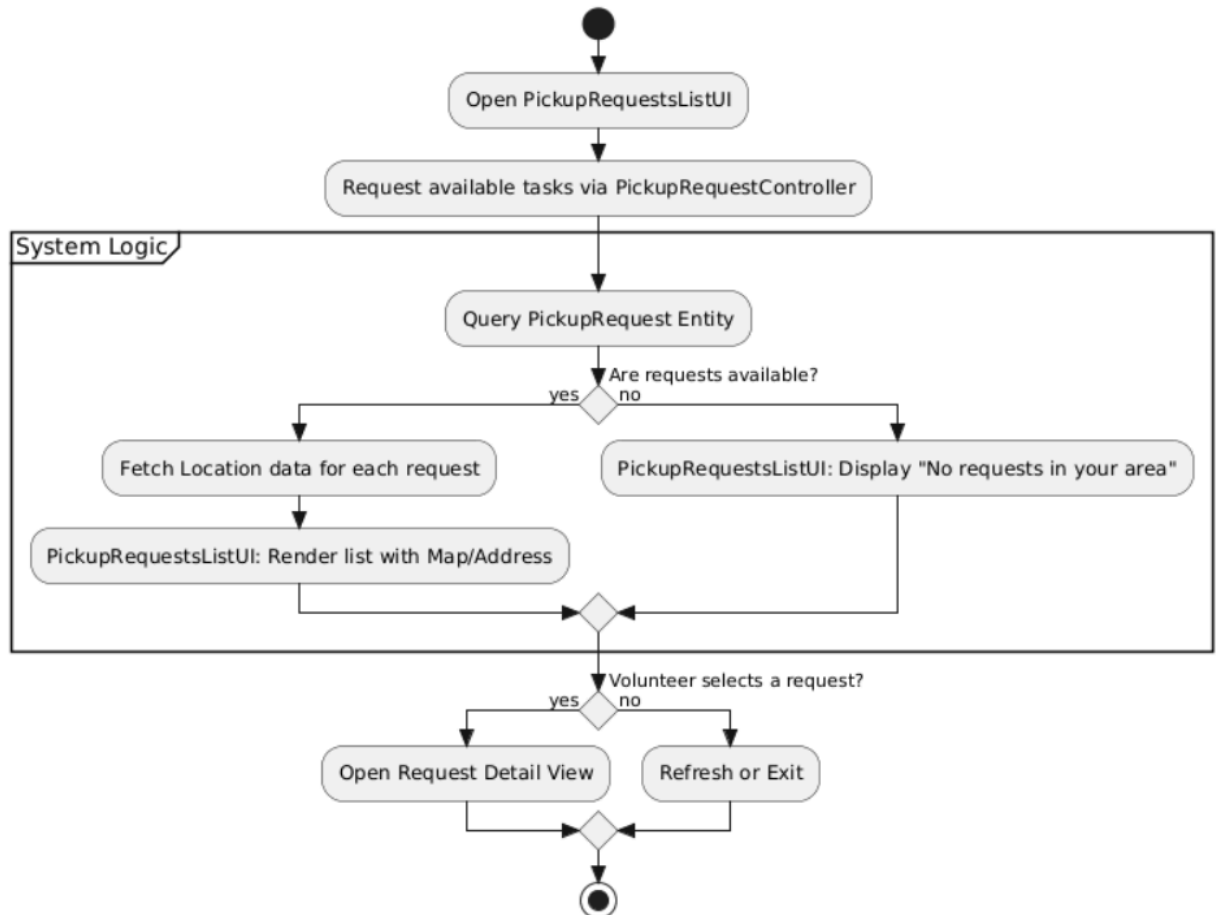
Save

3.3 View Pickup Requests

- SEQUENCE DIAGRAM -



- ACTIVITY DIAGRAM -

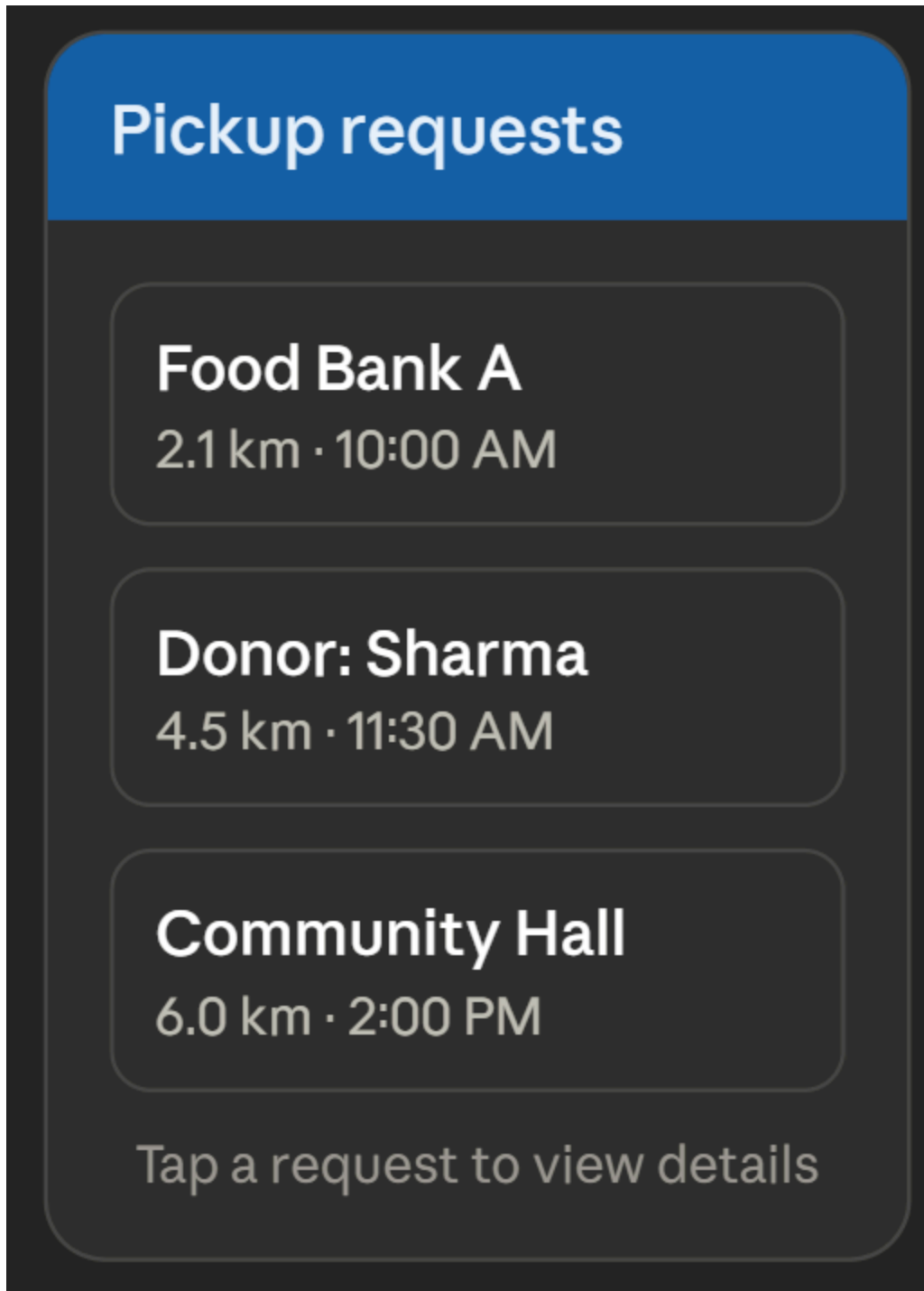


- CLASS DIAGRAM -

Class Diagram - View Pickup Requests

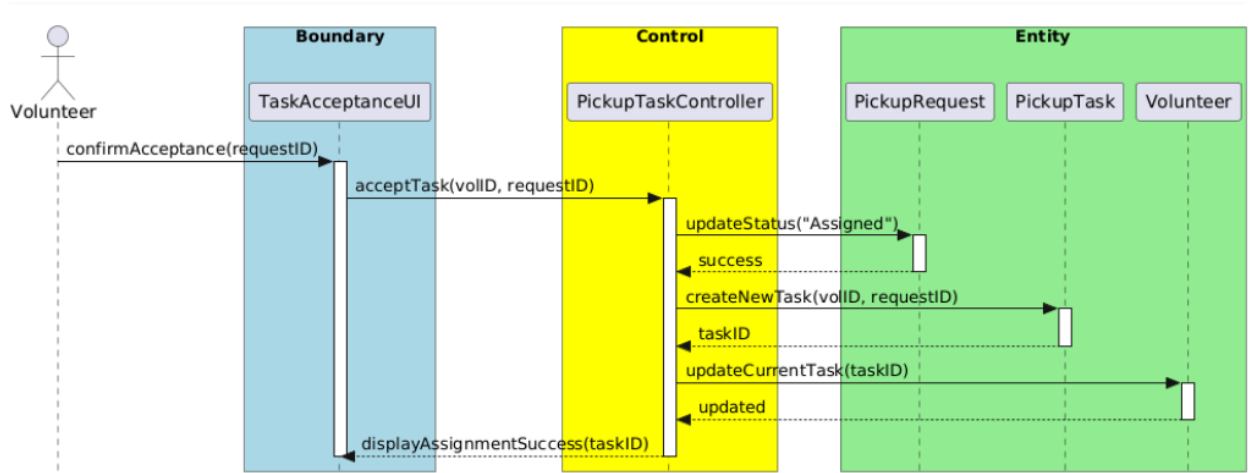


- UI -

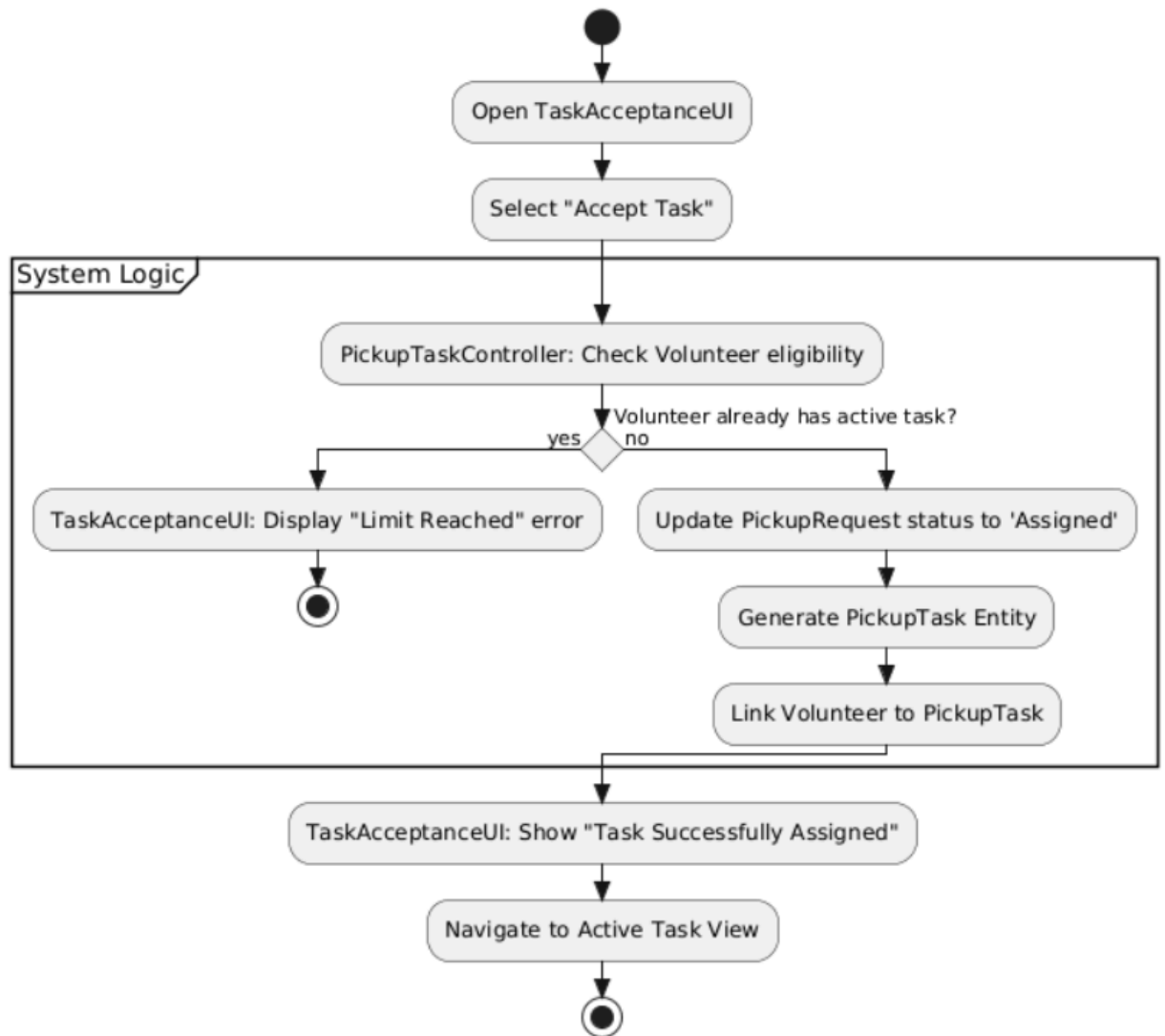


3.4 Accept Pickup Task

- SEQUENCE DIAGRAM -

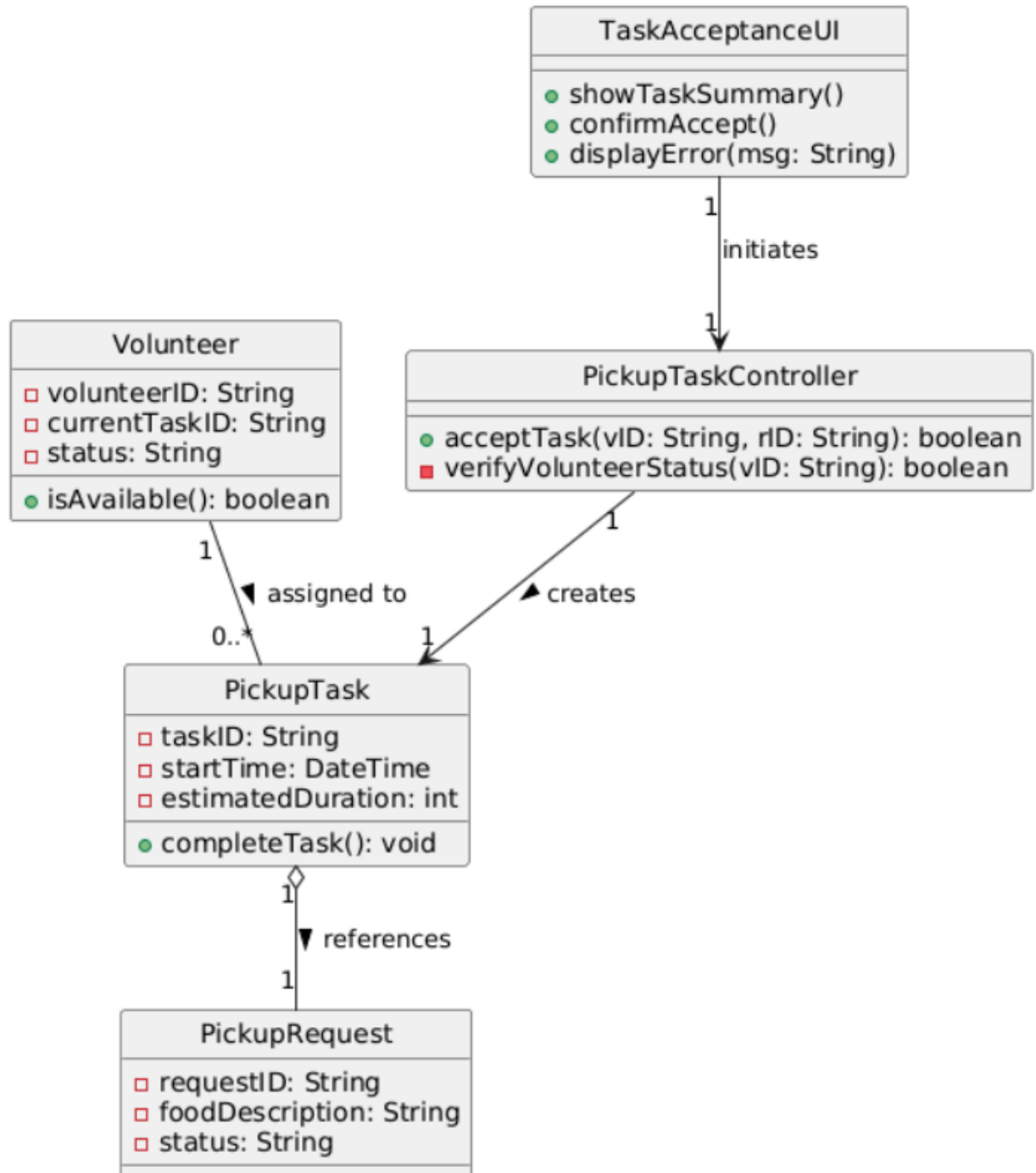


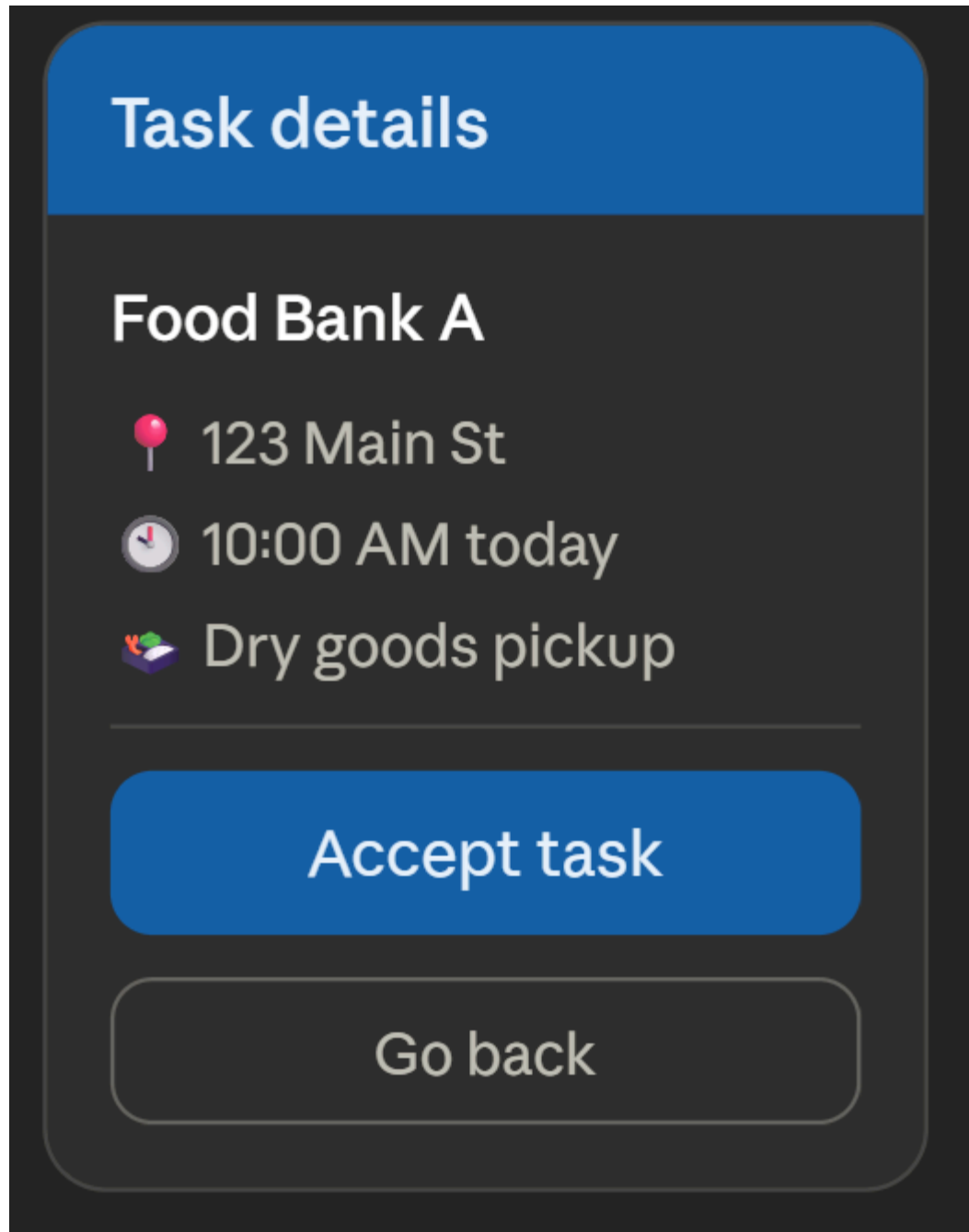
-
- **ACTIVITY DIAGRAM -**



- CLASS DIAGRAM -

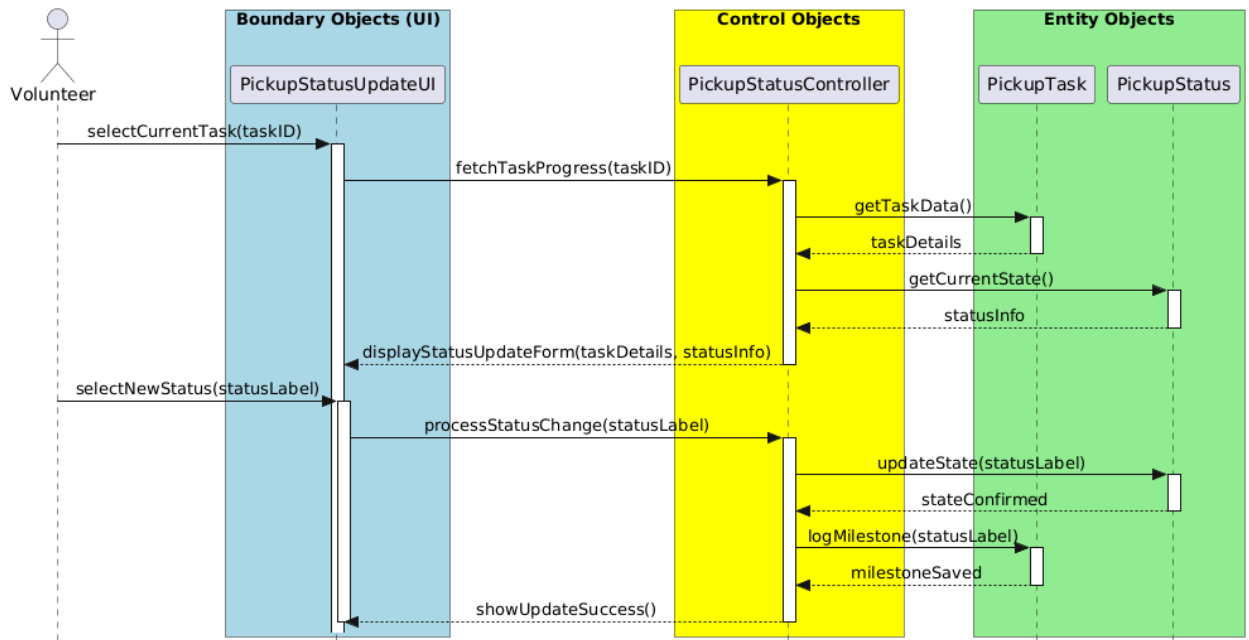
Class Diagram - Accept Pickup Task





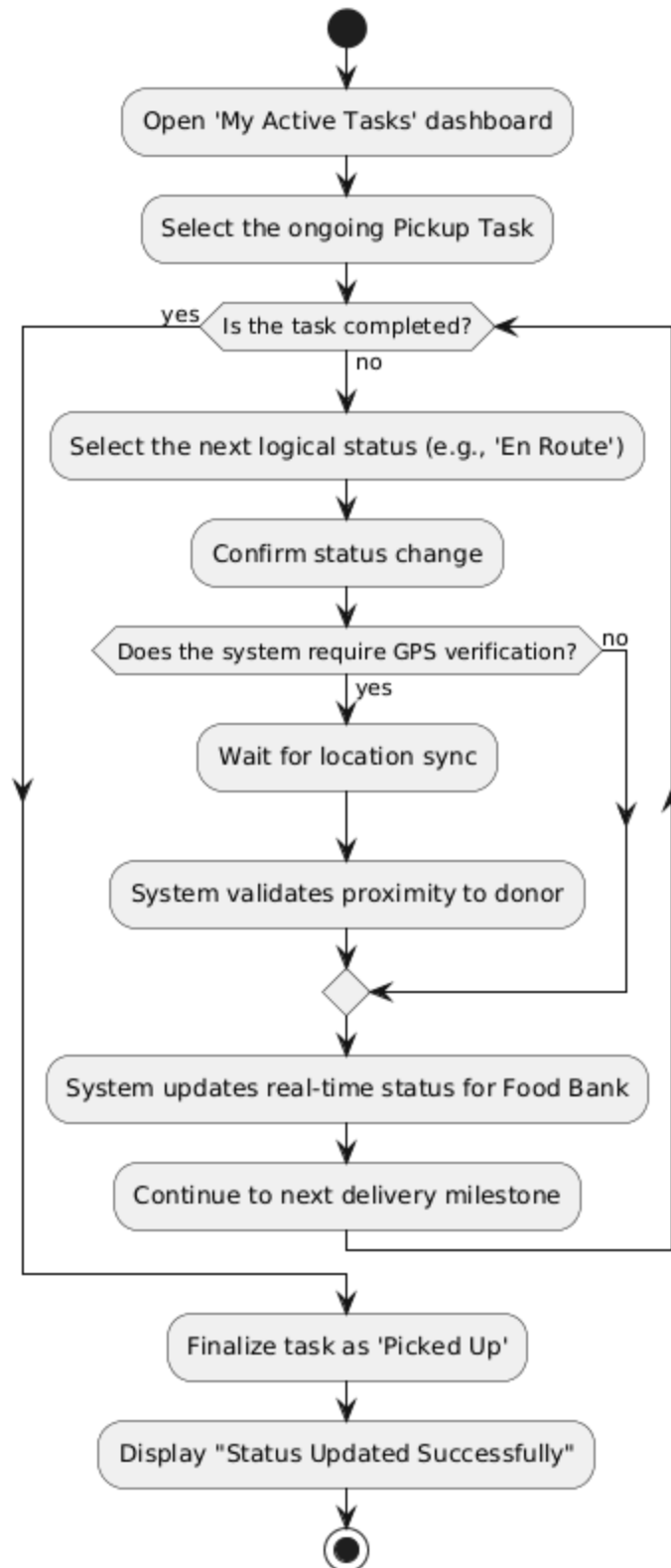
3.5 Update Pickup status

- SEQUENCE DIAGRAM -



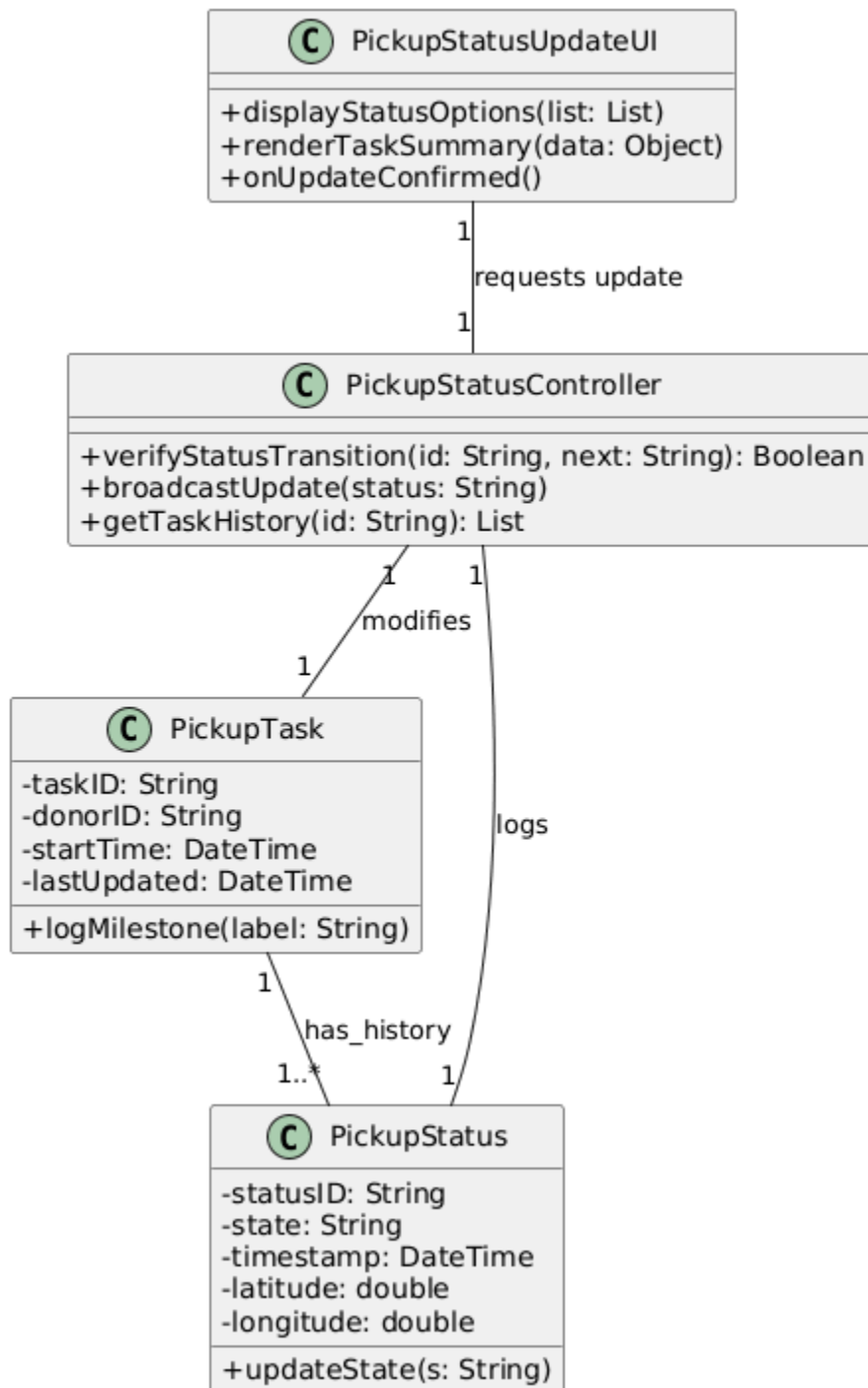
- **ACTIVITY DIAGRAM -**

Activity Diagram - Update Pickup Status (User Perspective)



- CLASS DIAGRAM -

Class Diagram: Update Pickup Status (Detailed)



Active task

Food Bank A pickup

Current status

✓ Assigned

▷ En Route

▷ Arrived

▷ Picked Up

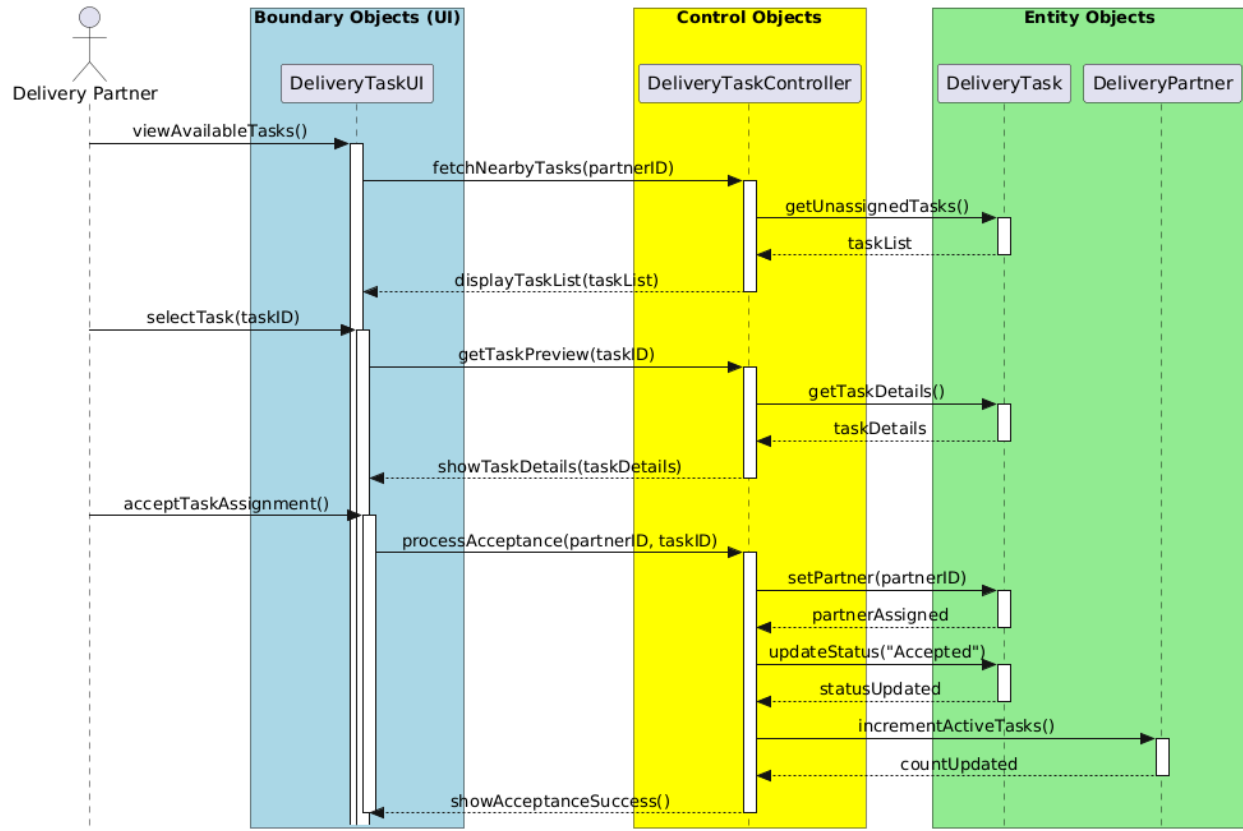
Update status

- UI -
-

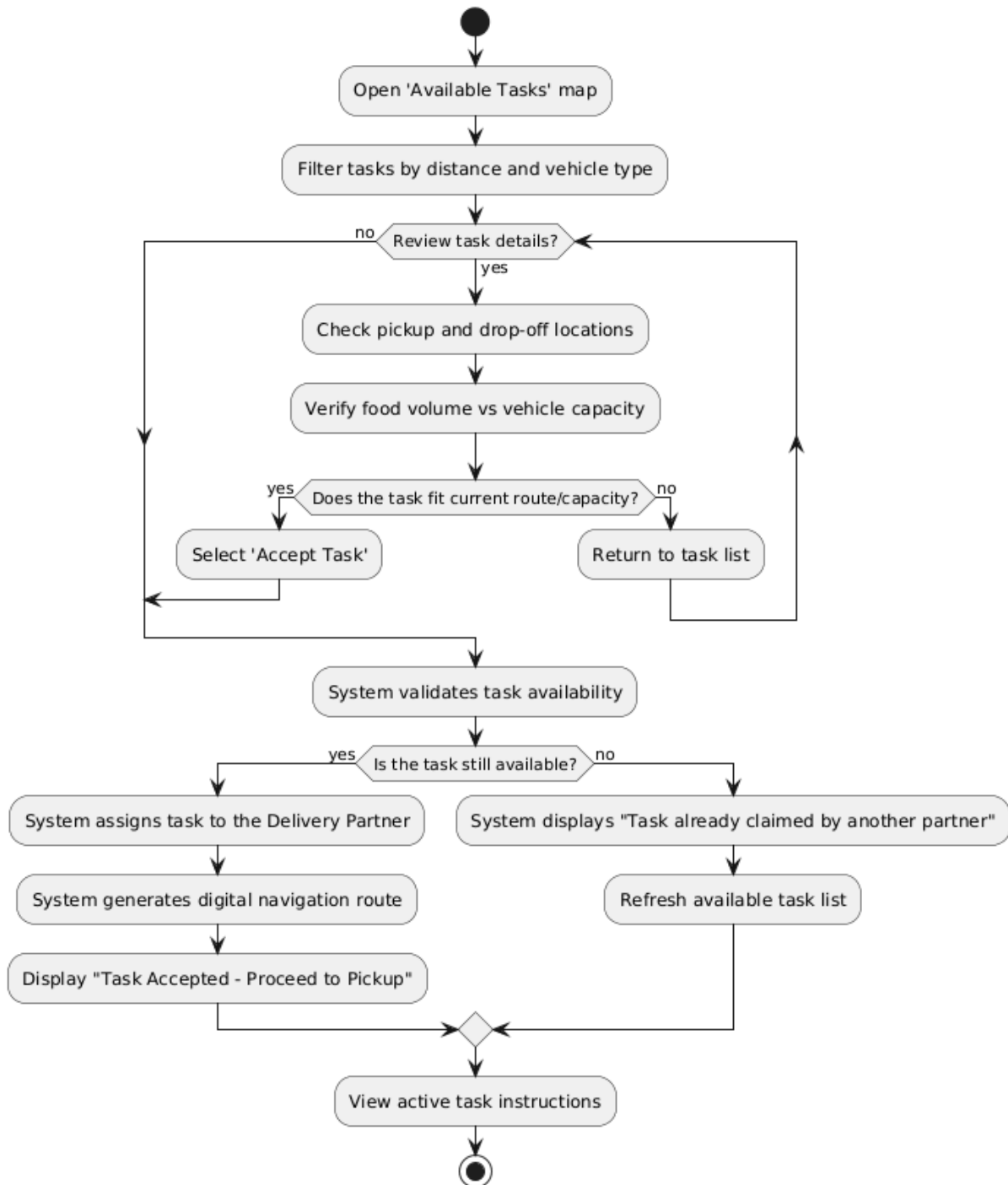
4. Delivery Partner Use Cases

4.1 Accept Delivery Task

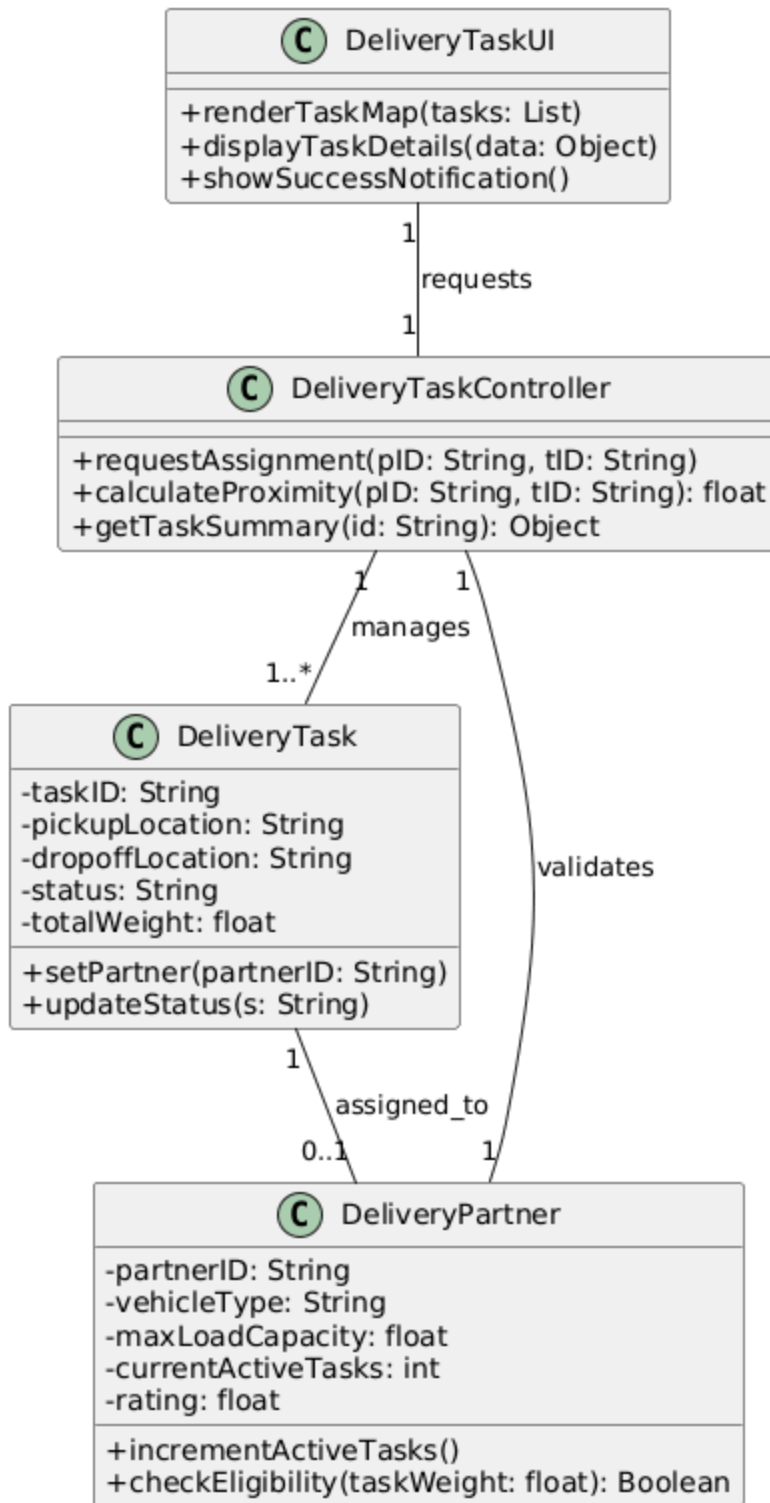
Sequence diagram -



Activity Diagram - Accept Delivery Task (User Perspective)



Class Diagram: Accept Delivery Task (Detailed)



Ui -

Available tasks

≤ 5 km

Van

Task #1042

Pickup: Food Bank A

Drop-off: NGO Centre

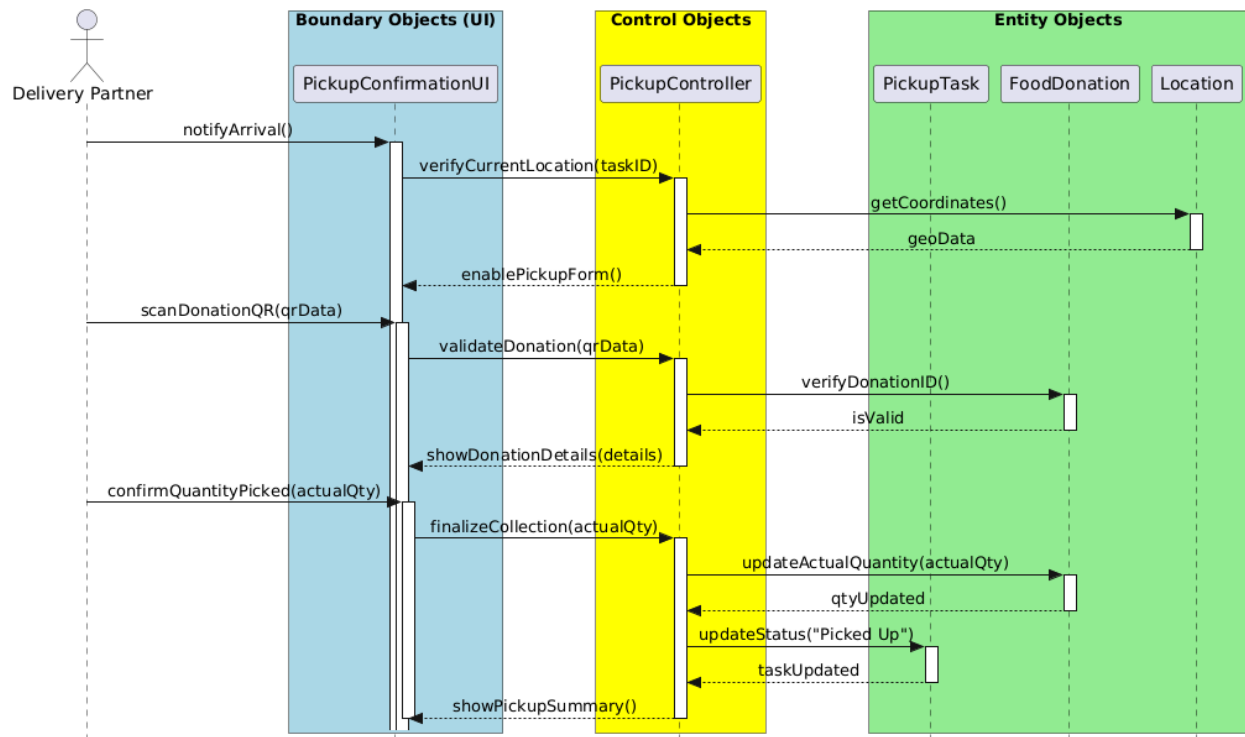
Accept

Skip

Task accepted — proceed
to pickup

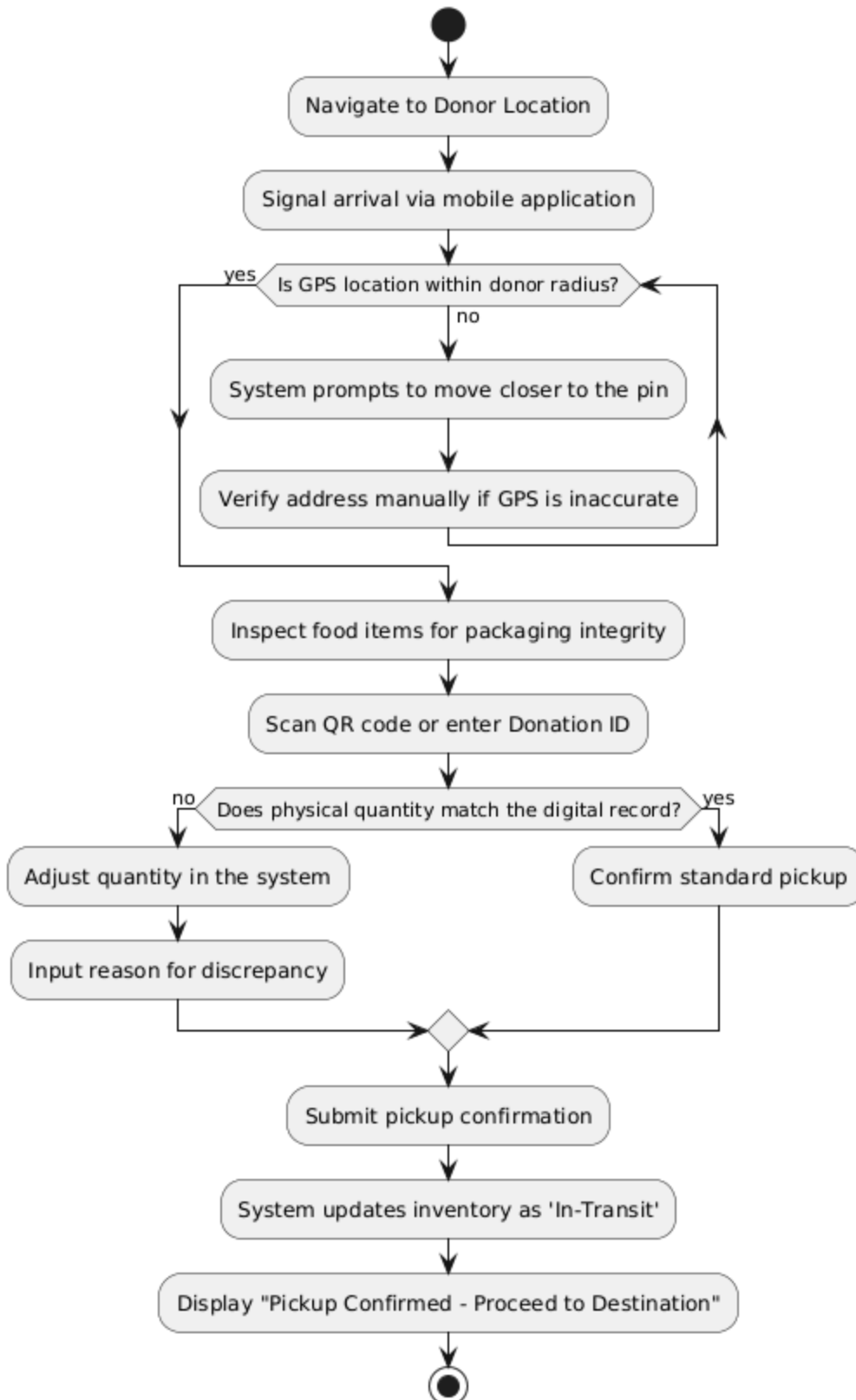
4.2 Pick Up Food

A. Sequence Diagram



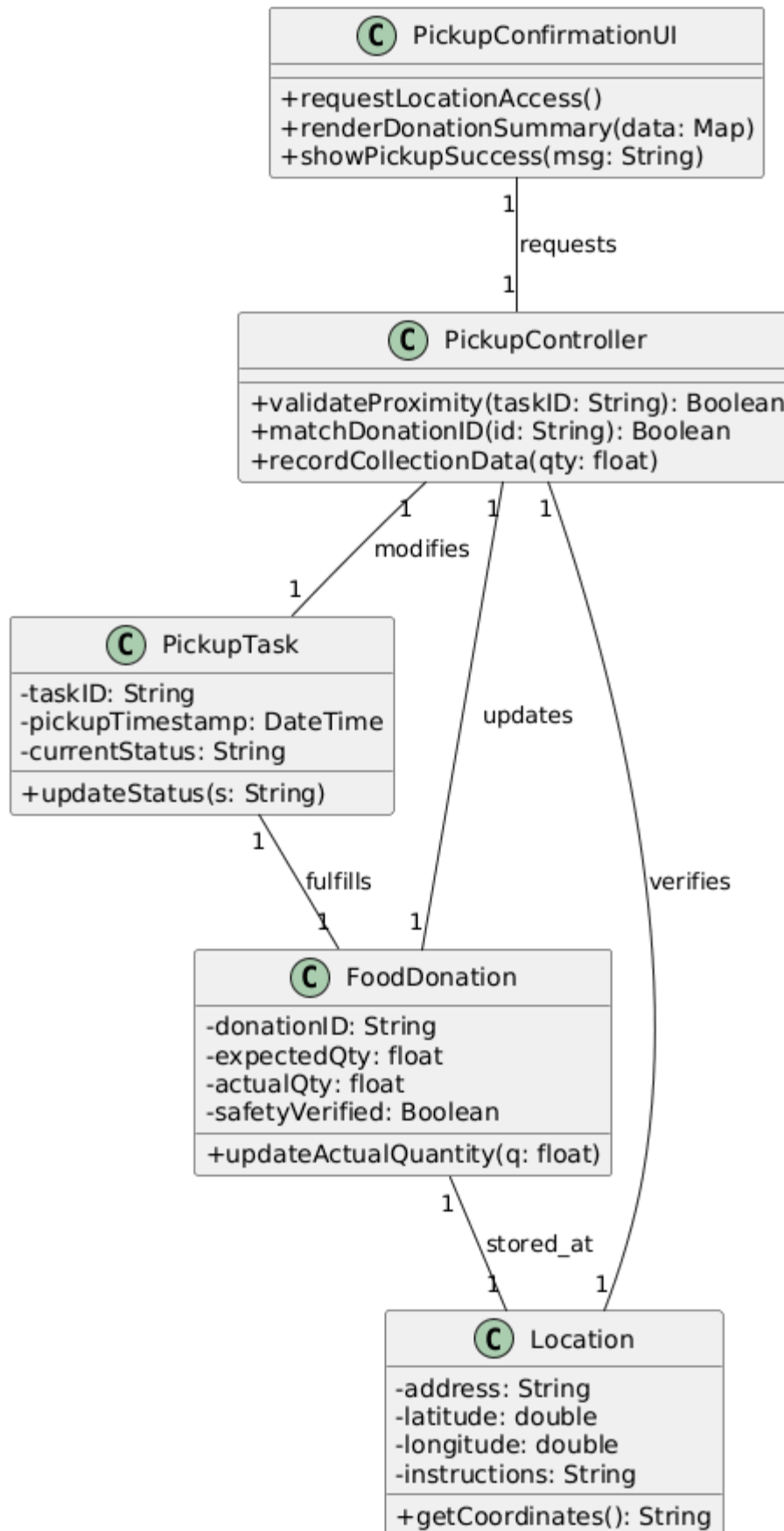
B. Activity Diagram

Activity Diagram - Pick Up Food (User Perspective)



C. Class Diagram

Class Diagram: Pick Up Food (Detailed)



D. UI

Pickup at donor

✓ GPS verified — within range

Scan QR / enter ID



Quantity match?

✓ Matches

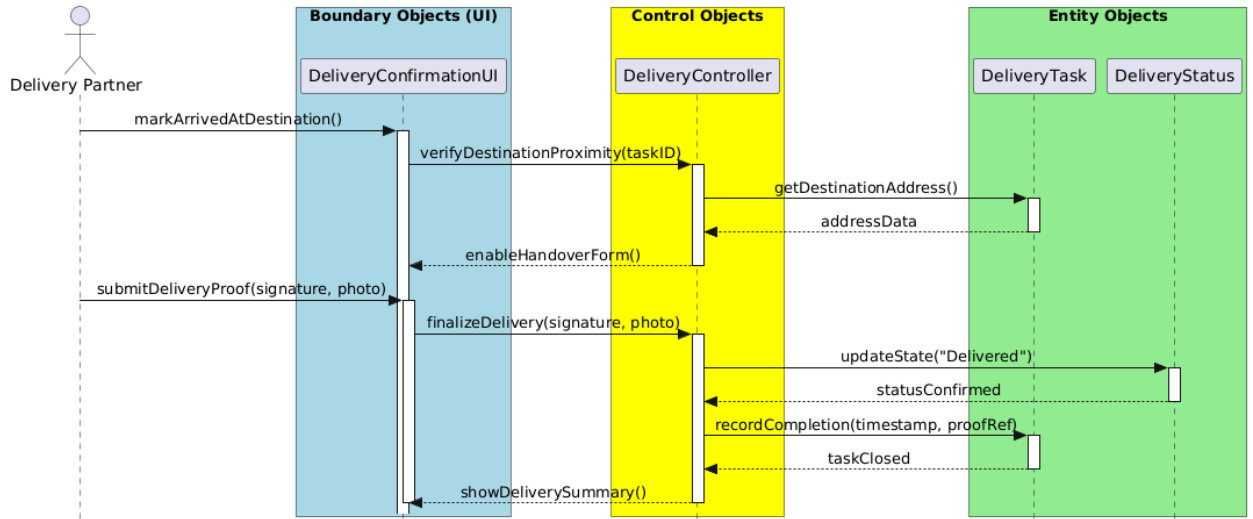
Adjust qty

Reason for
discrepancy...

Confirm pickup

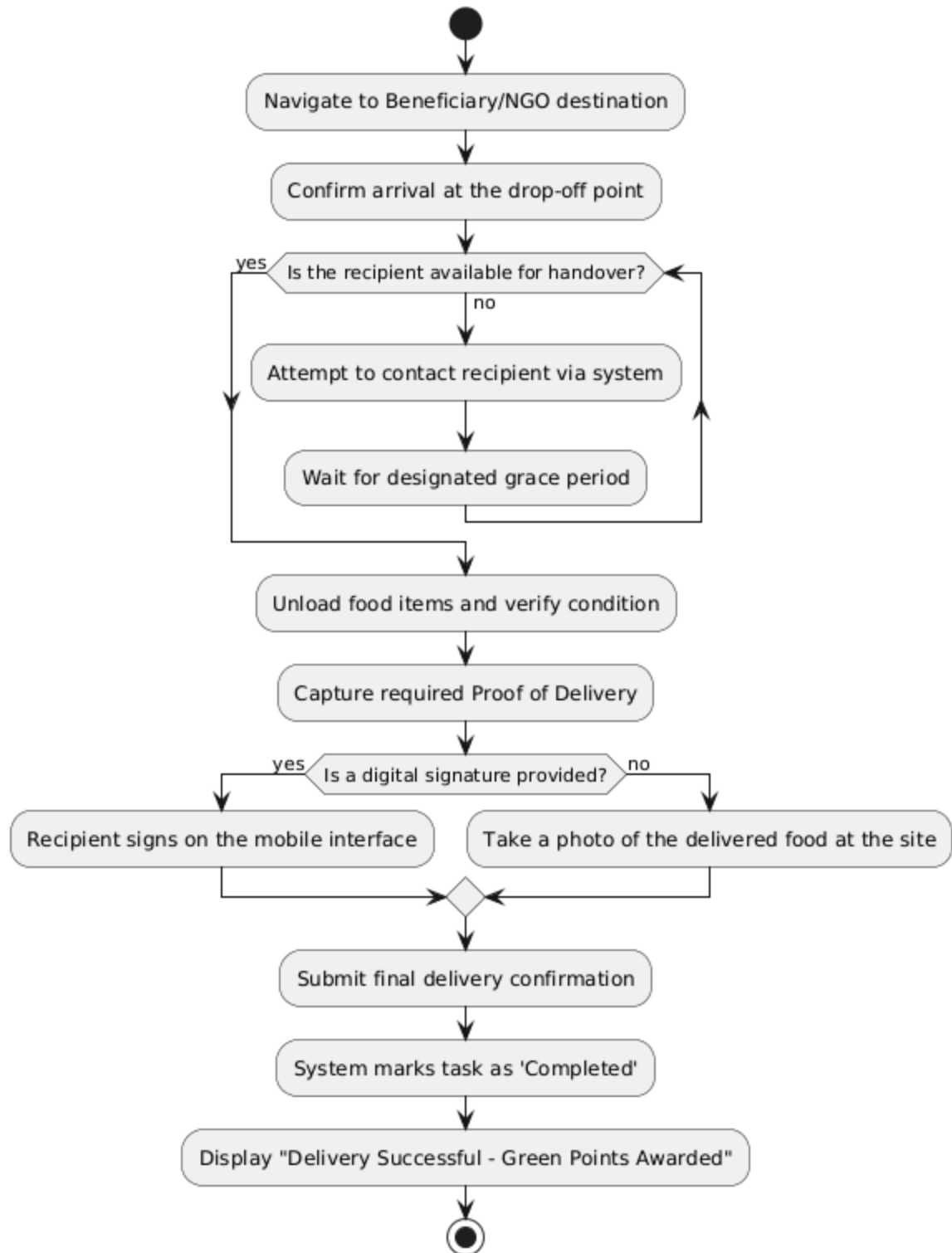
4.3 Deliver Food

A. Sequence Diagram



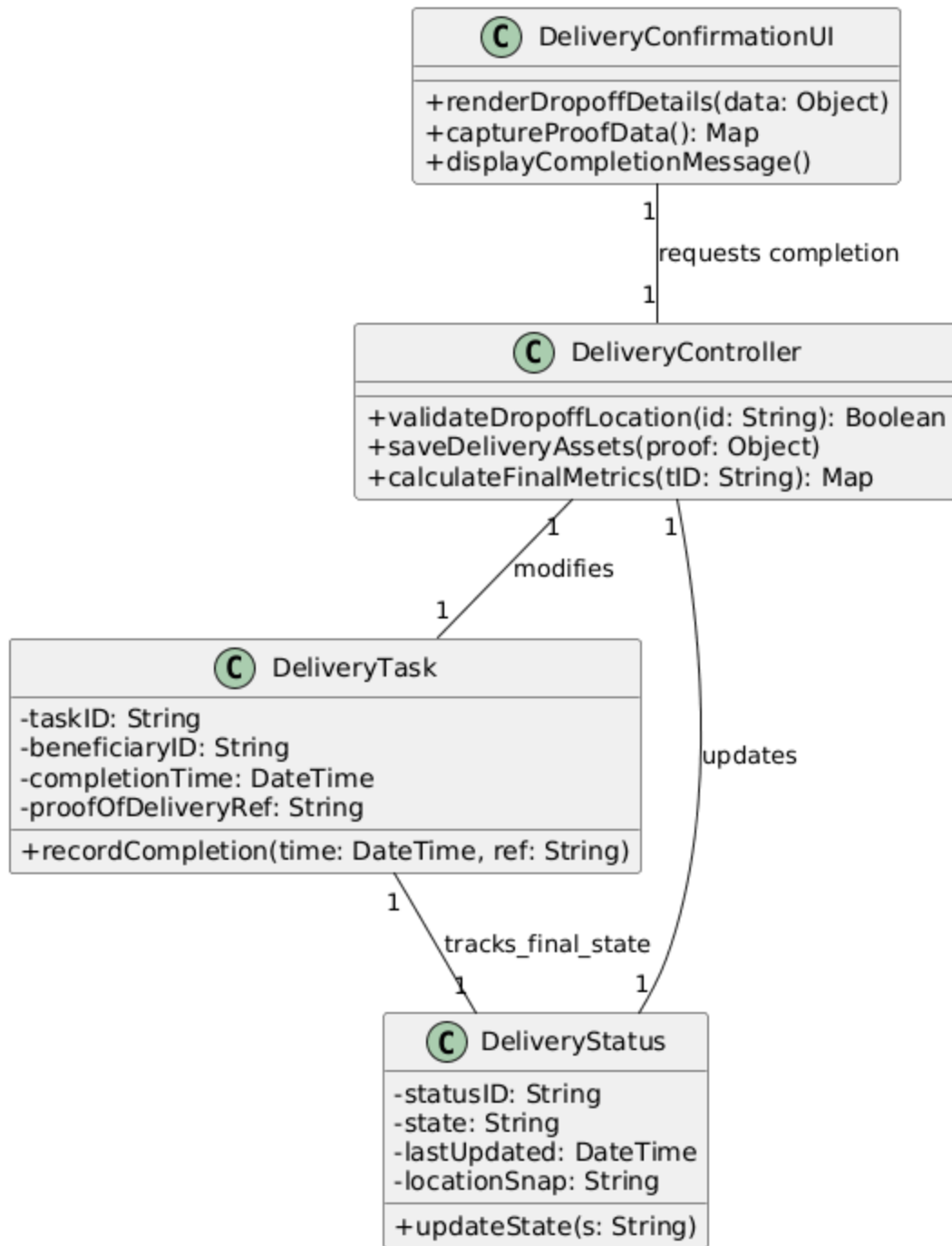
B. Activity Diagram

Activity Diagram - Deliver Food (User Perspective)



C. Class Diagram

Class Diagram: Deliver Food (Detailed)



Ui -

Delivery drop-off

NGO Centre

Confirm recipient available

Available

Contact
them

Proof of delivery



Sign



Photo

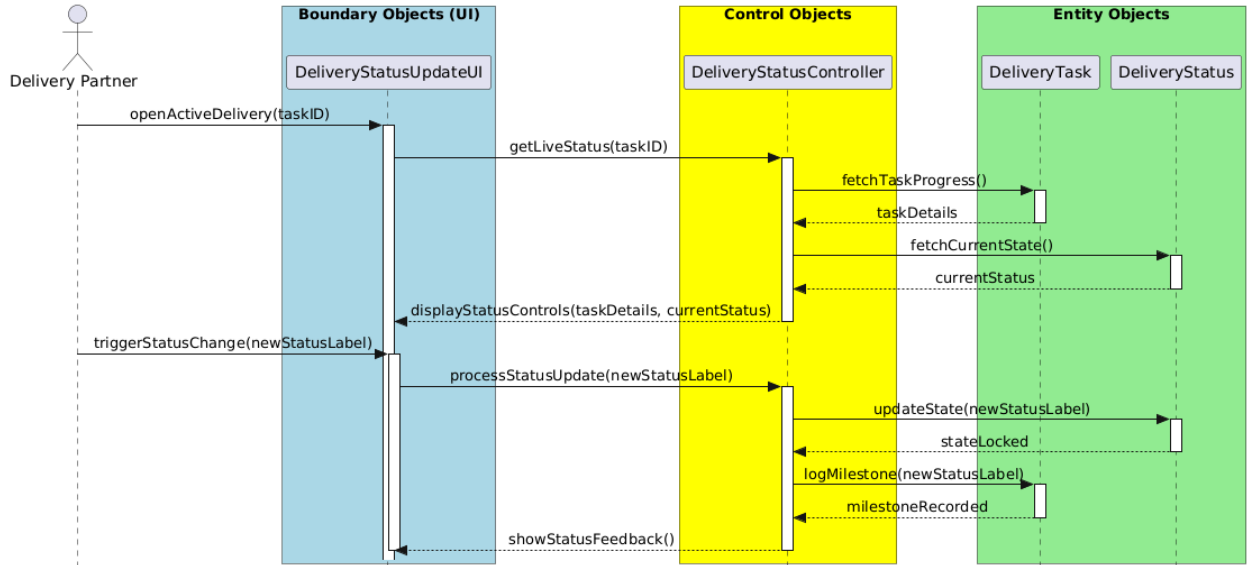
Submit delivery



Delivery successful —
points awarded

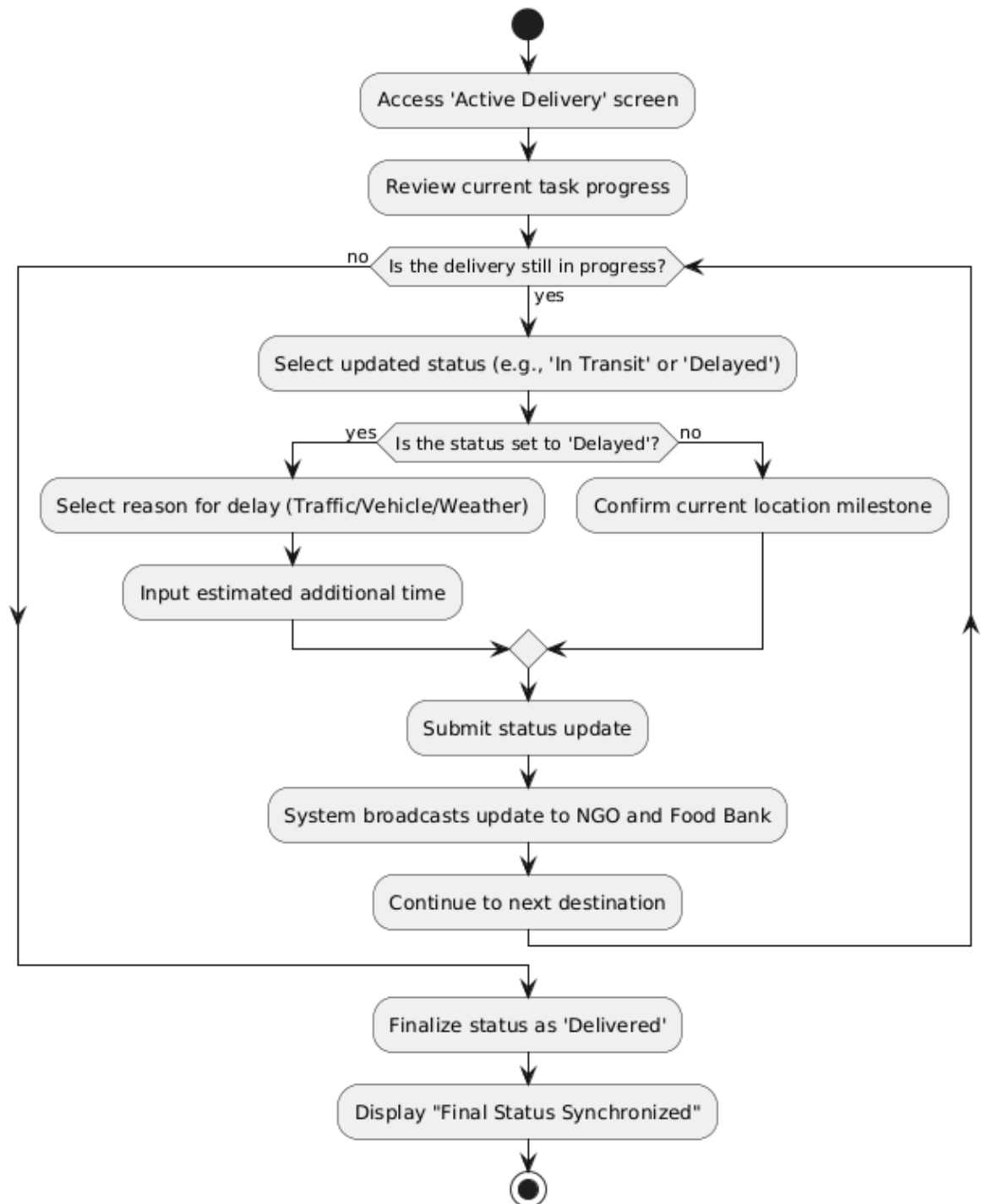
4.4 Update Delivery Status

A. Sequence Diagram



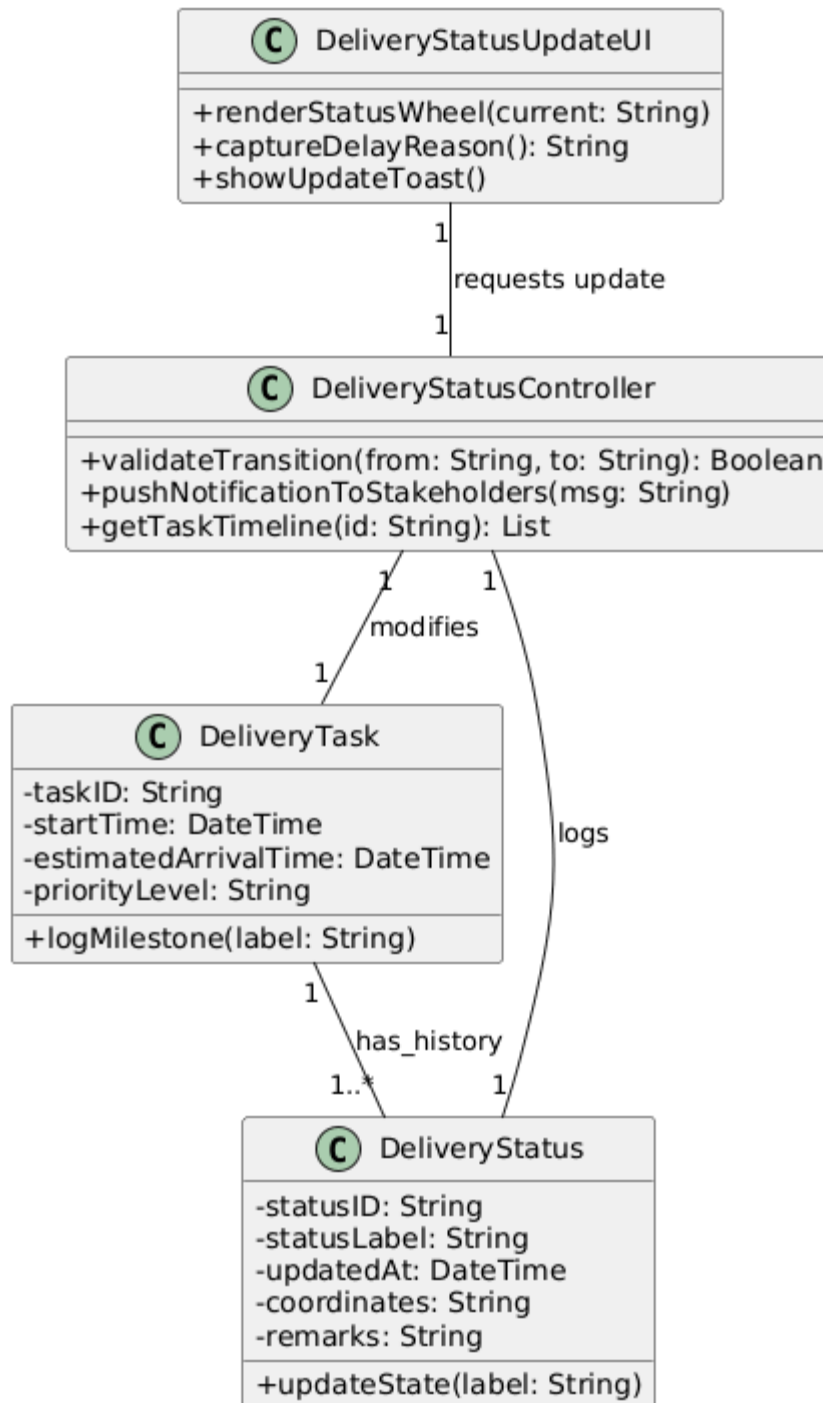
B. Activity Diagram

Activity Diagram - Update Delivery Status (User Perspective)



C. Class Diagram

Class Diagram: Update Delivery Status (Detailed)



D. UI

Active delivery

Task #1042 · in progress

Update status

☒ In Transit

☐ Delayed

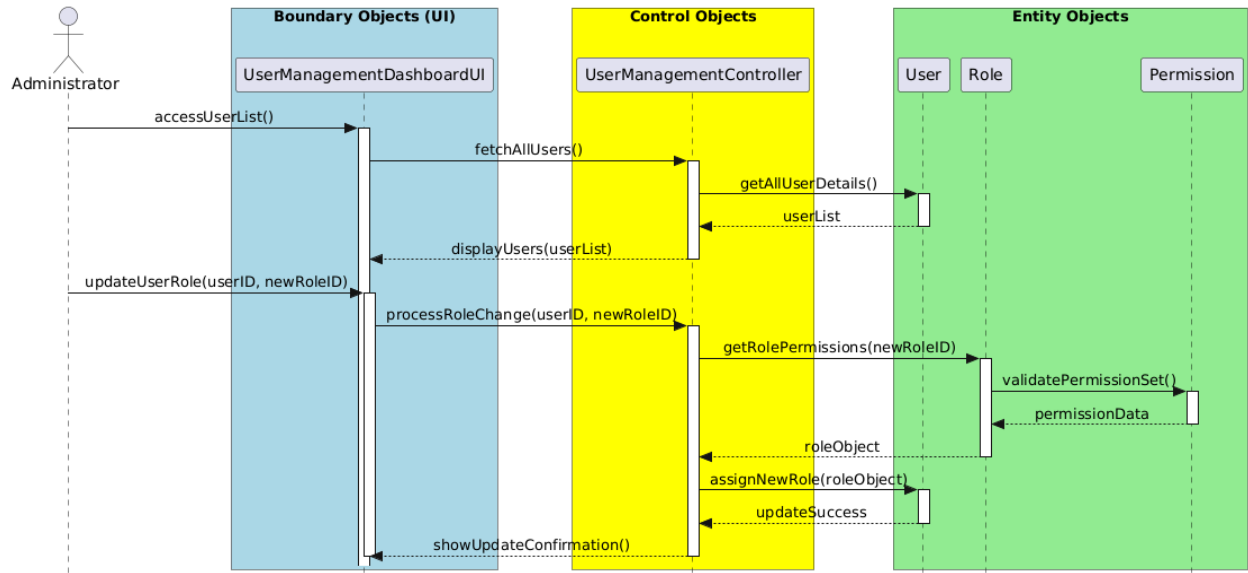
Delay reason... ▼

Est. extra time (mins)

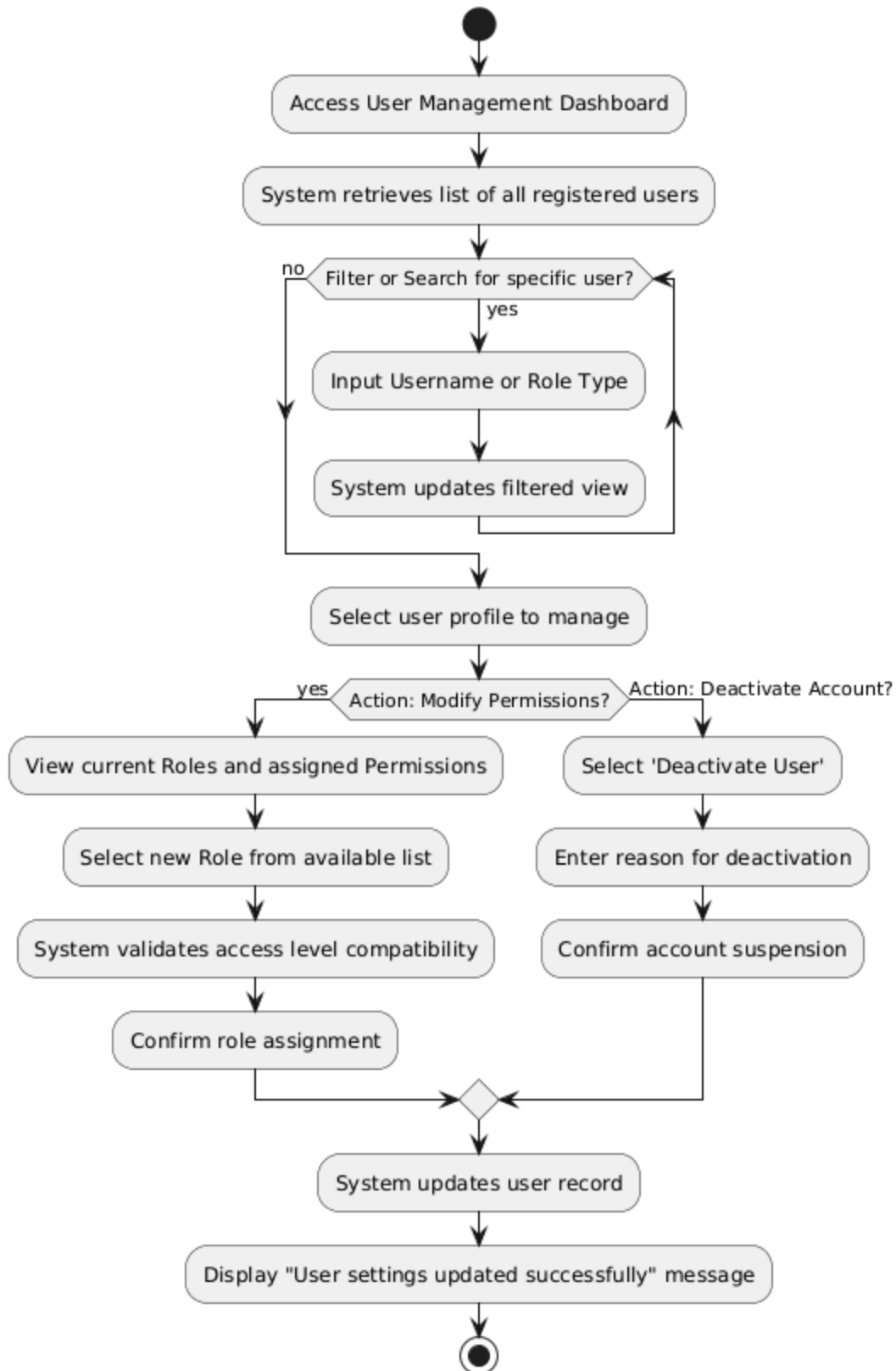
Submit update

5. System Administrator Use Cases

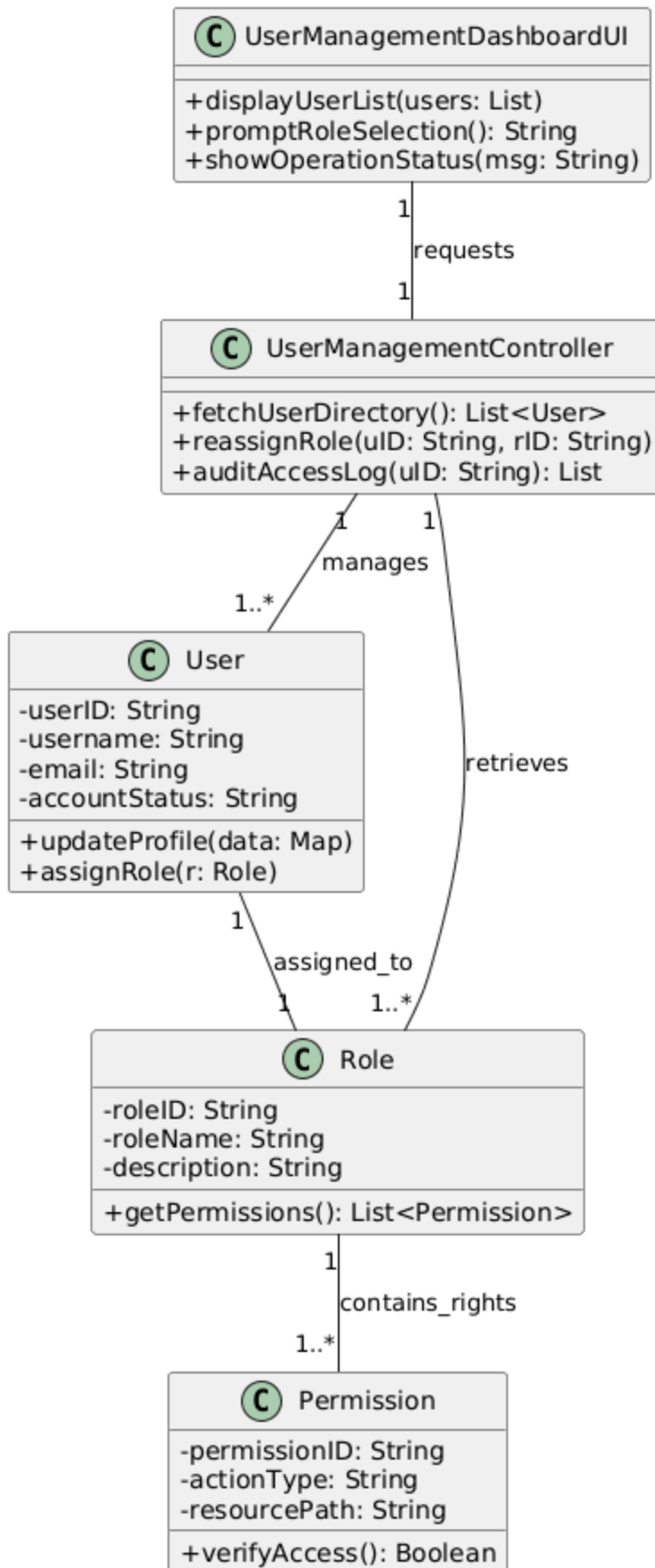
5.1 Manage Users



Activity Diagram - Manage Users (User Perspective)



Class Diagram: Manage Users (Detailed)



User management

Search name or role...

Arjun Mehta

Volunteer · Active

Priya Nair

Admin · Active

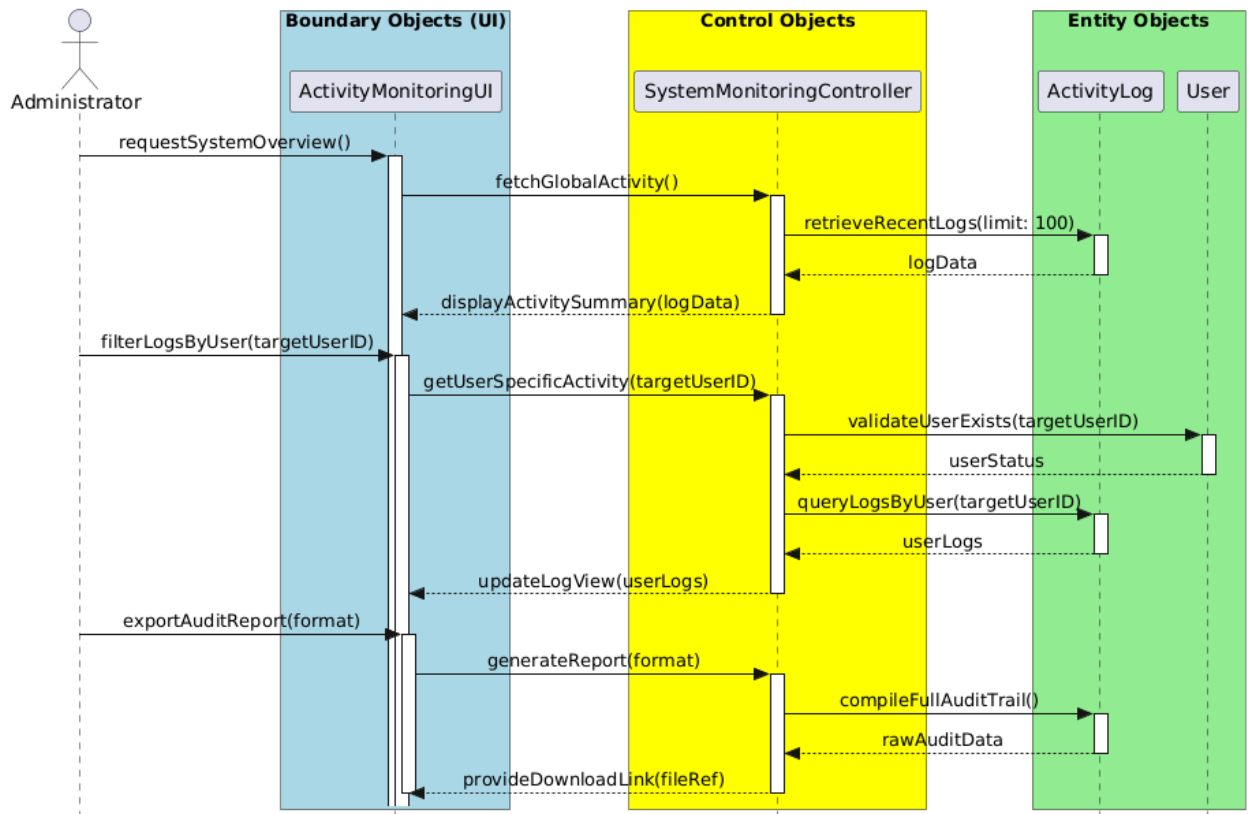
Change
role

Deactivate

User settings updated

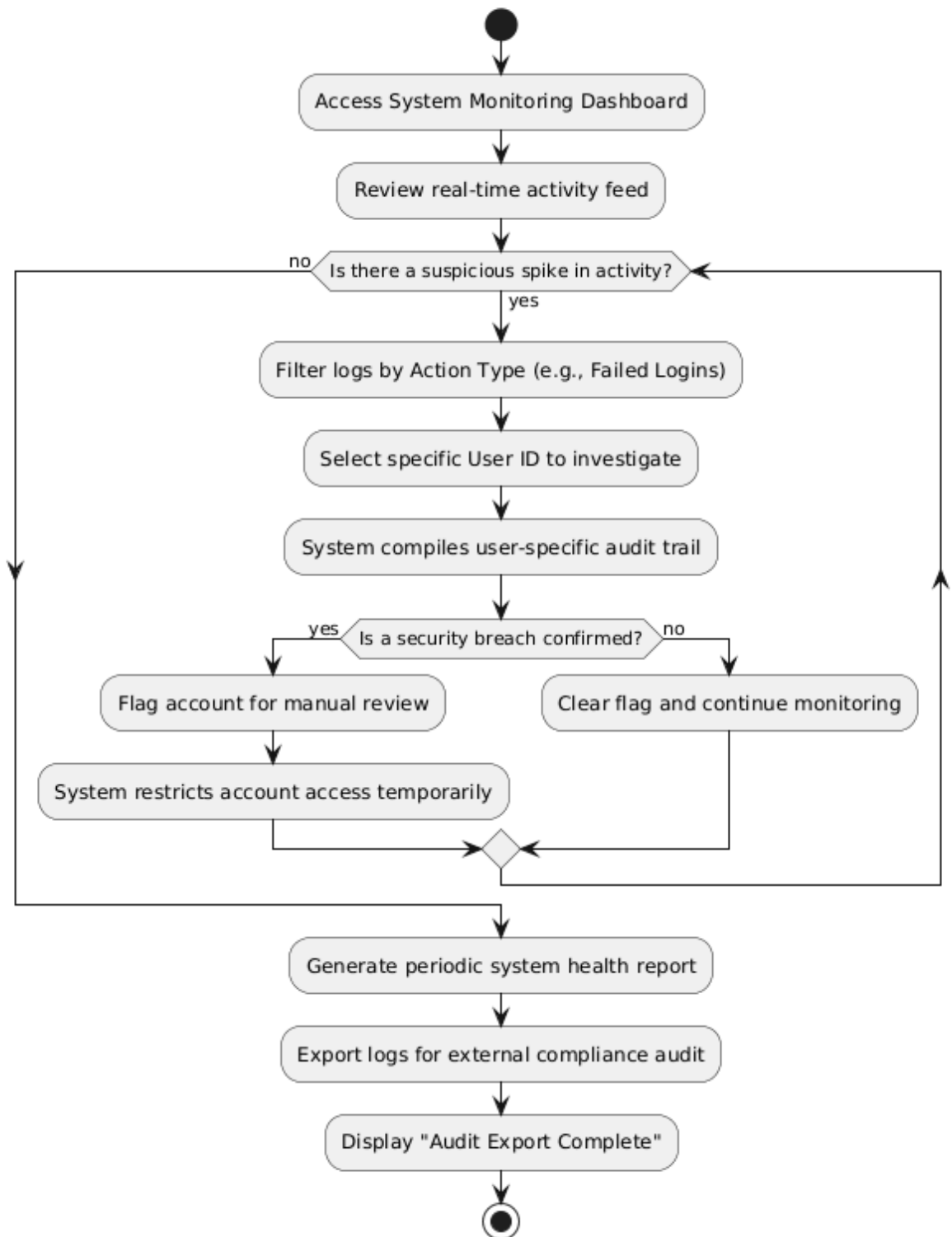
5.2 Monitor System Activity

A. Sequence Diagram



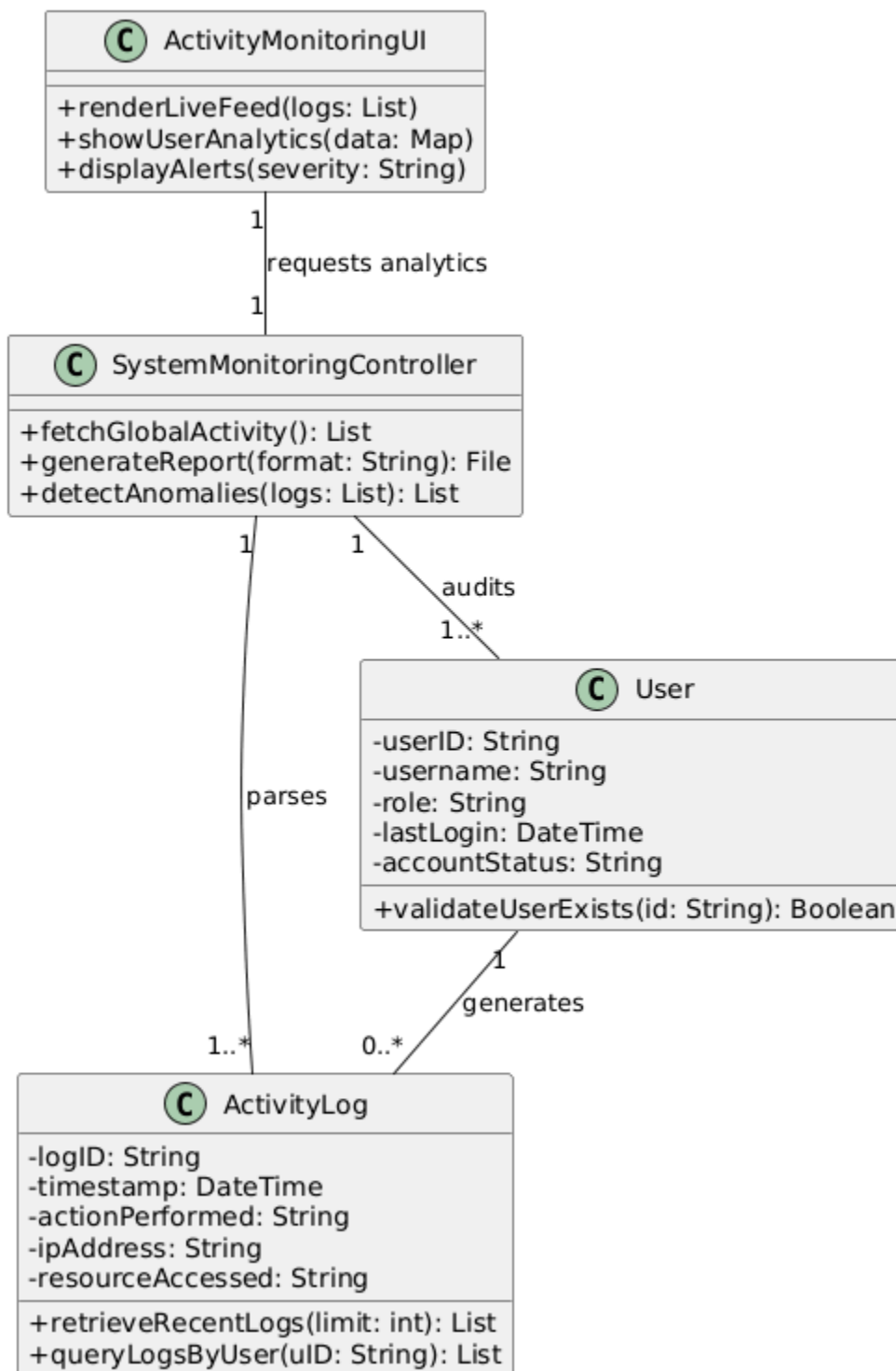
B. Activity Diagram

Activity Diagram - Monitor System Activity (User Perspective)

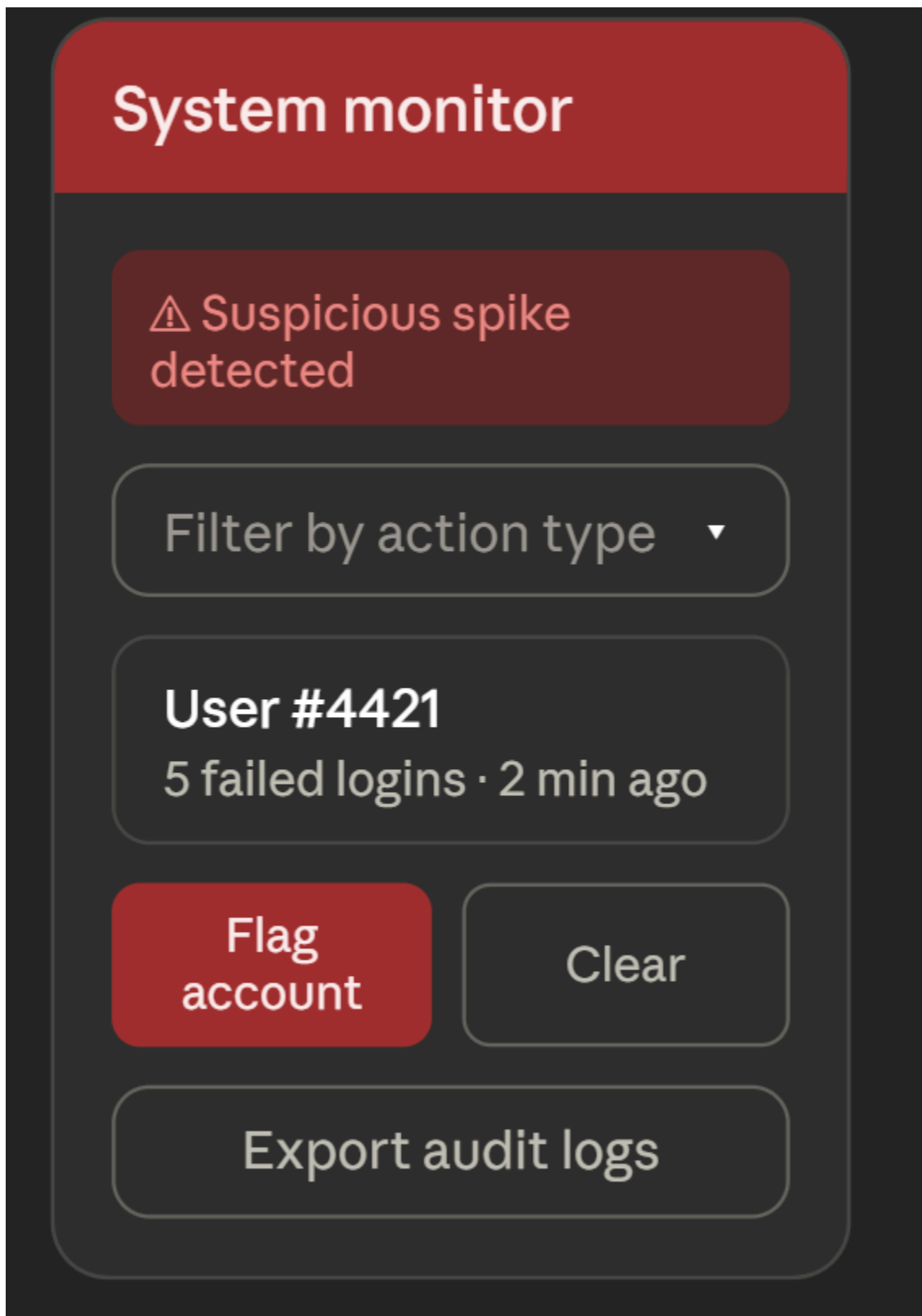


C. Class Diagram

Class Diagram: Monitor System Activity (Detailed)

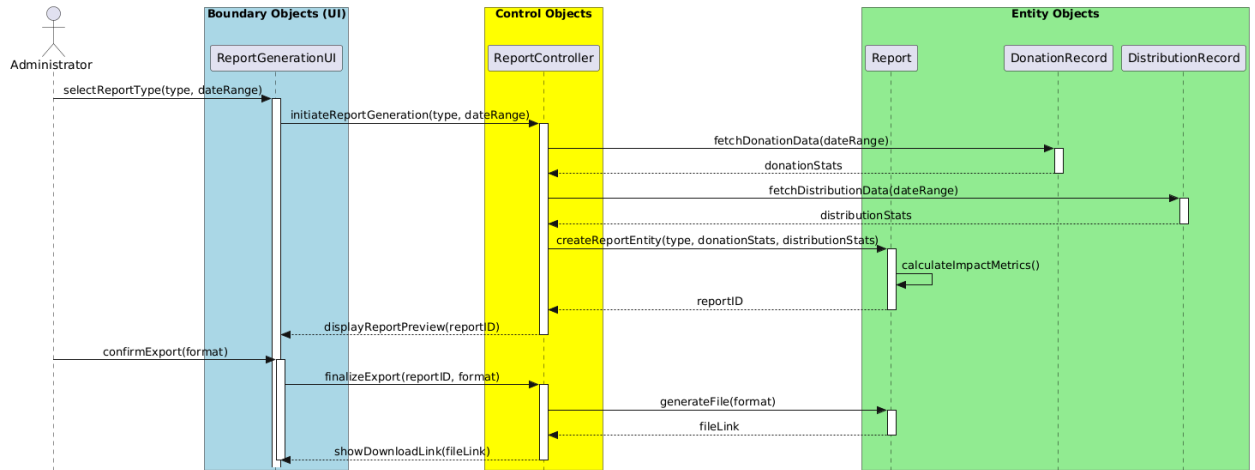


D. UI



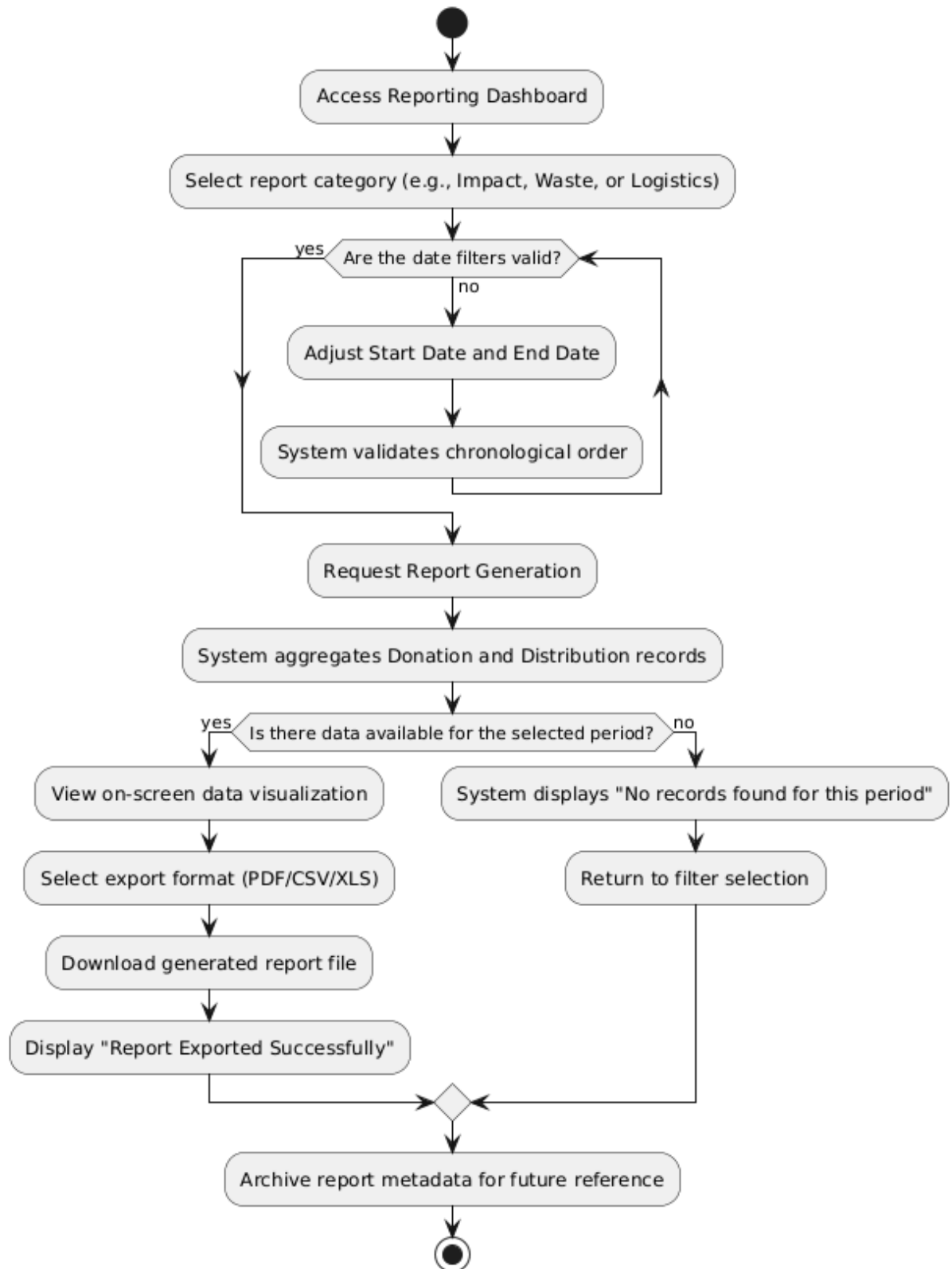
5.3 Generate Reports

A. Sequence Diagram



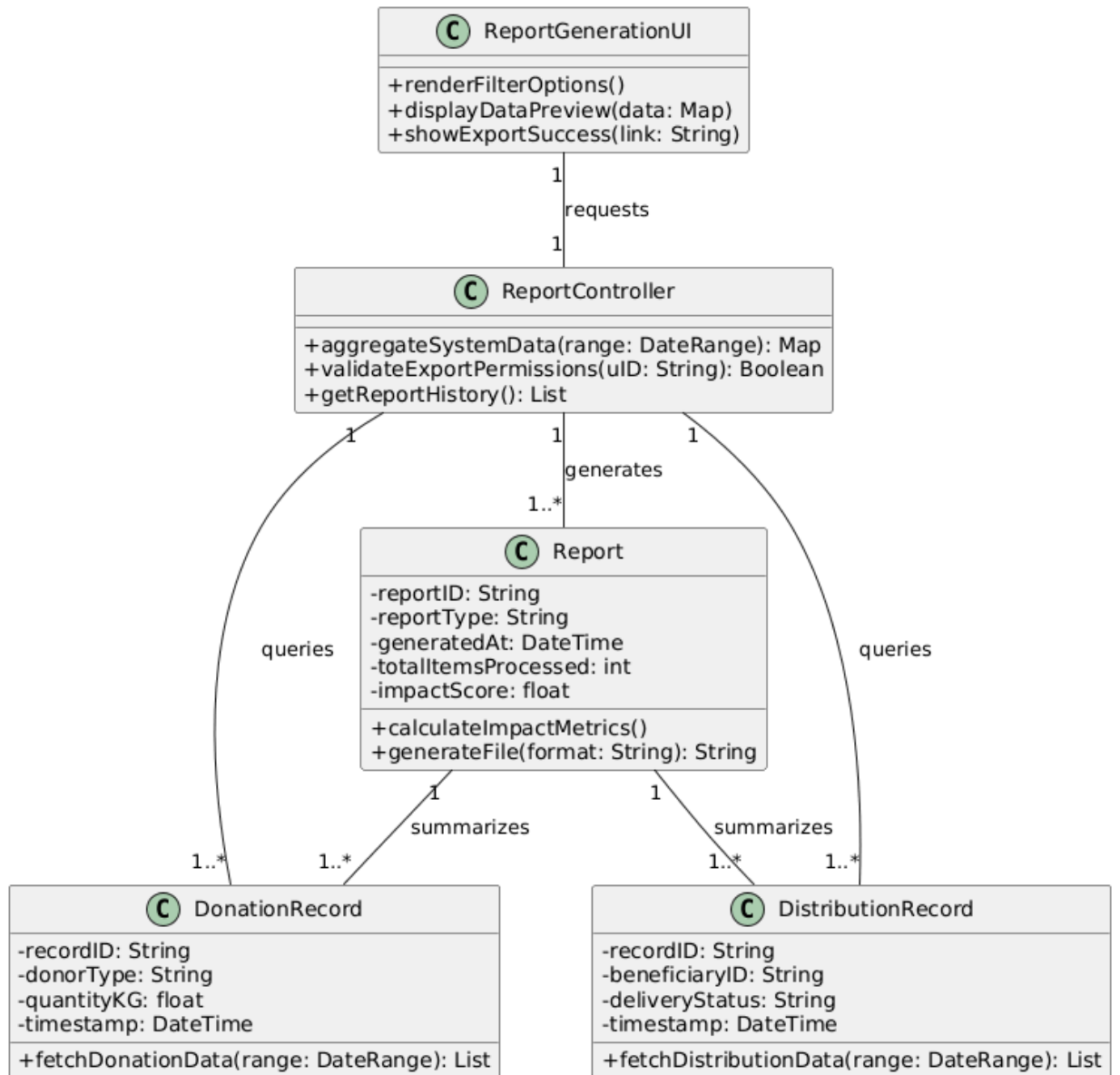
B. Activity Diagram

Activity Diagram - Generate Reports (User Perspective)



C. Class Diagram

Class Diagram: Generate Reports (Detailed)



D. UI

The image shows a UI mockup for a 'Reports' section. It features a dark gray background with a blue header bar at the top containing the word 'Reports' in white. Below the header, the text 'Report category' is displayed in a light gray font. Underneath this, there are three rounded rectangular buttons: 'Impact' (highlighted with a blue background and white text), 'Waste' (light gray background and text), and 'Logistics' (light gray background and text). Below these, there are two more rounded rectangular buttons: 'Start date' and 'End date', both with light gray backgrounds and text. A large, prominent blue button with white text labeled 'Generate report' is positioned below the date buttons. At the bottom, the text 'Export as' is shown in light gray, followed by three rounded rectangular buttons labeled 'PDF', 'CSV', and 'XLS', all with light gray backgrounds and text.

Reports

Report category

Impact Waste

Logistics

Start date End date

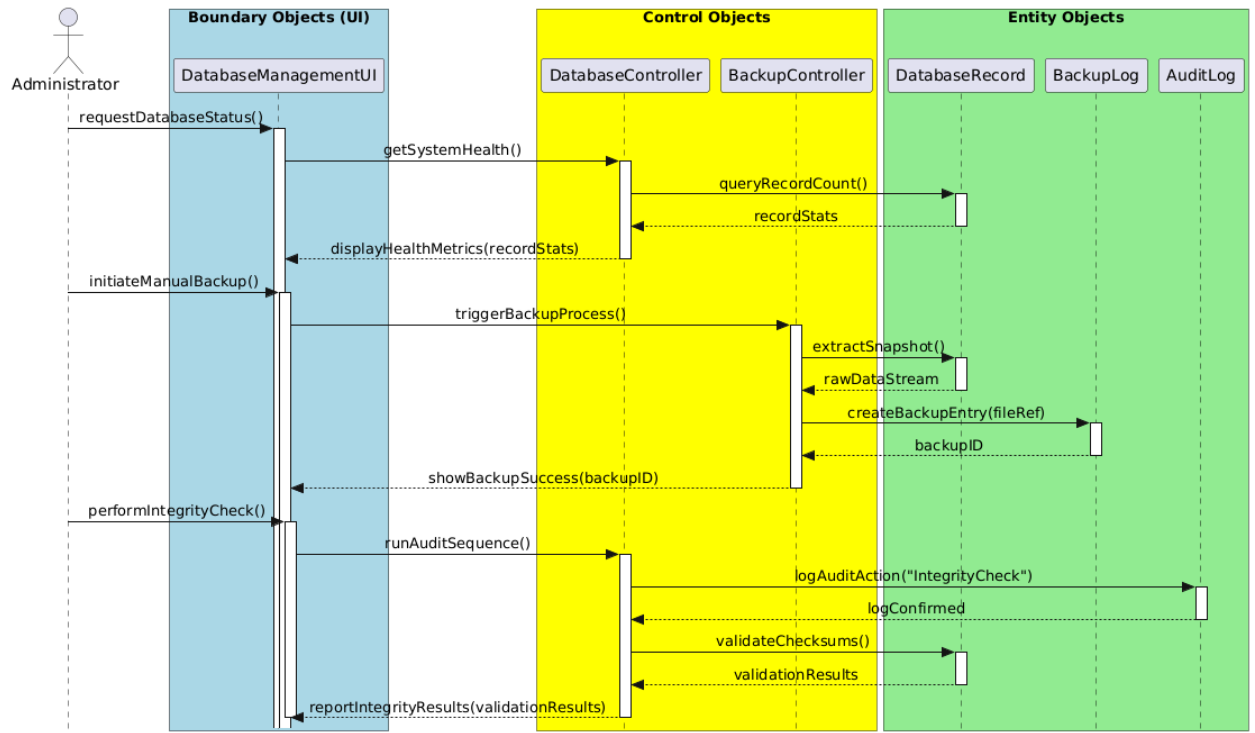
Generate report

Export as

PDF CSV XLS

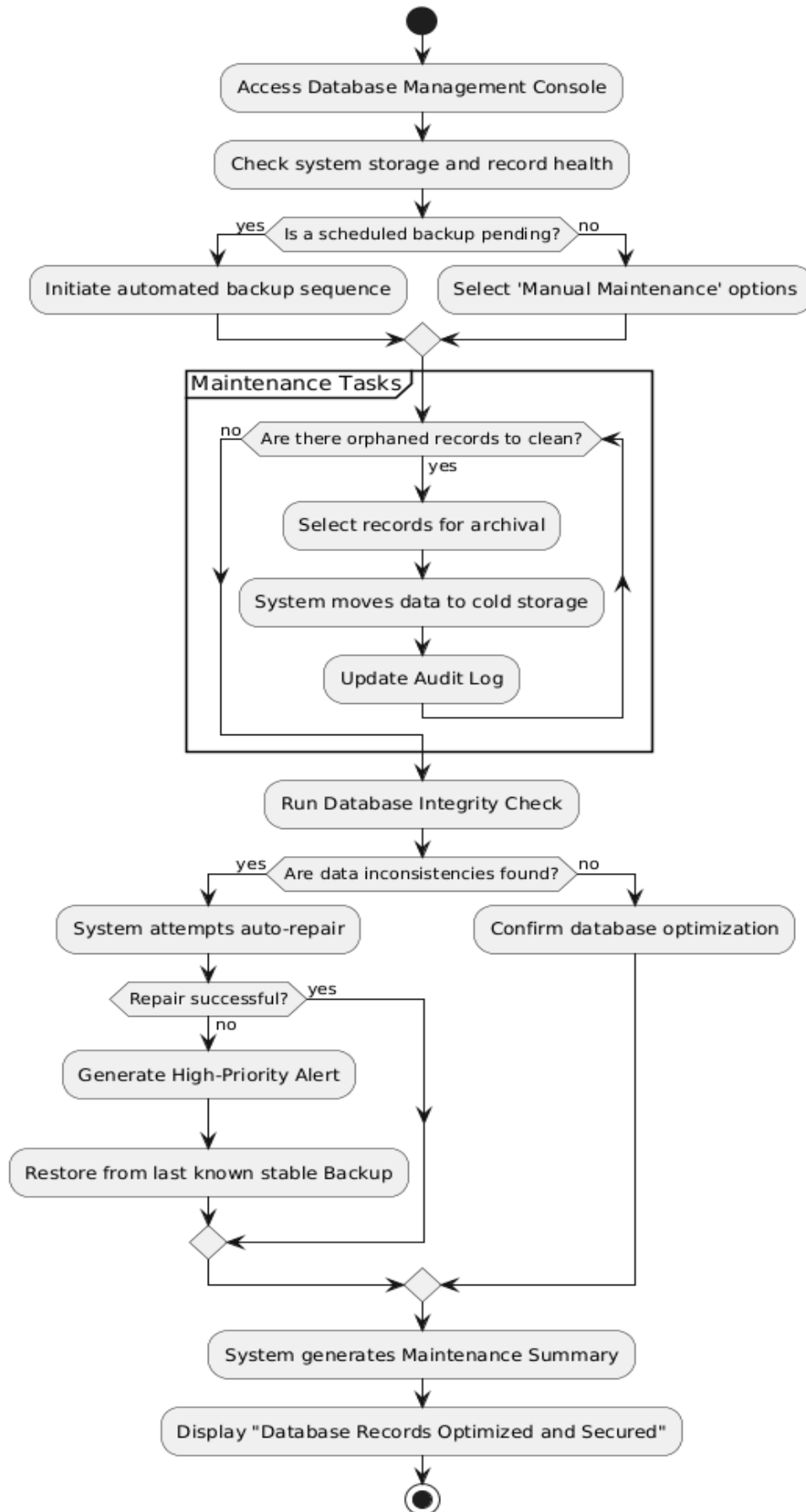
5.4 Manage Records

A. Sequence Diagram



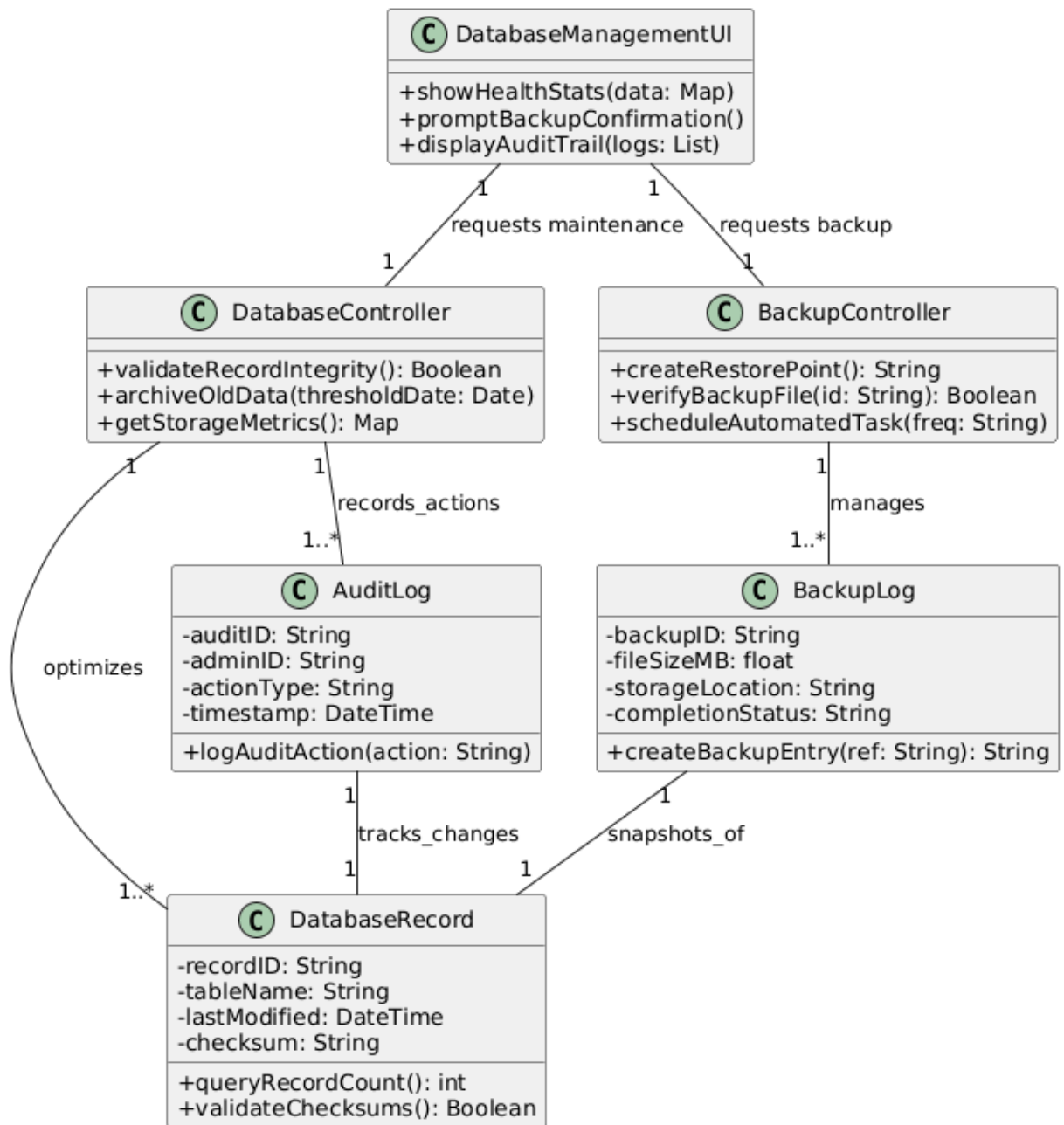
B. Activity Diagram

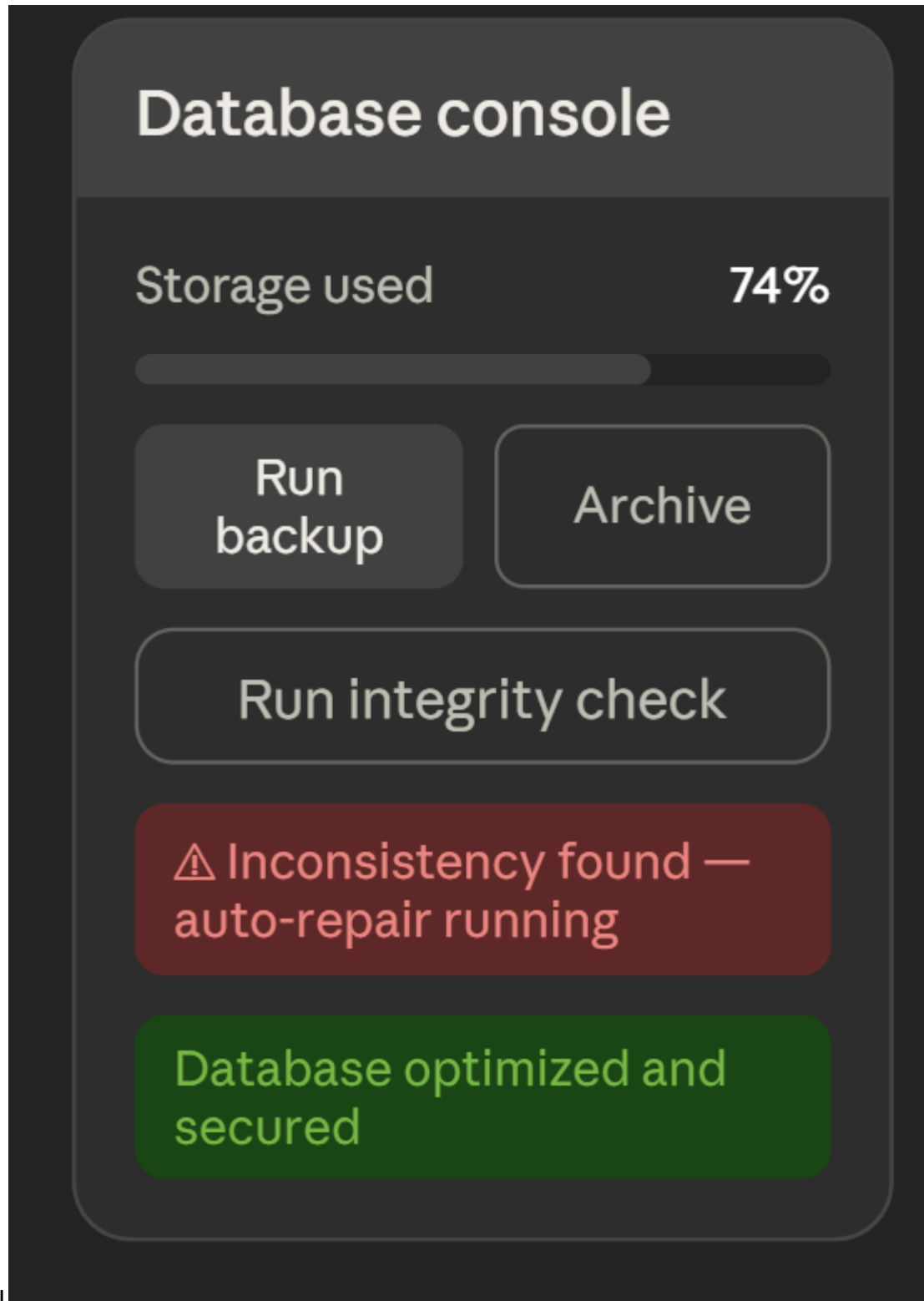
Activity Diagram - Manage Records (User Perspective)



C. Class Diagram

Class Diagram: Manage Records (Detailed)

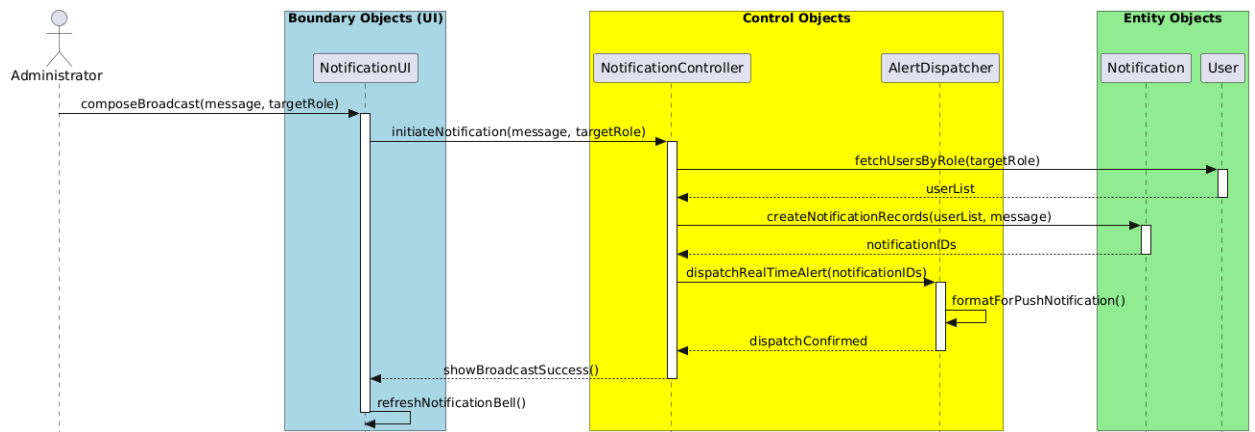




D. UI

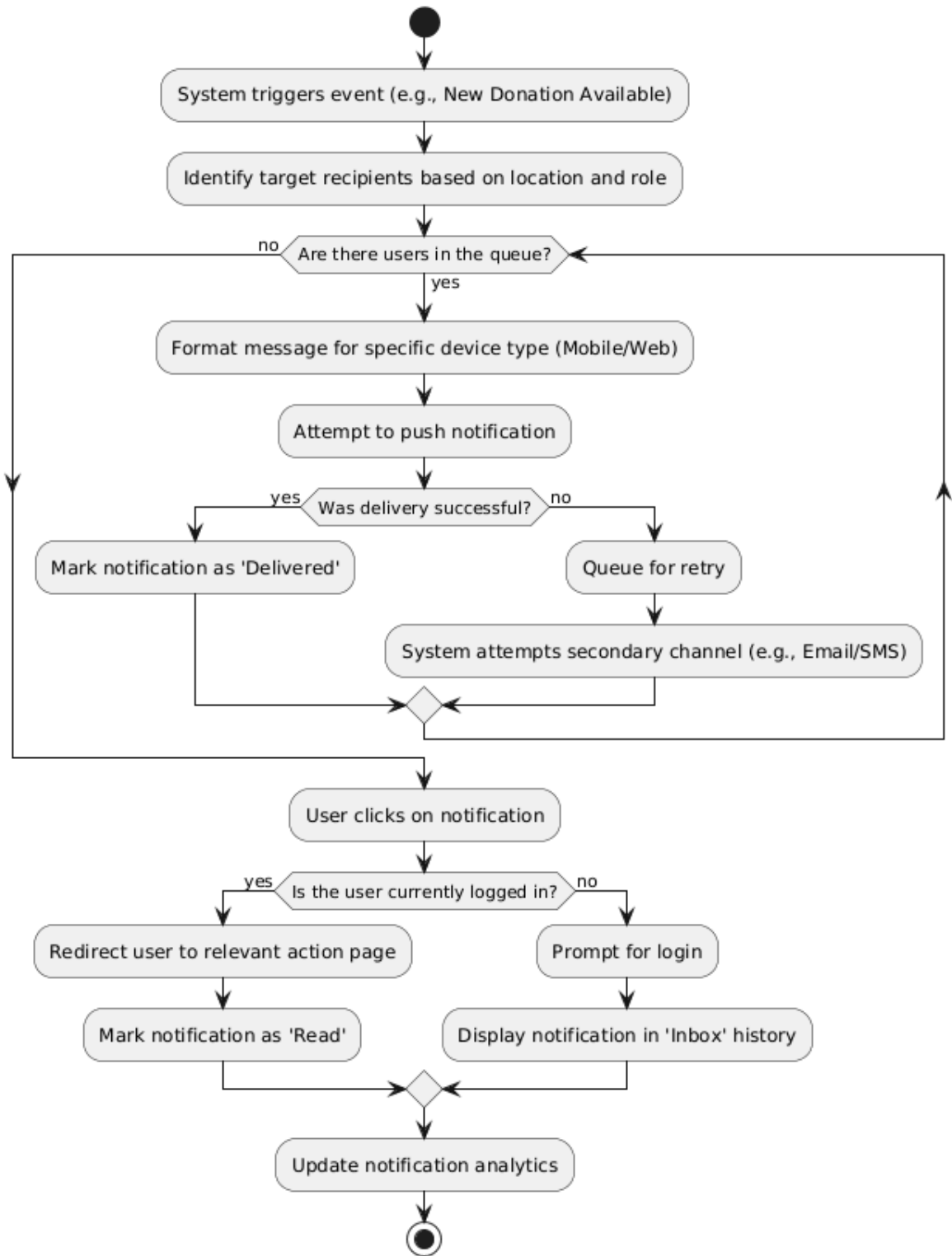
5.5 Real-Time Notification System

A. Sequence Diagram



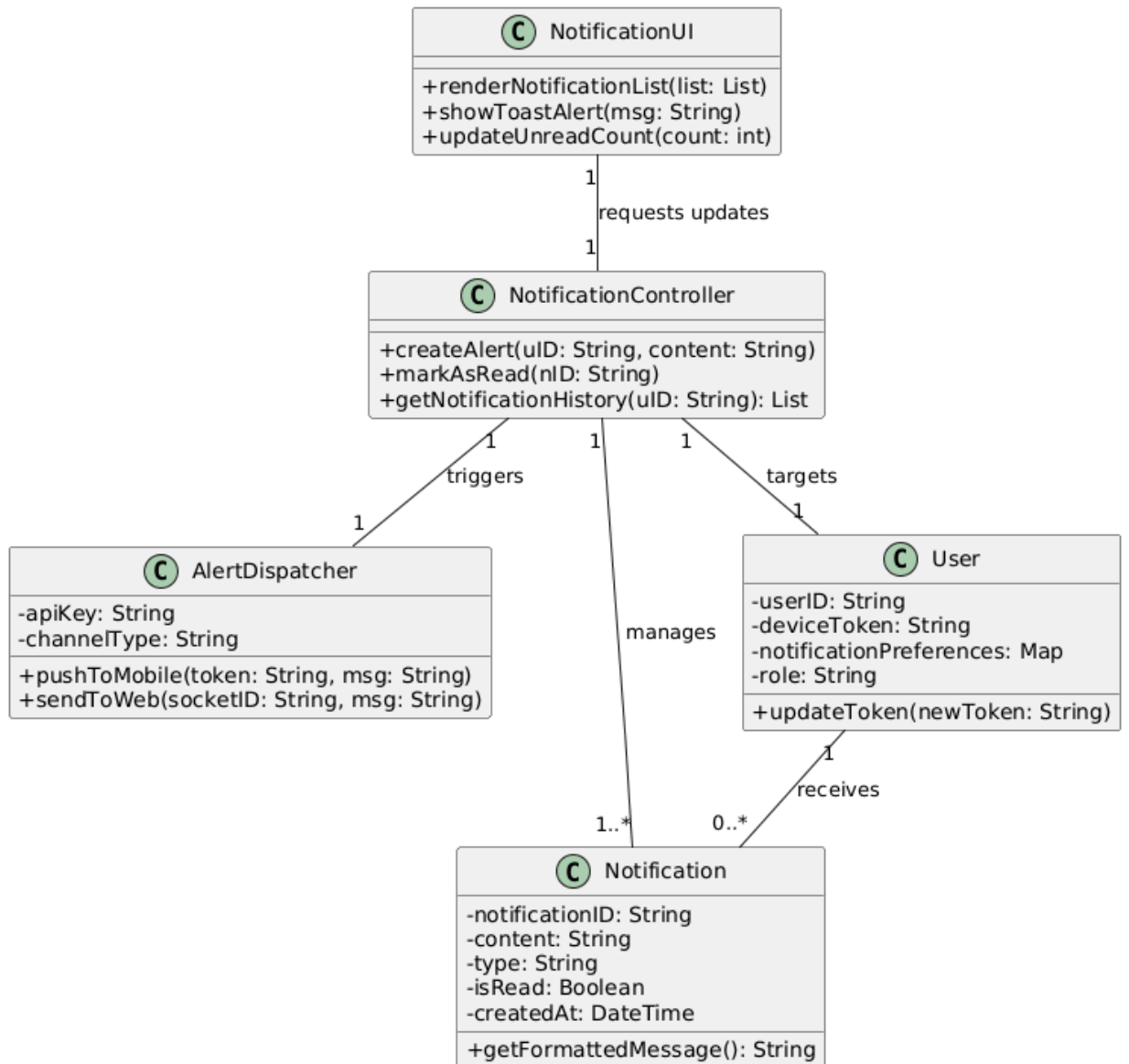
B. Activity Diagram

Activity Diagram - Real-Time Notification System (User Perspective)

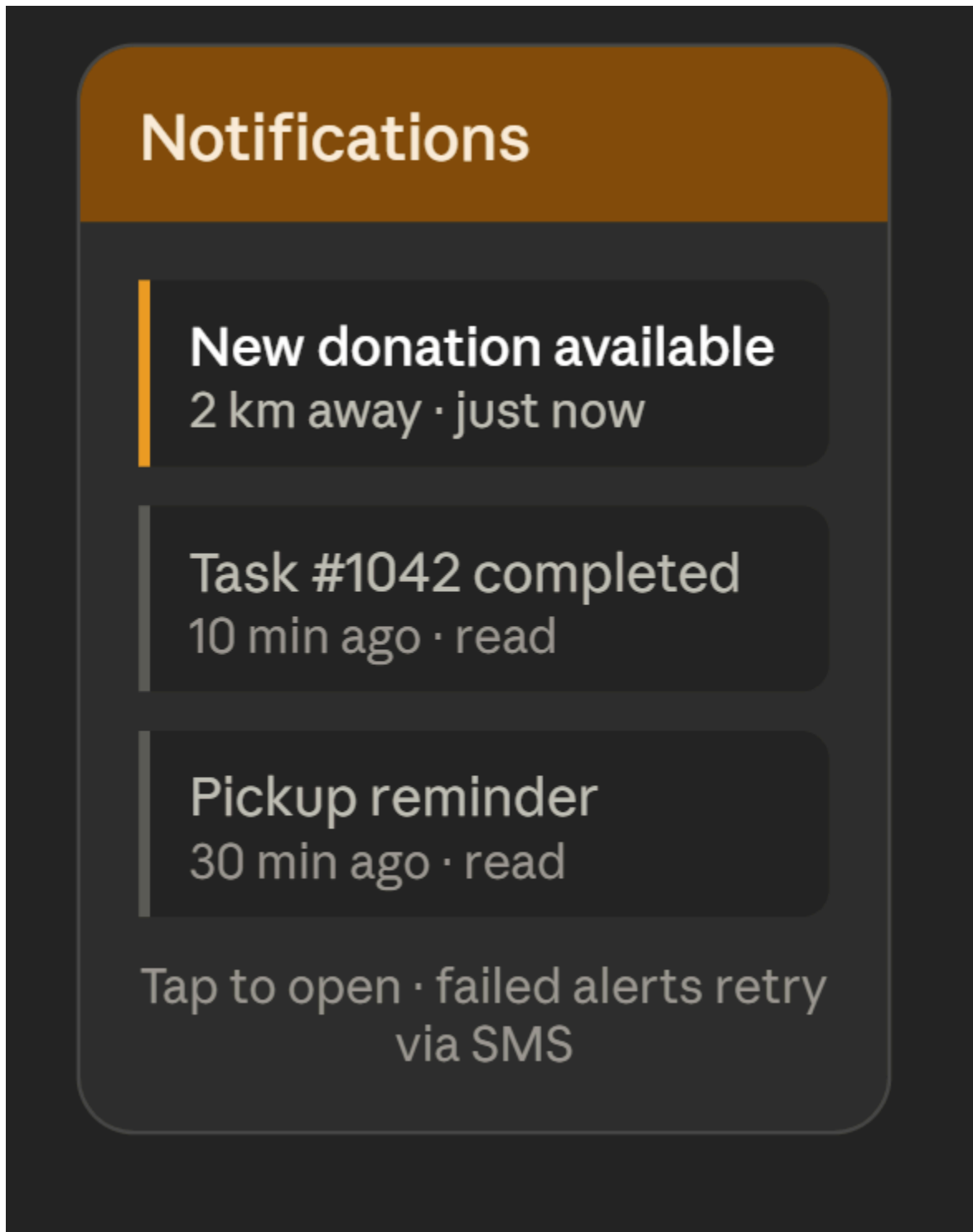


C. Class Diagram

Class Diagram: Real-Time Notification System (Detailed)

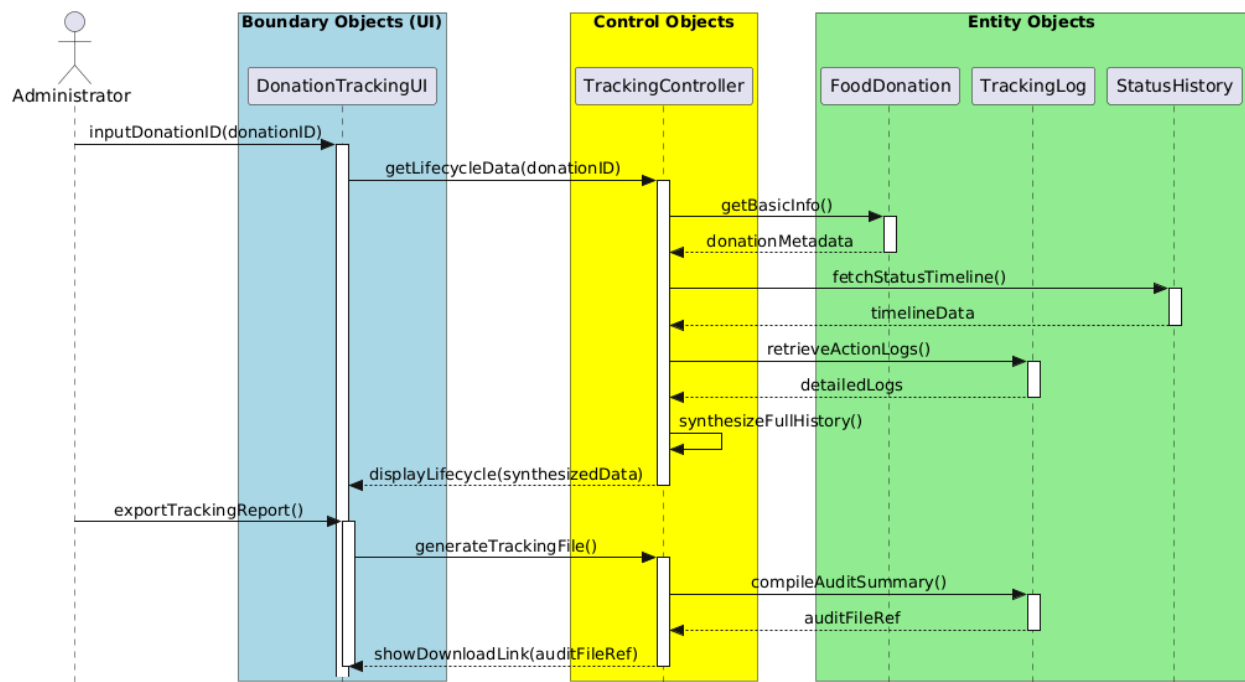


D. UI



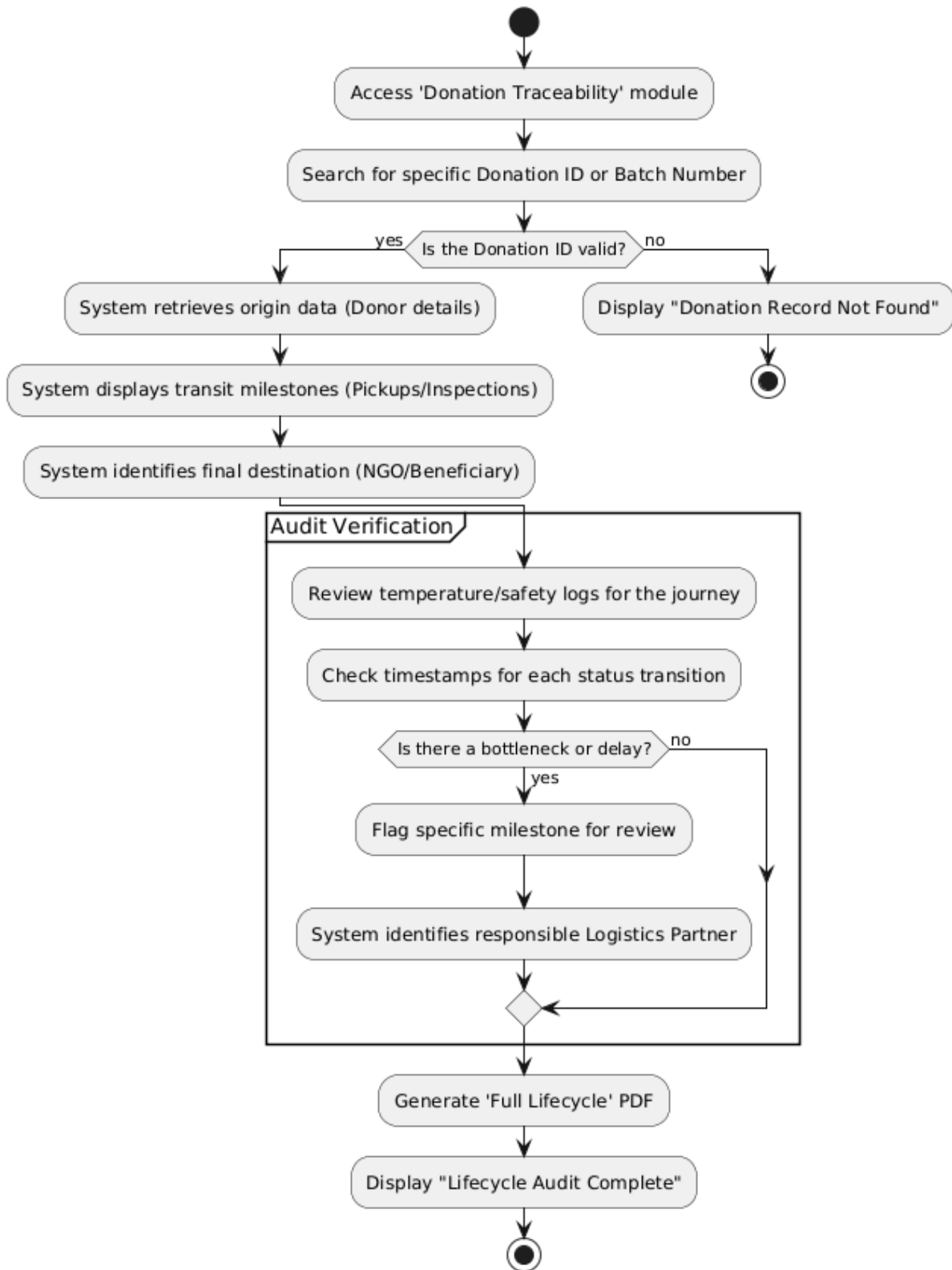
5.6 Track Donation Lifecycle

A. Sequence Diagram



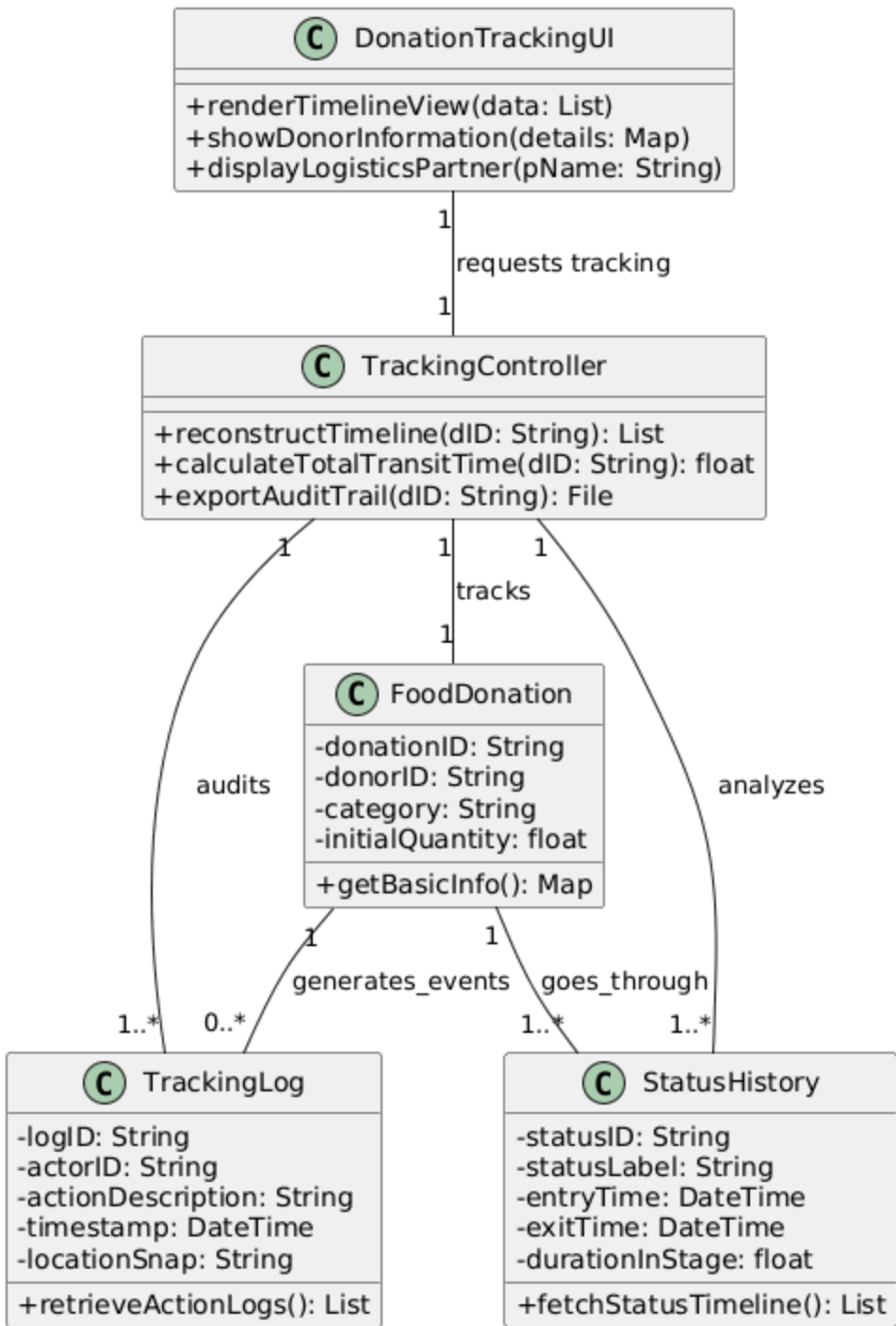
B. Activity Diagram

Activity Diagram - Track Donation Lifecycle (User Perspective)



C. Class Diagram

Class Diagram: Track Donation Lifecycle (Detailed)



Donation traceability

Enter Donation ID...

- Donor: Food Bank A
- Picked up · 10:15 AM
- In transit · verified
- NGO delivery pending

▮ Delay flagged at step 2

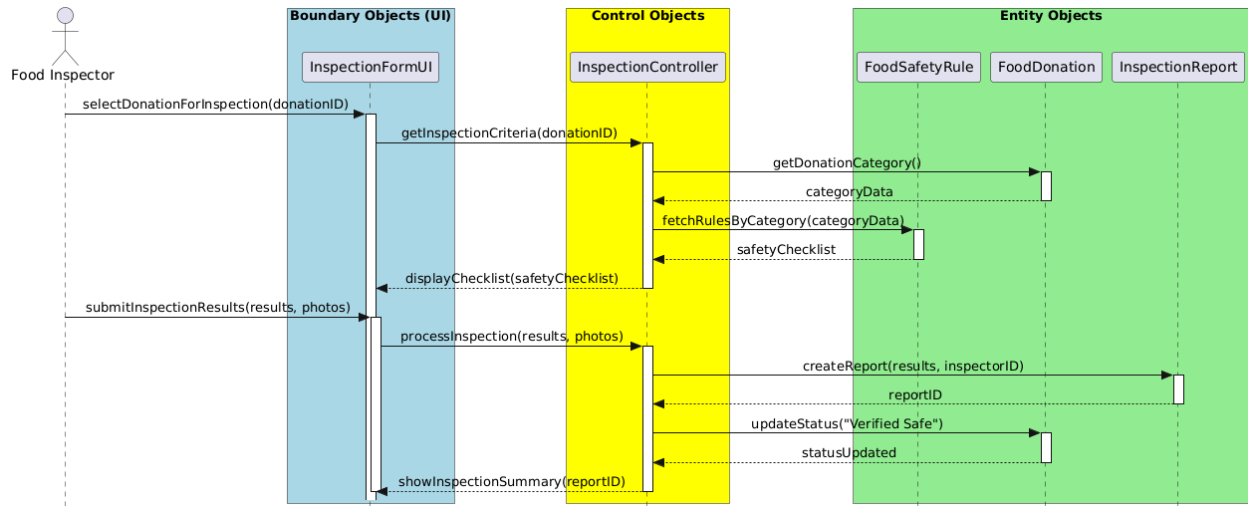
Export lifecycle PDF

D. UI

6. Food safety Inspector

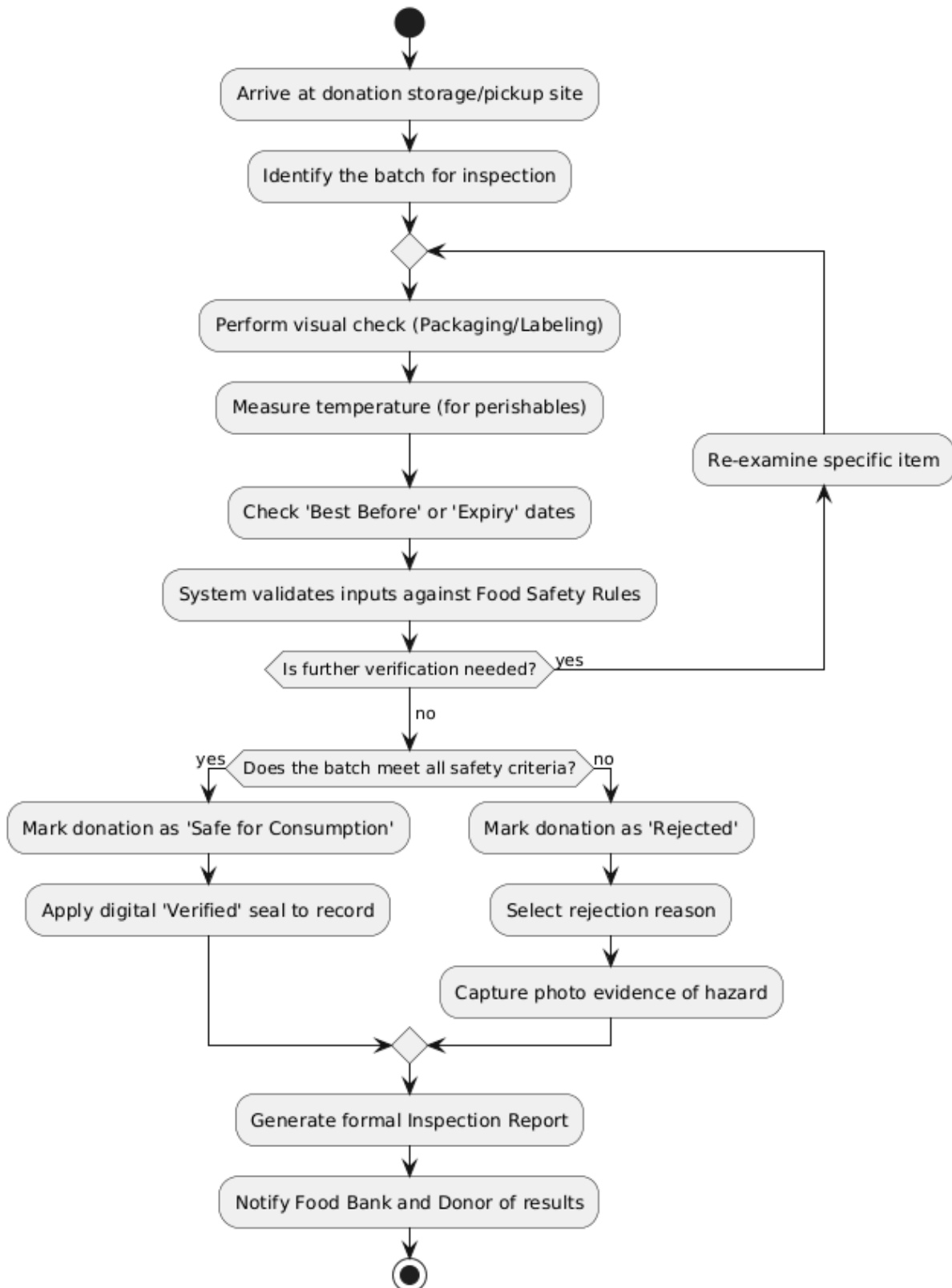
6.1 Inspect Food Donation

A. Sequence Diagram



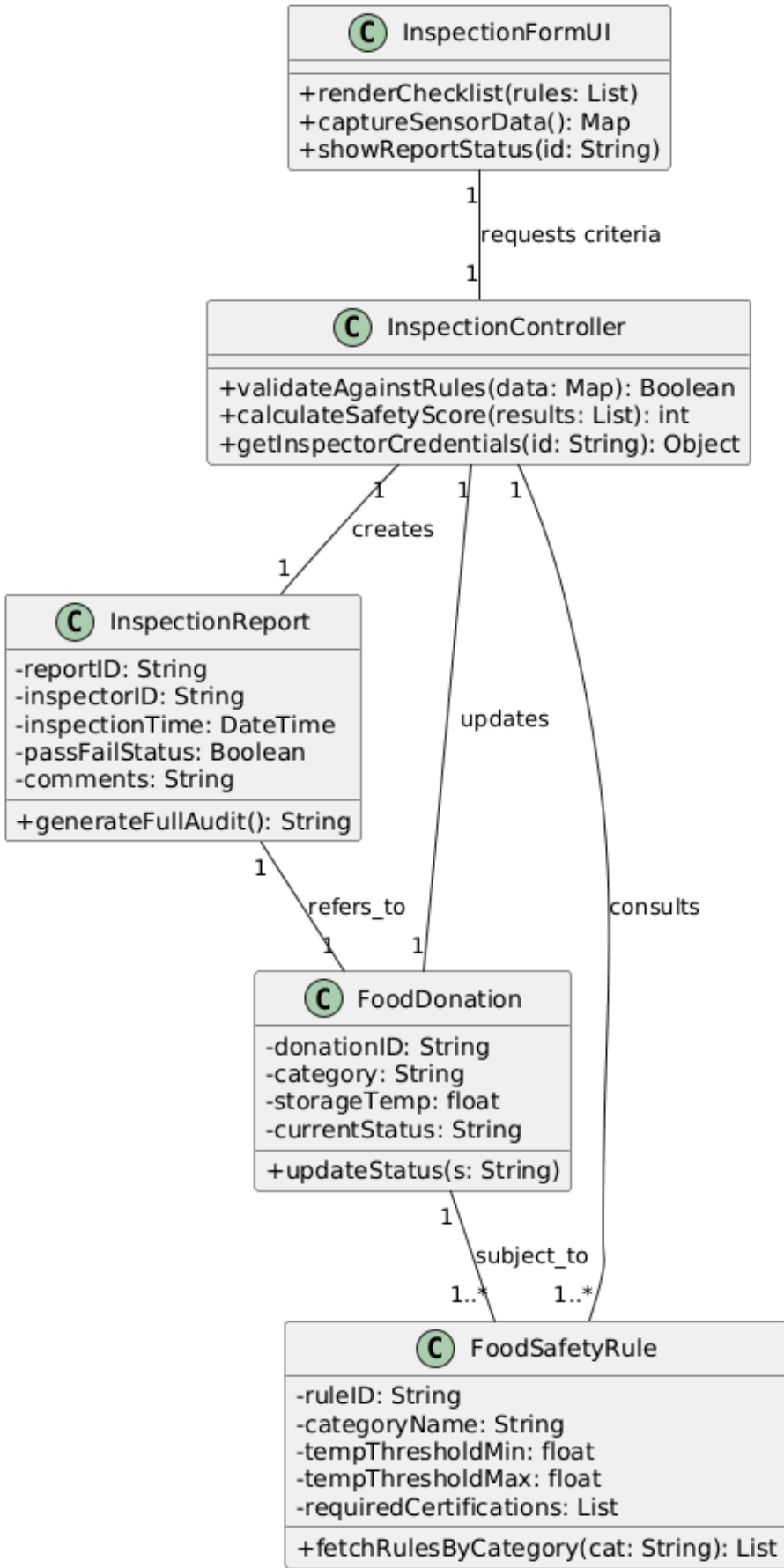
B. Activity Diagram

Activity Diagram - Inspect Food Donation (User Perspective)



C. Class Diagram

Class Diagram: Inspect Food Donation (Detailed)



Food inspection

Batch ID / scan...

☐ Packaging OK

☐ Temperature OK

☐ Expiry date valid

✓ Safe X Reject

Rejection reason ▼

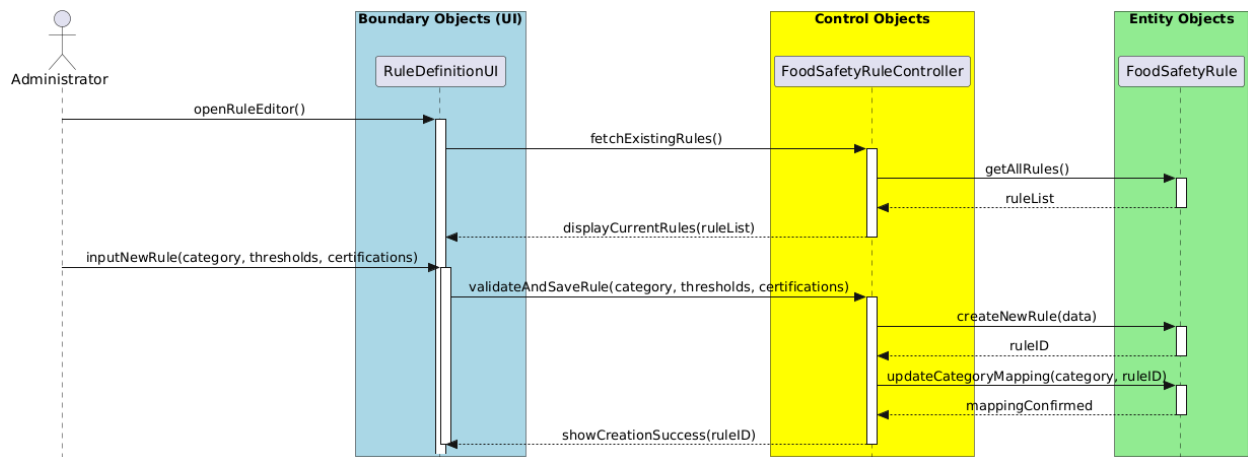
 Photo evidence

Generate report

D. UI

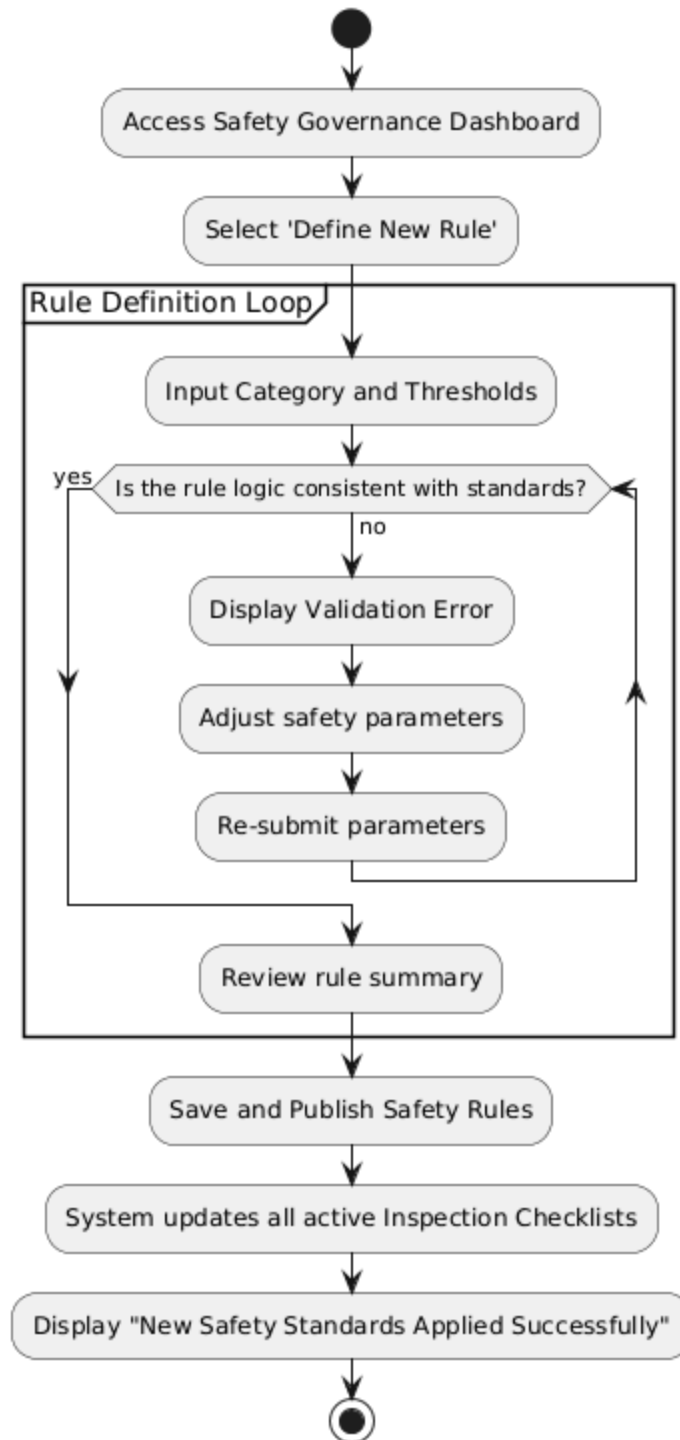
6.2 Define Food Safety Rules

A. Sequence Diagram



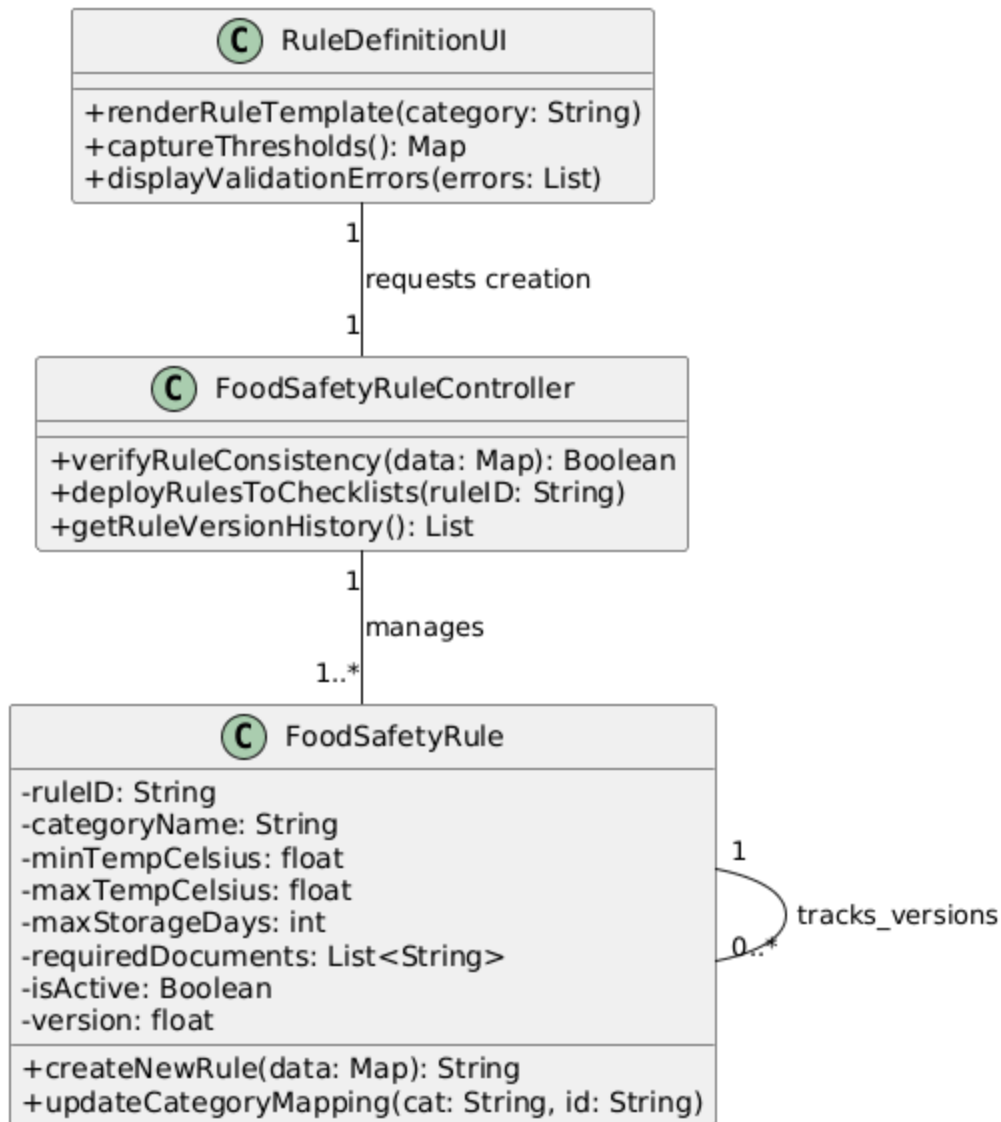
B. Activity Diagram

Activity Diagram - Define Food Safety Rules (User Perspective)



C. Class Diagram

Class Diagram: Define Food Safety Rules (Detailed)



D. UI

Safety governance

Category ▼

Max temp (°C)

Min shelf life (days)

⚠ Validation error —
adjust parameters

Rule summary

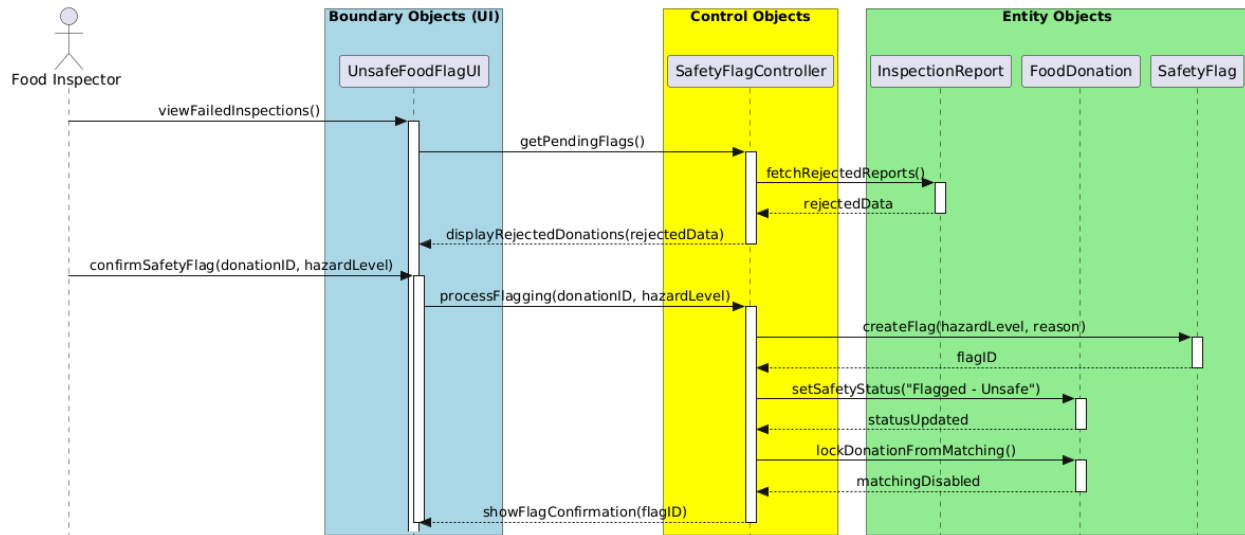
Perishables · $\leq 4^{\circ}\text{C}$ · 3 day
min

Save & publish

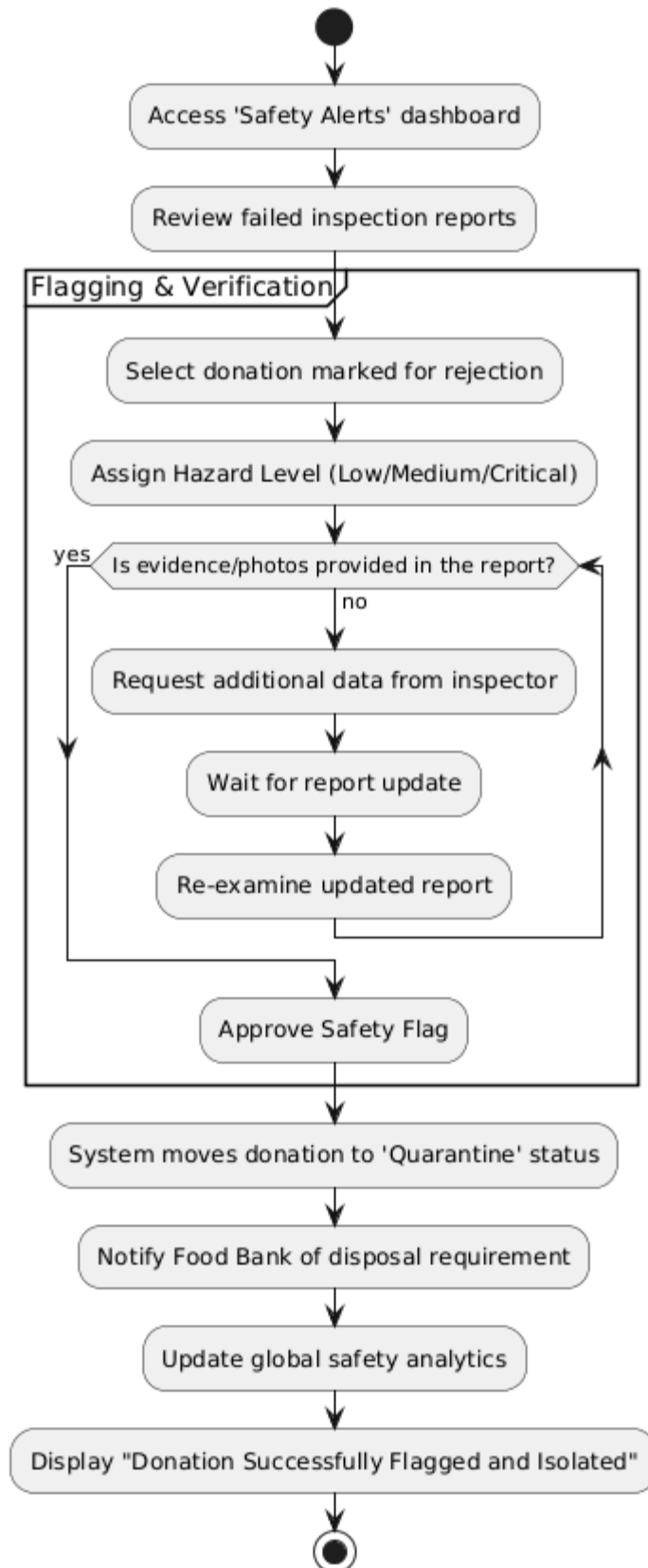
New safety standards
applied

6.3 Flag Unsafe Food

A. Sequence Diagram



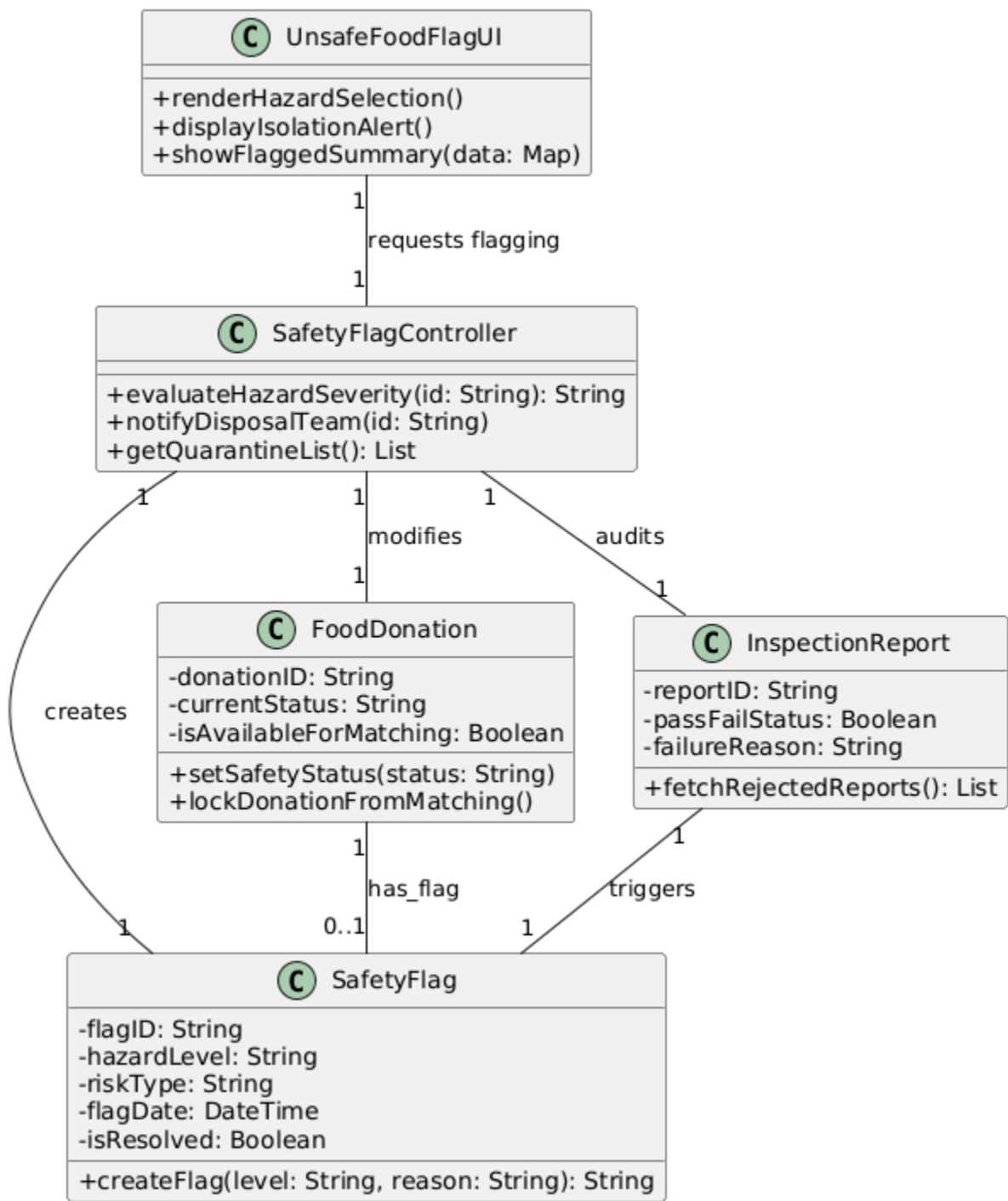
Activity Diagram - Flag Unsafe Food (User Perspective)



B. Activity Diagram

C. Class Diagram

Class Diagram: Flag Unsafe Food (Detailed)



D. UI

Safety alerts

**Batch #8821 —
rejected**
Expired labels · 3 items

Hazard level

Low

Medium

Critical

No photo attached —
request from inspector

Approve safety flag

Donation quarantined —
food bank notified

COMBINED CLASS DIAGRAM

Food Redistribution System - Detailed System-Wide Class Diagram

